

Technical Documentation
Turbocharger
Operating Instructions



Turbochargertype..... TCA66-2

6671-5 C1-5 EN

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General Safety Instructions

Safety Instructions

Safety instructions are always noted down prior to the description of any action. They are categorized in 3 hazard levels and differentiated by the degree of hazard seriousness for personnel or equipment.

Hazard Levels

The safety instructions for the hazard levels 1 - 3

- Danger,
- Warning,
- Caution

are indicated as follows:



When the respective caution measures are not complied with, there is imminent danger which will result in:

- Death or serious injury,
- Total damage to property or
- Serious environmental damage.



When the respective caution measures are not complied with, a potentially hazardous situation will be caused.

This can lead to:

- Death or serious injury,
- Serious damage to property or
- Serious environmental damage.



When the respective caution measures are not complied with, a hazardous situation can be caused.

This can lead to:

- Personal injury,
- Possible damage to property.



For notes concerning fault sources/handling errors; material damage can occur.

Additional notes, information and tips.

A symbol appropriate to the hazard is always depicted in the marginal column. It either indicates a hazard (hazard symbol) or the corresponding protective measure (safety symbols):

Safety Symbols

Wear
gloves



Wear
protective
goggles



Wear
hearing pro-
tection



Wear
hard hat



Wear
safety boots



Wear
protective
clothing



Read
manual

Danger Symbols

Warning of a
hazard loca-
tion



Danger of slip-
ping



Tripping haz-
ard



Danger of
electric shock



Corrosive
materials



Crush hazard



Inflammable
materials



Hot
surface



Danger of sus-
pended loads



Rotating
gear hazard

Bans

Fire,
open flames
and smoking
prohibited!



No access for
unauthorized
persons!

Safety Instruction Sign - Engine Room

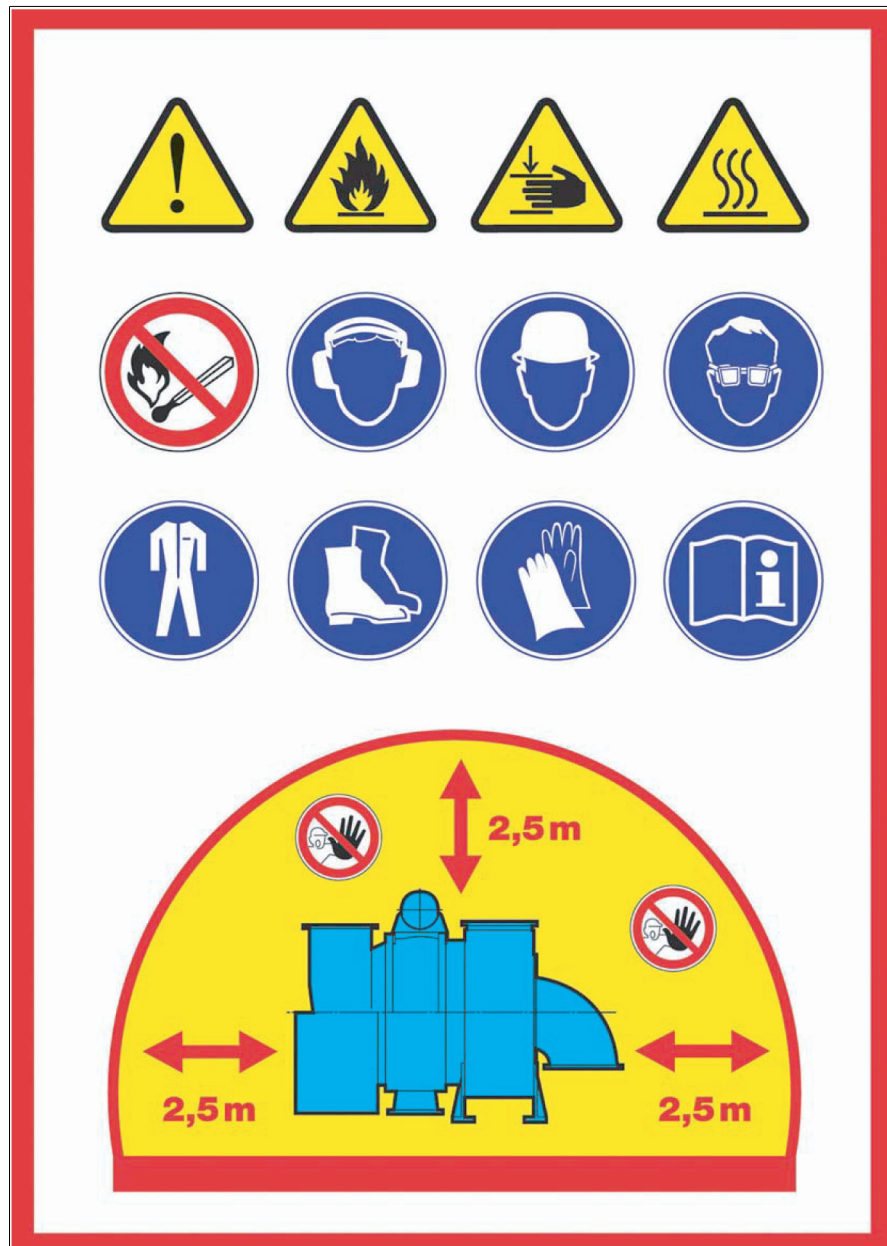


Figure 1: Safety Instruction Sign - Engine Room

NOTICE

To point out the most important regulations, bans and hazards during operation and maintenance, the safety instruction sign enclosed with each turbocharger must be posted at the entry of the engine room.

Persons, who for operational reasons are to be within the danger area of 2.5 m around the engine, are to be instructed of the predominant hazards. Access to the danger area is permitted when the turbocharger is in proper operating condition and only when wearing suitable protective equipment. It is prohibited to stay within the danger area without reason.

Product Liability

Personnel Requirements

Safe and economic operation of the engine system including the turbocharger requires extensive knowledge. Also, the operability can be maintained or regained only through maintenance and repair work when these are carried out with expertise and skill by trained personnel. The rules of proper craftsmanship are to be observed. Negligence is to be prevented.

Technical Documentation

The technical documentation supplements this ability through specialized information, draws attention to dangers and points out the safety regulations to be observed. The MAN Diesel SE asks you to observe the following guidelines:

WARNING



For improper usage of the turbocharger, the danger of accidents is given!

Improper usage of the turbocharger can lead to damage of the turbocharger or its individual components. Damaged components pose a potential risk of injury for the operating and maintenance personnel.

- Before putting into operation for the first time, carefully read the technical documentation at hand with the therein included safety instructions!
- Use the turbocharger only for the purpose intended for by the manufacturer!
- Do not unnecessarily stay in the vicinity of the turbocharger. Staying within the vicinity of the turbocharger is permitted for maintenance which is expressly to be carried out during operation!
- Carry out maintenance on the turbocharger regularly and properly within the specified intervals!
- Carry out any and all repair work exclusively with original MAN Diesel spare parts!

NOTICE

MAN Diesel SE disclaims all liability if the above-mentioned safety regulations are infringed.

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Addresses / Contact

Address

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Daegu, 704-946 Korea

<http://www.stxenpaco.co.kr>

Contact

Turbocharger Plant	Phone ++82 53 722 6100 Fax ++82 53 722 6699 E-mail : hj4018@onestx.com
After Sale Service	Phone ++82 53 722 6300 Fax ++82 53 722 6199 E-mail : mhpark@onestx.com

MAN Diesel | PrimeServ

Works in Germany

Contact persons

The table contains addresses for MAN Diesel in Germany, together with telephone and fax numbers for the departments responsible and ready to provide advice and support on request.

Location	Address	Telephone/Fax/e-mail/Internet	
Augsburg Works	MAN Diesel SE MAN Diesel PrimeServ Augsburg 86224 Augsburg Germany	Tel. Fax e-mail Internet	+49 821 322 0 (switchboard) +49 821-322-494180 primeserv-aug@mandiesel.com http://www.mandiesel.com
	PrimeServ Turbocharger Technical service	Tel. Tel. Fax e-mail Internet	+49 821 322 4010 (axial turbochargers) +49 821 322 4020 (radial turbochargers) +49 821 322 3998 primeserv-tc-technical@mandiesel.com http://www.mandiesel.com/primeserv
	PrimeServ Turbocharger Retrofit	e-mail Internet	primeserv-tc-retrofit@mandiesel.com http://www.mandiesel.com/primeserv
	PrimeServ Turbocharger Spare parts	Tel. Fax e-mail Internet	+49 821 322 4030 +49 821 322 3998 primeserv-tc-commercial@mandiesel.com http://www.mandiesel.com/primeserv
Hamburg Works	MAN Diesel SE MAN Diesel PrimeServ Hamburg Hachmannkai Rossweg 6 20457 Hamburg Germany	Tel. Fax e-mail Internet	+49 40 7409 0 +49 40 7409 104 primeserv-ham@mandiesel.com http://www.mandiesel.com/primeserv

Worldwide Service Addresses

Internet

MAN Diesel service addresses and authorized service partners (ASP) can be found on the Internet at:

<http://www.mandiesel.com/primeserv>

PrimeServ Worldwide

A printed brochure with the worldwide service addresses can be obtained from:

Turbocharger Service (administrative office)	
Tel.	+49 821 322 1198
Fax	+49 821 322 3998
e-mail	primeserv-tc-commercial@mandiesel.com

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3.1 Turbocharger

3.2 Operating Media/ Quality Requirements

Operating Data and Alarm Points

Turbocharger Type Plate

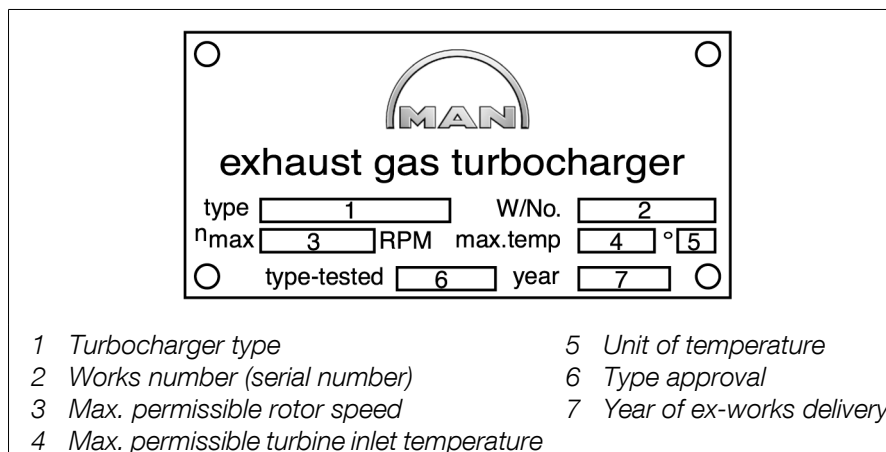


Figure 1: Type plate

NOTICE

The type plate is mounted on the pressure socket of the compressor casing or on the air intake casing (if provided).

Operating Parameters

Speed	Operating limit $n_{\max}$	see type plate
Exhaust gas temperature ahead of turbine	Operating limit $t_{\max}$	see type plate
Lube oil temperature	Inlet temperature Outlet temperature, max. Δ Inlet temperature/outlet temperature	40 — 70°C See engine start-up record ¹⁾ $\leq 40^\circ\text{C}$ ²⁾
Lube oil pressure	At regular operation and full load of engine Admissible lubricating oil pressure, max.	1.2 — 2.2 bar 2.2 (6.0) ³⁾ bar
Lube oil flow rate	with SAE 30 at 60 °C and 1.3 bar operating pressure	
	with SAE 30 at 60 °C and 2.2 bar operating pressure	
Noise development	Up to 105 dB (A) at a distance of 1 m from the turbocharger	
1) IBM = Putting into operation 2) Only for turbocharger retrofits 3) For engine lube oil pressures > 2.2 bar, an orifice or a pressure reducing valve is to be provided for in the oil supply to the turbocharger. See Chapter [4.2].		

Table 1: Operating parameters TCA66

Alarm Points

When exceeding the following operating data, an alarm is to be actuated by the engine control system:

Speed	Alarm actuates at	97% n _{max}
Exhaust gas temperature ahead of turbine	Alarm actuation at t _{max}	see type plate
Lube oil temperature	Alarm actuation at max. outlet temperature Alarm actuates when Δ inlet temperature/outlet temperature	See engine start-up record ¹⁾ ≥40°C ²⁾
Lube oil pressure	Alarm actuates at Reduction to engine half load (slow down) Engine shut down	<1.2bar <1.0bar <0.8bar
1) IBM = Putting into operation 2) Only for turbocharger retrofits		

Table 2: Alarm Points

Weights and Dimensions

Weights

500	Exhaust gas turbocharger, incl. silencer	5500kg
501	Gas-admission casing, complete 90°	380kg
506	Gas outlet casing, complete	1200kg
509	Gas outlet diffuser Shroud ring	232kg 20kg
513	Nozzle ring	37kg
517	Bearing casing, complete	910kg
518	Casing feet, complete	323kg
520	Rotating element, complete Turbine rotor Compressor wheel	200kg 130kg 66kg
540	Insert	198kg
542	Diffuser	64kg
544	Silencer, complete (if existing)	800kg
545	Intake casing, complete (if existing) axial 90°	237kg 384kg
546	Compressor casing, complete single outlet double outlet	829kg 813kg
554	Emergency and post-lubrication system, complete	140kg
595	Spare parts with box – see „List of Spare Parts“	See chapter 9.1
596	Tools with box – see „List of Tools“	See chapter 9.2

Table 1: Weights (approx. values)

Dimensions and Connections

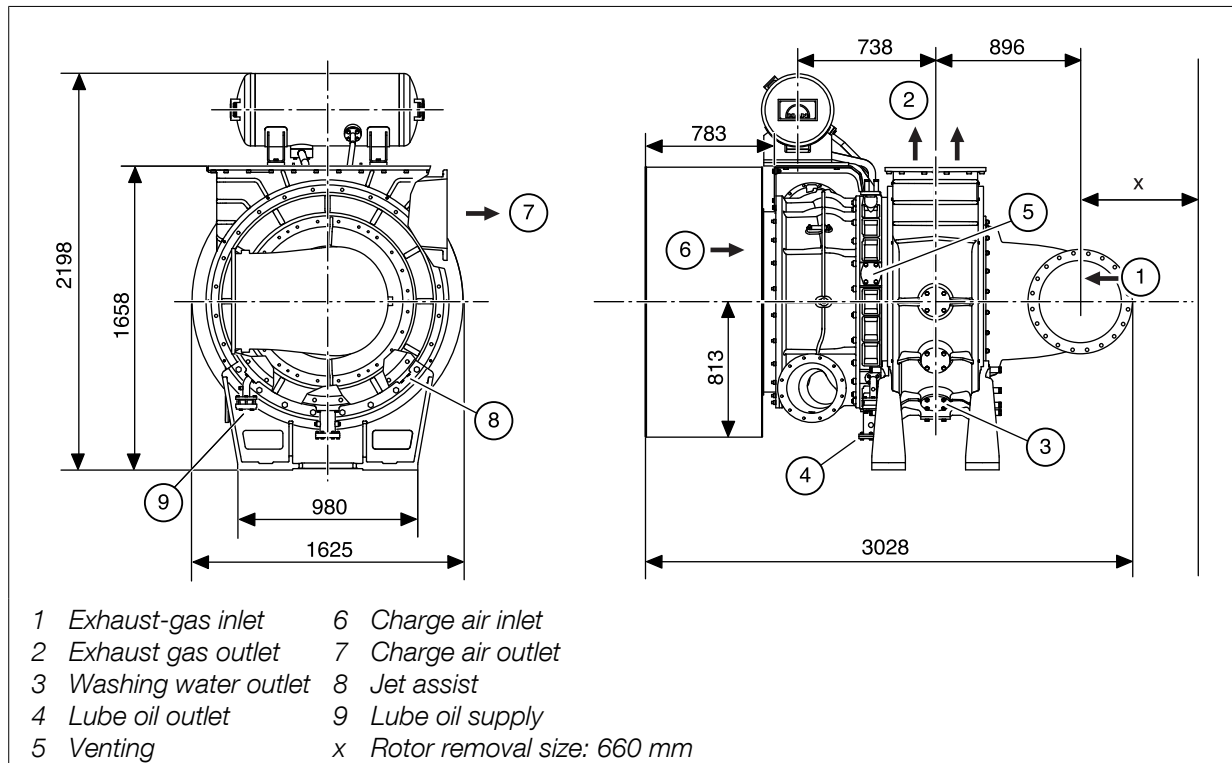


Figure 1: Main dimensions, TCA66 with 90° gas-admission casing

Allowable Inclinations

Allowable Inclinations

Installation position

The exhaust gas turbochargers of the TCA Series require horizontal assembly with respect to the axis of the rotating element.

Ship Applications

For operation in ships, however, where the installation position is crosswise to the longitudinal axis, inclination angles occur that can influence the operating ability of the exhaust gas turbocharger.

The following inclination angles can be handled by the exhaust gas turbocharger without problems.

Inclination	Continuous	Short-term
α in °	± 15	± 22.5
β in °	± 15	± 22.5

Table 1: Possible inclines for installation positions crosswise to the longitudinal axis

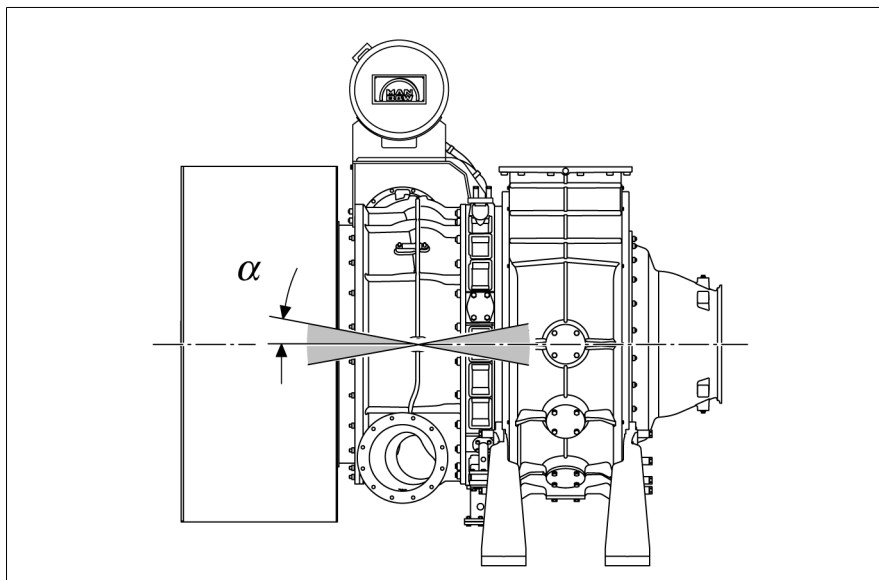


Figure 1: Turbocharger side view, installation crosswise to the longitudinal axis of the vessel

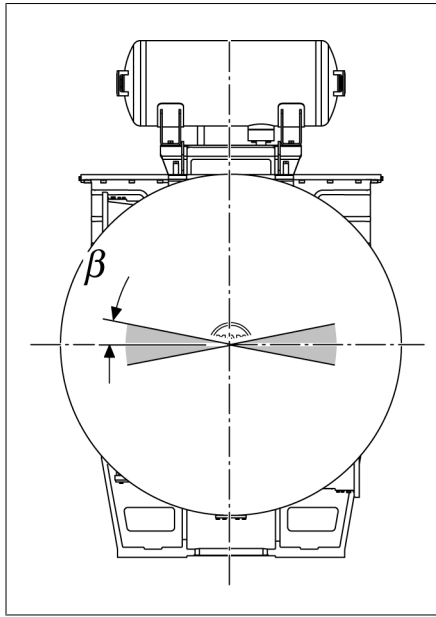


Figure 2: Turbocharger side view, installation crosswise to the longitudinal axis of the vessel

With the installation position in longitudinal direction of the ship, these limit values are not reached even under unfavorable exterior conditions.

3.1 Turbocharger

3.2 Operating Media/ Quality Requirements

Fuels for the Engine

Fuels for the Engine

Engine Fuel Affects Exhaust Gas Quality

The quality of the fuel with which the engine is operated affects the composition of the exhaust gas that flows through the turbocharger. Impurities in the fuel can lead to residues in the exhaust gas, which can effect the turbocharger in an abrasive or corrosive manner.

Quality Requirements for Marine Diesel Oil (MDO)

MDO is also known as diesel fuel oil, diesel oil, bunker diesel oil or marine diesel fuel.

MDO is offered exclusively for shipping as heavy distillate (ISO-F-DMB or as a mixture of distillate and low amounts of residual oil (ISO-F- DMC). The term “blended MDO” is common for the dark brown to black colored mixture. MDO is produced from crude oil and must be free of organic acids.

The usability of the fuel depends on the design of the engine and the cleaning device as well as if the characteristics mentioned below, which are based on the condition at the time of supply, are observed.

The characteristics determined were based on the ISO 8217-1996 and CIMAC-2003 specifications. The characteristics refer to the mentioned testing methods.

Properties/Characteristics		Unit	Testing Method	Characteristic	
ISO-F specification				DMB	DMC
Density	at 15 °C	kg/m ³	ISO 3675	900	920
Kinematic viscosity	at 40 °C	mm ² /s = cSt	ISO 3104	<11	<14
Pour point	Winter quality	°C	ISO 3016	<0	<0
	Summer quality	°C	ISO 3016	<6	<6
Flash point	Pensky Martens	°C	ISO 2719	>60	>60
Total sediment content		Wt. %	ISO CD 10307	0.10	0.10
Water content		Vol. %	ISO 3733	<0.3	<0.3
Sulphur content		Wt. %	ISO 8754	<2.0	<2.0
Ash content		Wt. %	ISO 6245	<0.01	<0.03
Carbon residue (MCR)		Wt. %	ISO CD 10370	>0.3	<2.5
Cetane number		–	ISO 5165	>35	>35
Copper strip test		–	ISO 2160	<1	<1
Vanadium content		mg/kg	DIN 51790 T2	0	< 100
Aluminum and silicium content		mg/kg	ISO CD 10478	0	<25
Visual check		–		*	–
Other specifications:					
British Standard BS MA 100-1987				Class M2	Class M3
ASTM D 975				2D	4D
ASTM D 396				No. 2	No. 4
* In good light and at room temperature, the fuel should appear clear and transparent.					

Table 1: Marine Diesel Oil (MDO) – Characteristics to be met

NOTICE

Mixing fuels can lead to a reduction of quality. This can result in combustion with a high degree of residuals, which can cause heavy contamination of the turbocharger. Extreme contamination can damage the turbocharger.

Therefore the following points are to be observed:

- In loading plants and during transportation, MDO is treated as residual oil. Mixing with, e. g. high-viscous fuel oil or Interfuel remaining in a bunker vessel is possible and can lead to a considerable reduction in quality.
- Different bunker batches of blended MDO (ISO-F DMC) can be incompatible and therefore should not be mixed. For this reason, the respective fuel tank should be emptied to the greatest possible extent before a new batch is filled up.

NOTICE

Ocean water in the fuel aides corrosion in the turbocharger and leads to high-residual combustion. Solid foreign matter increases the mechanical wear in the nozzle ring and the turbine of the turbocharger.

Therefore the following is to be observed:

- If blended MDO (ISO-F DMC) is primarily being used, we recommend installing a centrifugal separator in front of the fuel filter. This widely separates solid particles (sand, rust, catalyst residue, catfines) and water, so that the cleaning intervals for the filter inserts can also be prolonged.

Operating data:

- Separator admission 65%, with reference to the rated throughput capacity,
- Separation temperature 40-50 °C.

Quality requirements for
Heavy Fuel Oil (HFO)

MAN Diesel turbochargers can be operated on engines that run on crude-oil based heavy fuel oil, when the engine and the processing system are designed accordingly.

The fuels used must meet the fuel specifications in accordance with the Table "Fuel specifications and corresponding characteristics".

The respective limit values to be observed are listed in the Table "Fuel specifications and corresponding characteristics". The limit values that influence the engine operation are to be specified when ordering fuel, e. g. in the bunker or charter clause.

NOTICE

Adding motor oil (waste oil), mineral-oil-foreign materials (e. g. coal oil) and remainders from refining or other processes (e. g. solvent) is prohibited! This ban is specifically to be pointed out in the fuel order, as it is not yet part of the standard fuel specifications.

Such additions lead to combustion with high residue and increased wear and corrosion on components of the turbocharger. Adding motor oil (waste/old oil) is particularly critical, as the lube oil additives cause emulsions to form and keep debris, water and catalyst particles finely distributed in poise. This impedes or avoids the required fuel cleaning.

NOTICE

The heavy fuel oils ISO F-RMK 35/45/55, with a maximum density of 1010 kg/m³, can be used only when respectively modern separators are available.

Fuel Specification									
CIMAC 2003			A10	B/C10	D15	E/F25	G/H35	H45	H55
BS MA-100				M4	M5	M7	8/9	M8/–	M9/–
ISO F-RM			A10	B/C10	D15	E/F25	G/H35	H45	H55
System-relevant Characteristics									
Viscosity (at 50 °C)	mm²/s (cSt)	max.	40	40	80	180	380	500	700
Viscosity (at 100 °C)	mm²/s (cSt)	max.	10	10	15	25	35	45	55
Density (at 15 °C)	g/ml	max.	0.975	0.981	0.985	0.991			
Flash point	°C	min.	60						
Pour point (summer)	°C	max.	6	24	30	30			
Pour point (winter)	°C	min.	0	24	30	30			
Engine-relevant Characteristics									
Carbon residue (Conrad-son)	Wt.%	max.	10	10/14	14	15/20	18/22	22	22
Sulphur	Wt.%	max.	3.5	3.5	4	5		5	
Ash	Wt.%	max.	0.10			0.15		0.20	
Vanadium	mg/kg	max.	150	150/300	350	200/500	300/600	600	
Water	Vol.%	max.	0.5	0.5	0.8	1	1	1	1
Sediment (potential)	Wt.%	max.	0.1						
Supplementary Characteristics									
Aluminum + silicium	mg/kg	max.	80						
Asphaltenes	Wt.%	max.	2/3 of the carbon residue (Conradson)						
Sodium	mg/kg		Sodium< 1/3 vanadium, sodium< 100						
Cetane number of the fluid mixing component: min. 35									
Fuel to be free of mineral-oil-foreign additives such as coal oil or vegetable oil. Free of creosol and lubricating oil (waste oil)									

Table 2: Fuel specifications and corresponding characteristics (HFO)

NOTICE

Thorough processing of heavy fuel oil is required for trouble-free engine operation.

The following points are to be observed for this:

- Heavily abrasive inorganic, solid foreign substances (catfines, rust, sand) must be separated to the greatest possible extent.
- With an aluminum content >10 mg/kg, the abrasive wear in the turbo-charger increases heavily.
- Use only separators of the latest generation which are fully effective over a large density range without any adjustment, and separate water with

an HFO density of 1.01 g/ml at 15 °C. The cleaning effect is controlled by the separator itself.

- The HFO purification is to be designed in such a manner that the characteristics in the Table "Achievable foreign matter and water content after separation" are reached:

Characteristic	Particle Size	Amount
Inorganic, solid foreign substances (incl. catfines)	<5µm	<20mg/kg (AlSi content < 15 mg/kg)
Water	–	<0.2 vol.%

Table 3: Achievable foreign matter and water content after separation

- With unfavorable vanadium-sodium ratio, the melting temperature of the HFO ash drops to the range of the exhaust valve temperature, which causes hot corrosion. By pre-cleaning the HFO in the settling tank and in the centrifugal separators, the water and thus the water-soluble sodium compounds can be removed to the largest extent. With a sodium content exceeding 100 mg/kg, increasing salt deposits in the turbine are to be expected. This jeopardizes the turbocharger operation (by pumping of the turbocharger, among other things). When using PTG, the sodium content must be limited to 50 mg/kg. Under certain conditions, hot corrosion can be avoided with a fuel additive that increases the melting temperature of the HFO ash.
- Heavy fuel oils with a high ash content in form of foreign substances, e. g. sand, rust, catfines, increase the mechanical wear in the turbocharger. Heavy fuel oils from catalytic cracking plants can contain catfines. Generally, these are aluminum silicate, which effects high wear in the turbocharger. The determined aluminum content multiplied by 5-8 (depending on catalyst composition) approximately amounts to the content of catalyst material in the HFO.

Lubricating Oil

Lubricating Oil

Base Oil

The base oil must fulfil the following limit values, particularly concerning the aging stability:

Properties/Characteristics	Unit	Testing Method	Characteristic
Structure	–	–	Preferably paraffin-based
Behaviour in cold - still fluid	°C	ASTM-D2500	-15
Flash point acc. to Cleveland	°C	ASTM-D92	>200
Ash content (oxide ash)	Wt. %	ASTM-D482	<0.02
Carbon residue acc. to Conradson	Wt. %	ASTM-D189	<0.50
Aging tendency after 100 h of heating to 135 °C	–	MAN aging cabinet	–
n-heptane insoluble	Wt. %	ASTM-D4055 resp. DIN 51592	<0.2
Evaporation loss	Wt. %	–	<2
Drop test (filter paper)	–	MAN test	Must not show separation of resinous and asphaltic aging products.

Table 1: Lubricating oil – Characteristics to be met

Additives

Additives must be dissolved in oil and of such a structure that as little ash as possible results during combustion. The ash must have a soft structure. If this is not the case, increased formation of residue in the bearing casing of the turbocharger must be taken into account. Hard additive ash aids increased mechanical wear.

Additives may not aid a clogging of the filter inserts, neither in the active or processed condition.

Viscosity

SAE Class	Viscosity mm ² /s at 40 °C resp. 100°C
30 40	Preferred in the upper range of the SAE class assigned to the turbocharger

Table 2: Viscosity (SAE class) of lubricating oils

Lubricating Oil Additives

We urgently advise against the later addition of additives to the lubricating oil or mixing different lube oil brands together, as this can destroy the function of the additive package matched for the base oil. Such action also annuls the supplier's responsibility for the lubricating oil.

Intake Air

Intake Air

Intake Air Characteristics

The state as well as the condition of the intake air has a decisive influence on the performance of the turbocharger. Not only is the atmospheric condition of great importance, but also the degree of solid and gaseous impurities.

Mineral dust particles in the intake air have a wear-aiding effect. Chemical/gaseous components, on the other hand, have a corrosion-aiding effect.

For this reason, effective cleaning of the intake air and regular maintenance/cleaning of the silencer air filter mat is required.

The particle size in the intake air must not exceed 5 µm after the silencer/air intake casing or ahead of the compressor inlet.

The following maximum concentrations in the intake air may not be exceeded:

Properties/Characteristics	Concentration in mg/Nm ³ 1)
Dust (sand, cement, CaO, Al ₂ O ₃ etc.)	5
Chlorine	1.5
Sulphur dioxide (SO ₂)	1.25
Hydrogen sulphide (H ₂ S)	15
1) Standard cubic meter in Nm ³	

Table 1: Maximum concentration of particles in the intake air

When dimensioning the intake air system, pay attention not to exceed a total pressure loss (filter, silencer, piping) of 20 mbar.

Cleaning Agents for Compressor and Turbine

Wet Cleaning of the Compressor and Turbine

NOTICE

Use freshwater only.

Salt water causes corrosion on the turbocharger components.

Wet Cleaning

For wet cleaning of the compressor and turbine during operation, use only freshwater without chemical additives.

Dry Cleaning of the Turbine

Dry Cleaning

For dry cleaning of the turbine during operation, commercial granules are used that must meet the following specifications:

Granulate of:

- Nutshells,
- Activated charcoal (soft).

Particle size:

- 1.0mm (max. 1.5mm).

NOTICE

Store and mark granulates in such a manner that they can not be mistaken for other materials (salt, sand, etc.).

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Turbocharger

General

Economical operation of modern large-scale engines is not imaginable without exhaust gas turbochargers. The already high requirements for propulsion systems and energy producing units concerning efficiency and longevity are being continuously increased under the aspects of fuel utilization and environmental load.

In this, the components of exhaust gas turbochargers are subject to extreme operating conditions.

- Exhaust gases of up to 650 °C continuously flow through the turbine and heat up its components, without an own counteractive cooling system. Especially the shaft bearing must withstand the high operating temperatures without the lubricating film ever breaking.
- On the compressor side, the air is heated to over 200 °C.
- The high temperatures lead to extreme thermal loads of the material at many locations.
- Speeds are extremely high: The MAN Diesel exhaust gas turbochargers are operated with speeds ranging from 10 000 to 35 000 rpm, depending on size.. In this, circumferential velocities of 530 m/s and more are reached at the compressor wheel, which amounts to 1.7 times the speed of sound or 2 000 km/h.
- The centrifugal forces are extremely high: Forces of several 100 kN can easily apply at the foot of the turbine blade.
- The complete gas exchange of the engine is performed by the exhaust gas turbocharger. For this machine, the throughput of combustion air can amount to 24 m³/s.
- Simplified, it can be said that approx. 1/3 of the power produced by the engine is converted on minute space within the exhaust gas turbocharger.

These requirements can be fulfilled only with use of the most recent material and manufacturing technologies, introduced into the series by MAN Diesel with use of the latest developmental results, and based on decades of experience in building diesel engines and exhaust gas turbochargers.

NOTICE

Prescribed maintenance must be carried out precisely and in time using the prescribed tools, in order to keep the turbocharger in good condition and not damage it.
See Maintenance Schedule.



Subassemblies

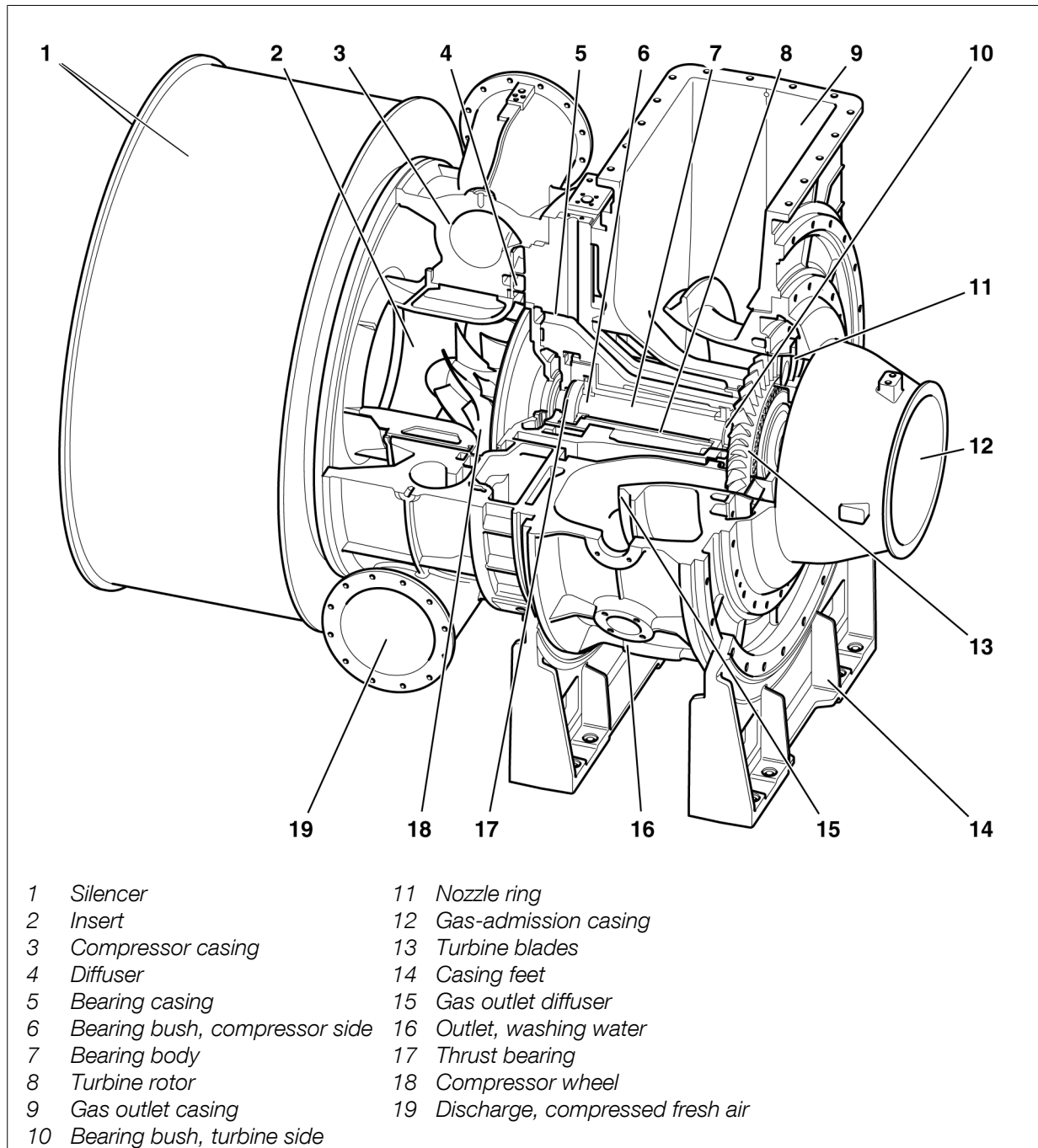


Figure 1: Cutaway view, TCA66

Turbochargers consist mainly of a turbine and a compressor, which are seated on the same shaft. The exhaust gas of the engine drives the turbine; the compressor draws in fresh air and compresses it.

The turbocharger consists of the following main subassemblies:

- Rotating element:

Turbine wheel and shaft are firmly connected together; the turbine blades are individually set into the turbine wheel. The compressor wheel is mounted on the shaft and can be removed.

- **Bearing casing:**
The interior bearing of the rotating element consists of two bearing bushes and a thrust bearing. Lubrication of the bearing is carried out via the lube oil circuit of the engine. Lubricating oil pipes, lube oil venting and sealing air pipes are integrated in the bearing casing.
- **Gas-admission casing:**
The nozzle ring is built into the gas-admission casing. It enables optimum adaptation of the turbocharger to the engine.
- **Gas outlet casing:**
The gas outlet diffuser in the outlet casing is flow-technically optimized. The outlet casing is fitted with 5 offset connections for the washing water outlet. Depending on the build-in position of the turbocharger, the connection positioned lowest is used.
The outlet casing is designed so that together with the flanged-on gas-admission casing, it offers optimum burst protection for the turbine wheel.
- **Silencer or air intake casing**
- **Compressor casing optional with one or two discharge connections.**
The compressor casing houses the diffuser, which allows for optimum adaptation of the turbocharger to the engine. Additionally, the diffuser functions as burst protection.

Functional Description

The exhaust gas of the engine flows through the gas-admission casing and the nozzle ring, and runs axially onto the turbine wheel. The exhaust gas drives the turbine wheel; in this process, the energy contained in the exhaust gas is transformed into mechanical rotation energy at the turbine wheel. As the turbine wheel and the compressor wheel are seated on the same shaft, the compressor wheel is driven at the same time. The exhaust gas exits the turbocharger through the gas outlet diffuser and the gas outlet casing.

The compressor wheel draws in fresh air through the silencer or the intake casing and the insert. The fresh air is compressed in the compressor wheel, diffuser and compressor casing. The compressed fresh air is forced into the cylinders of the engine via charge air cooler and charge air pipe.

The rotating element of the turbocharger is led radially by two bearing bushes, which are situated in the bearing casing between turbine wheel and compressor wheel. The thrust bearing positioned on the compressor side not only handles the axial guidance, but also transfers the thrust in axial direction. A bearing body holds the bearing seat and at the same time is used as insulation against the hot exhaust-gas side of the turbocharger.

Intended Use

The turbocharger is intended for use on diesel and heavy-fuel-oil engines. It produces the required amount of air for engine operation and the charge pressure. Any application beyond this must be discussed with and approved by MAN Diesel, otherwise MAN Diesel will refuse any and all liability.

Each turbocharger is specially adapted to the engine with which it will be operated. This means the turbocharger may be operated only on this engine or one that is absolutely comparable. (Also see scope of supply, technical specifications and order confirmation).

In addition to this, the intended use includes as follows:

- The build-in location must afford sufficient space for maintenance and assembly/disassembly of the turbocharger. Usage of lifting equipment must be possible.
- Supply, measuring, control and regulating systems must enable trouble-free operation (design/lay-out in accordance with MAN Diesel recommendation, resp. state of art).
- Observing the operating parameters: Admissible exhaust gas temperature, admissible lube oil temperature and lube oil pressure, use of the specified operating materials. and See chapter 3.2.
- The turbocharger may be put into operation only when in perfect condition.
- Operation may be carried out only by trained personnel which is familiar with the system; operation through unauthorized personnel is to be prevented with suitable means. For operation, the operating manuals and the general safety and operational regulations are to be observed.
- The operation must be controlled by means of suitable measures.
- Malfunctions that could influence the safety are to be corrected immediately.
- Performing the service and maintenance in accordance with the maintenance schedule.
- For repairs, use original spare parts only.

Not Intended Use

The following points specifically do not comply with the intended use. MAN Diesel shall not be held liable for any or all damages resulting therefrom.

- Unauthorized changes/alteration on the turbocharger.
- Installation of foreign (non-original) parts.
- Operation not within the defined operating parameters.
- Bridging/shutting off alarm systems, and thereby enabling operation in spite of limiting values being exceeded.
- Operation when supply systems have failed (e. g. lubricating oil system).
- Insufficient maintenance.
- Longer-term shut downs without preservation measures.

Lube Oil System

Lube Oil Diagram

Description

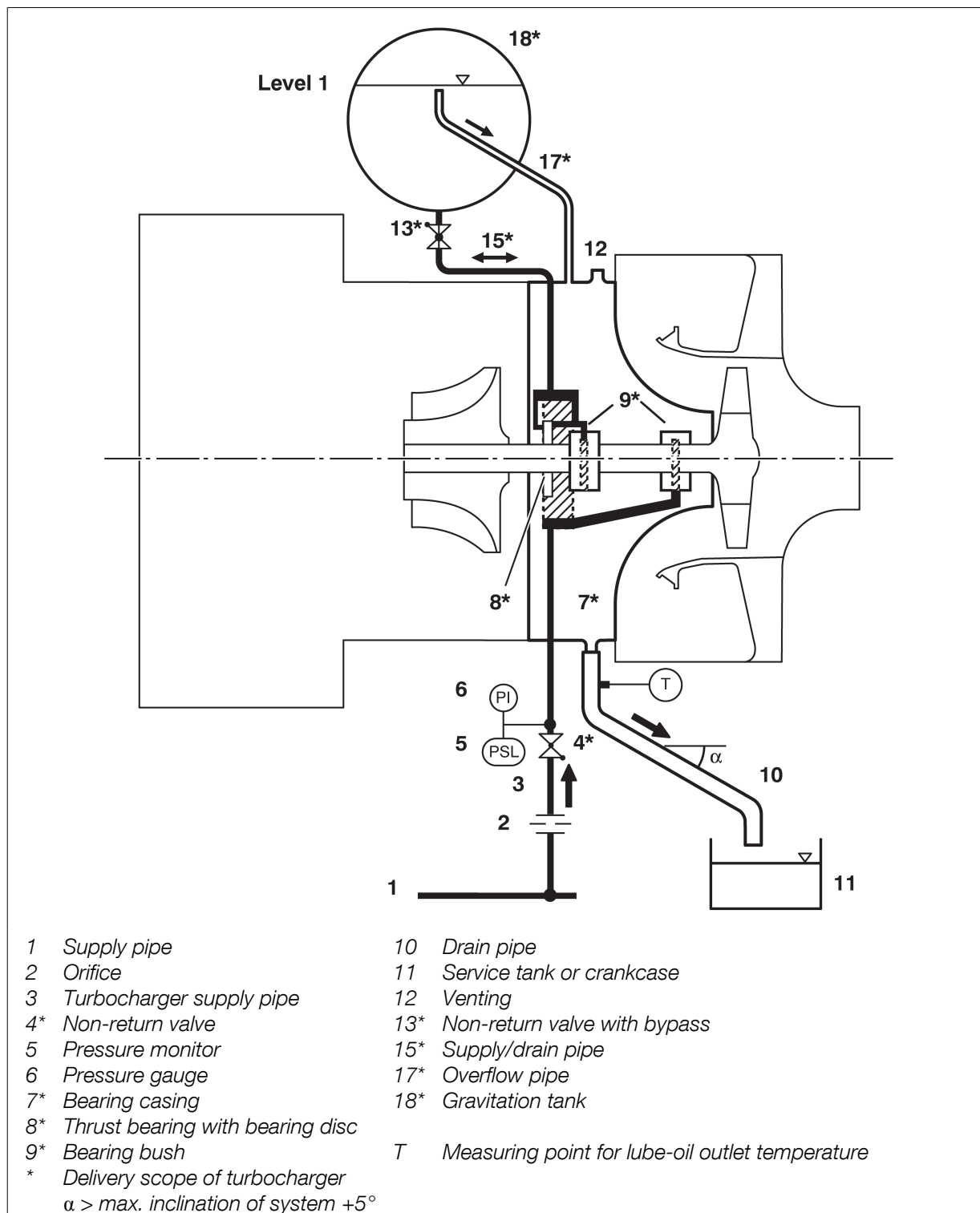


Figure 1: Lube Oil Diagram

Description
TCA

Functional Description

Lube Oil Circuit

The lubrication and cooling of the high-stressed bearing bushes in the turbocharger takes place by means of a lube oil system, which is integrated mainly in the bearing casing.

The lubricating oil is supplied from the lube oil system of the engine to the lube oil system of the turbocharger via a supply pipe (1). An orifice (2) adjusts the required lube oil pressure. The lube oil pressure is controlled behind the non-return valve (4) by means of a pressure monitor (5) and a pressure gauge (6).

NOTICE

Air bubbles may not be contained in the oil being supplied. This could lead to bearing damage at lubrication of the turbocharger.

The lubrication oil flows through the non-return valve (4) into the turbocharger casing, from where it reaches the thrust bearing (8) and the bearing bushes (9) via passages in the bearing casing (7) and the bearing body. The lubricating oil flows to the gap between bearing and shaft as well as to the face-sided lubrication point of the thrust bearing via bores in the bearing bushes. The lubricating oil leaves the gap between the bearing and the shaft and is splashed against the wall of the bearing casing by the rotation of the shaft. The lubricating oil exits the bearing casing through the drain pipe (10) and flows back into the lube oil system of the engine (11).

CAUTION



Danger of burning/scalding from running-out lube oil!

A backup in the lube oil outlet can lead to an oil-flood in the turbocharger. In the worst case, the oil can come out of the turbine-side and cause a fire in the exhaust pipe.

- The pipe for the lube oil drain must be installed with sufficient large diameter and inclination, so that drainage of the lube oil is readily possible without backing up.

The lube oil outlet must be installed with an inclination, which is calculated as follows (see Fig. [Lube-oil system](#)):

$$\text{Inclination } \alpha > \text{max. possible system inclination} + 5^\circ$$

Venting

Due to constructive measures, this turbocharger does not require a separate venting tank. Lubricating oil and air are separated from each other within the bearing casing. The connection for the venting pipe (12) is attached directly at the bearing casing.

Gravitations Tank

The gravitation tank (18) above the turbocharger ensures the lubrication of the bearings after the engine is switched off, while the rotating element runs out and the bearings still must cool down.

During operation the non-return valve (13) is shut. The gravitation tank is filled with lubricating oil via a small bypass bore in the valve plate of the non-return valve (13). As soon as the lube oil level reaches the height of the overflow pipe (17) (level 1), lubricating oil flows back into the bearing casing through the overflow pipe.

Shaft Sealing	If the engine is shut off, the pressure in the lube oil system is reduced. This causes the non-return valve (13) to open and the non-return valve (4) to shut. The lubrication oil flows out of the gravitation tank back into the turbocharger and cools the bearing points in the process. In case of a „blackout“, the gravitation tank serves as an oil reservoir for approx. 10 seconds of emergency operation of the turbocharger without lube-oil supply. The bearing casing is sealed with labyrinth sealings on the rotor shaft, both on the turbine and compressor side. The radial labyrinth clearance is dimensioned so that during the first operating phase, the rotating labyrinth tips lightly embed into the softer layer of the cover seal. At higher speeds, the rotating element is lightly elevated according to the thickness of the lubricating film. The labyrinth tips run freely then. When standing still, the rotating element is lowered again. The labyrinth tips embed into the grooves of the cover seal, through which a better sealing effect is achieved during pre and post lubrication. Run-in grooves at the circumference of the sealing cover bores are thus intentional and not a reason for replacement of the parts. On the turbine side, the shaft is additionally sealed by means of sealing air (see Chapter [4.3]).
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Operating Conditions

Lube Oil Quality	Commercial engine lubricating oils SAE 30 and SAE 40.
Lube Oil Filtration	High-alkaline cylinder oils are not suitable. See Chapter [3.2.2] The turbocharger does not require any separate lube oil filters. The filtration and processing of the lubrication oil through the engine system is generally sufficient. The following values must be observed: ▪ Lube oil filter ≤ 0.050 mm, ▪ Accumulation with water $< 0.2\%$ amount of weight
Lube oil temperature	Inlet temperature in the turbocharger: See Chapter [3.1.1]
Lube oil pressure	Admissible lube oil pressures in operation: See Chapter [3.1.1] For engine lube oil pressures > 2.2 bar to 6.0 bar max., we recommend installing an orifice(2). For this, the measuring connection for control of the lube oil pressure must be provided for behind the non-return valve (4) (see Fig. Lube oil diagram).

NOTICE

Differences in height between the indicating instrument and the turbocharger centreline are to be taken into account with ± 0.1 bar per 1 m height difference (indicator installed higher – lower pressure; indicator installed lower – higher pressure).

Alarm Points	Lube oil pressure at engine start and cold lube oil: For a short time up to 4 bar. Limit values for control of the lube oil pressure: See Chapter [3.1.1]
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NOTICE

When the limit values are reached, the alarm must be triggered and the engine control must react at the same time (without delay) in order to avoid damage to the turbocharger.

Lube Oil Flow Rate	The required lube oil flow rate depends on the viscosity and temperature of the lubrication oil.
Pre-lubrication	Lube oil throughput, see technical data, Chapter [3.1.1]. Before starting the engine, the bearing points of the turbocharger must be pre-lubricated. This is done automatically together with the pre-lubrication of the engine, as the lube oil system of the turbocharger is connected to that of the engine. Depending on the engine system, pre-lubrication occurs directly before starting, either through interval or continuous pre-lubrication. Pre-lubrication before starting: ▪ 10 minutes min., oil pressure 0.6 - 2.2 bar.
Post Lubrication	Occurs automatically via the gravitation tank.

Connecting Several Turbochargers to the Engine Lube Oil System

If an engine operates with several turbochargers, then these are connected parallel to the lube oil system of the engine (see Fig. Lube oil system – Connection of several turbochargers).

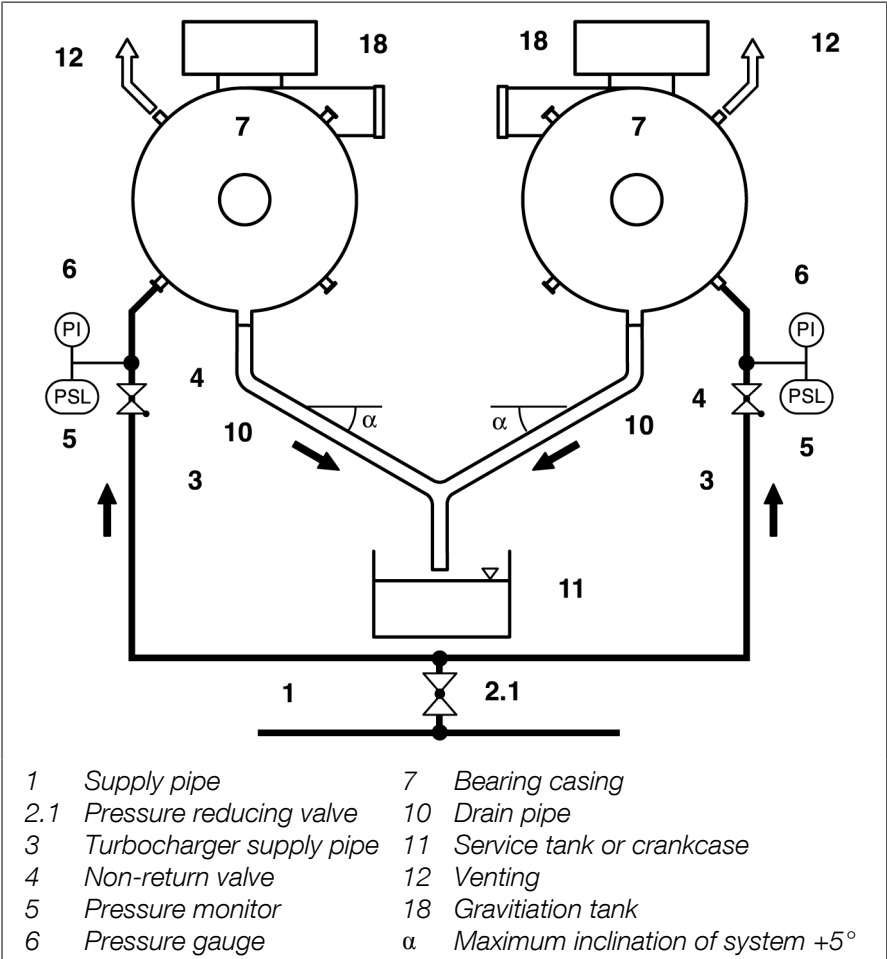


Figure 2: Lube oil system – Connection of several turbochargers

2009-05-07



Sealing Air System

Sealing Air Diagram

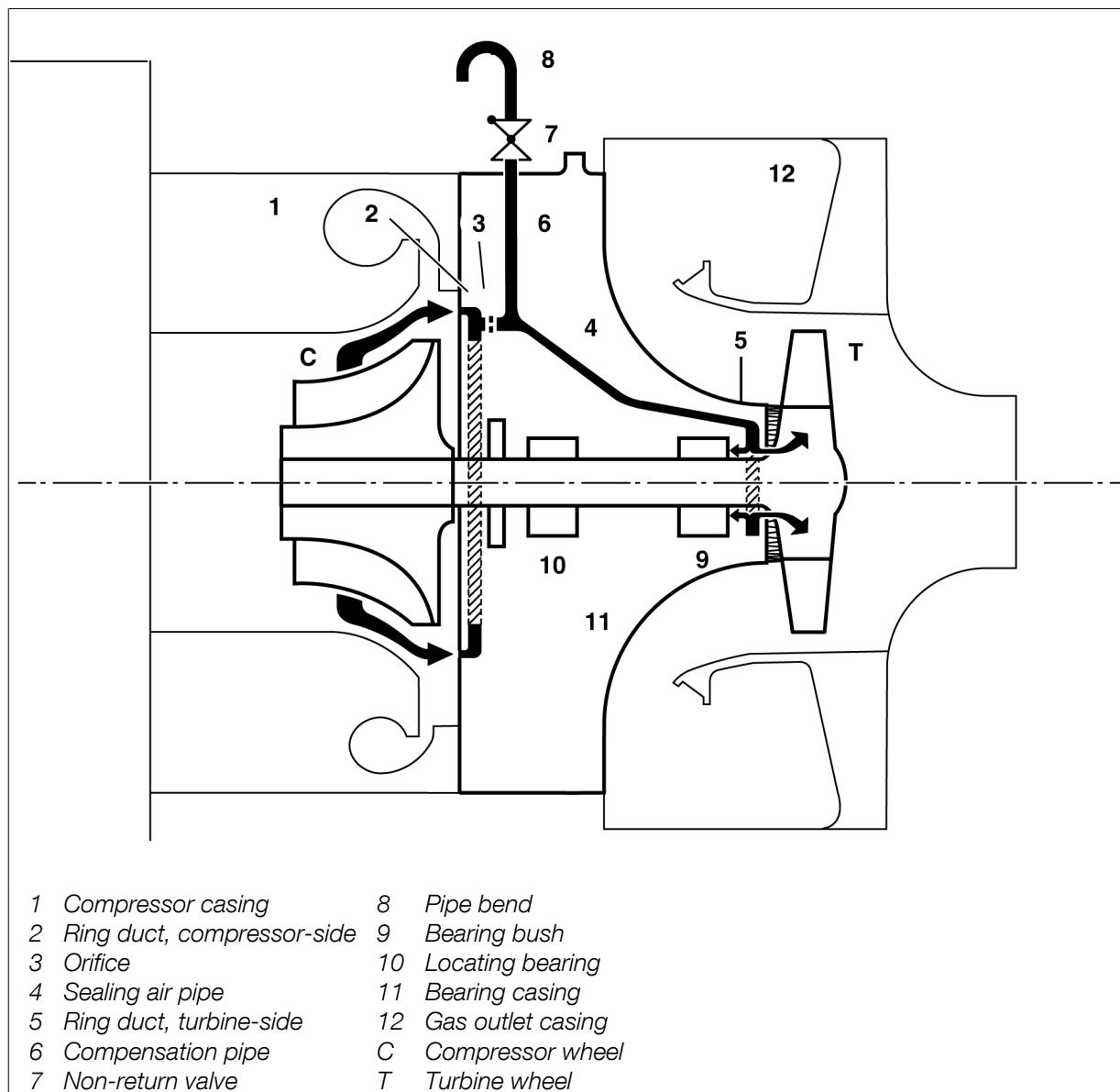


Figure 1: Sealing Air Diagram TCA66

Functional Description

Sealing Air System

The sealing air prevents hot exhaust gas from entering the bearing casing and the lubricating oil from seeping into the turbine (oil coke). Additionally, undesirable axial thrust on the bearing bush is reduced.

The sealing air system is fully integrated in the bearing casing (11). A part of the air compressed by the compressor wheel (C) is diverted and flows out of the compressor casing (1) into a ring duct (2) in the bearing casing. From there, the air is led into the sealing air pipe (4), whereby an orifice (3) reduces the pressure to the required sealing air pressure. The air is led to a ring duct

Description

Description
TCA66-2

(5) on the turbine side of the bearing casing. There, the sealing air emerges between shaft and turbine labyrinth.

- A small amount of the sealing air flows back into the bearing casing, presses against the turbine-sided bearing bush (9), and thus holds back the lubricating oil.
- The other part of the sealing air is led past the rotor shaft, through the labyrinth seal on the turbine side, and into the gas outlet casing (12).

This develops a thrust-compensating piston between the labyrinths on the turbine disc and bearing casing.

The sealing air pressure is factory set via the orifice (3) and must not be controlled or readjusted by the user.

Measures in Case of Vacuum

With four-stroke engines, a vacuum can develop on the compressor side at reduced partial load (suction operation). In this case, a compensation pipe (6) between sealing air pipe and ambient air prevents lubricating oil and exhaust gas from being drawn into the compressor casing.

A non-return valve (7) blocks the compensation pipe during regular operation. In case of a vacuum in the sealing air system, the spring-free non-return valve opens and outside air is drawn in through the compensation pipe.

Acceleration System „Jet Assist“ (Auxiliary Air Drive)

Diagram „Jet Assist“

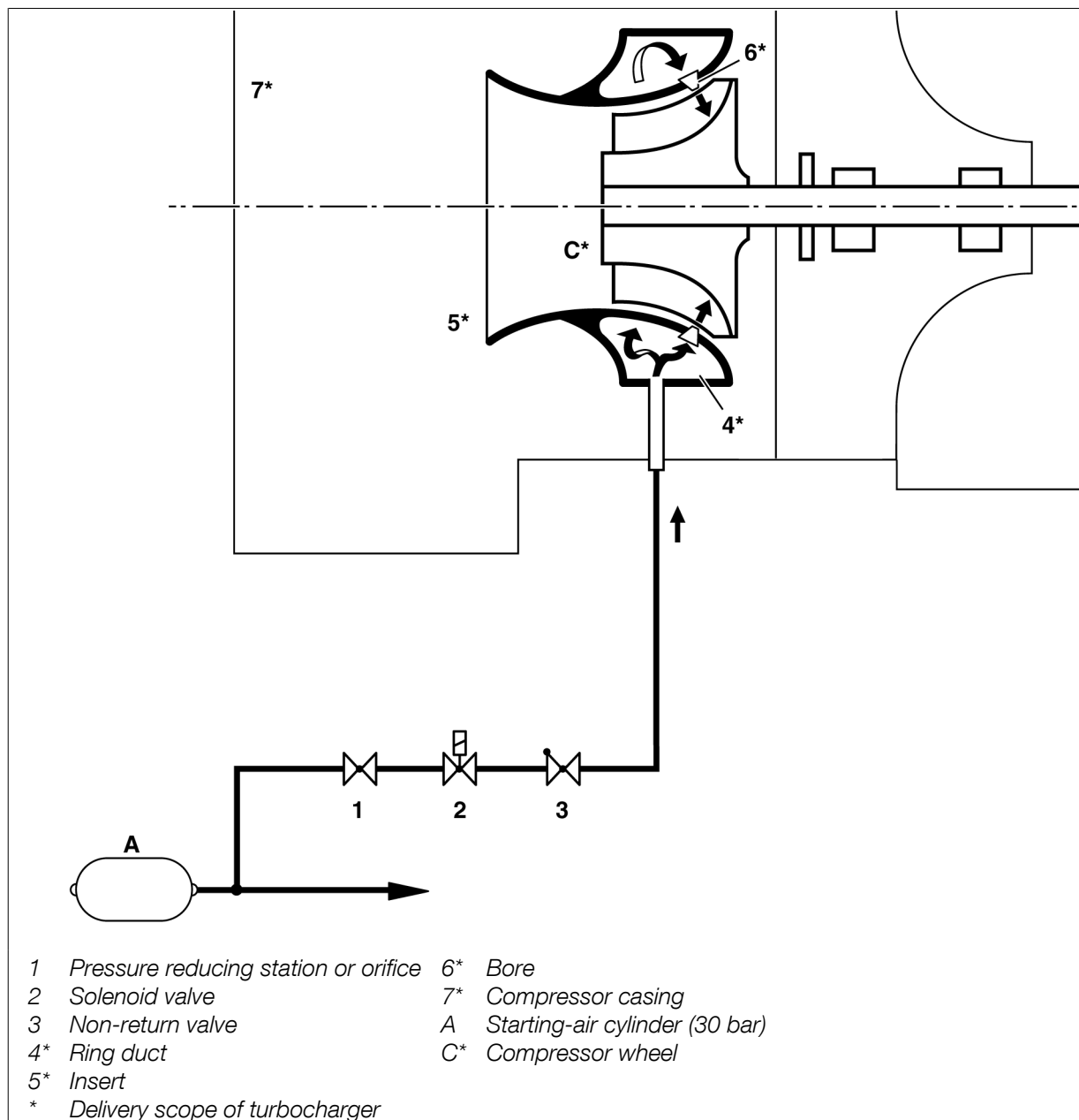


Figure 1: Jet assist diagram, TCA66

Functional Description

"Jet Assist"

The „Jet Assist“ acceleration system is used when special requirements are made towards swift and possibly soot-free acceleration, and/ or towards the load applications of the engine.

The engine control actuates the solenoid valve (2). Compressed air of 30 bar now flows from the starting-air cylinder through the pressure reducing station or orifice (1), where it is reduced to a maximum of 4 bar. The com-

Description
TCA66

pressed air reaches the compressor casing (7) via a non-return valve (3) at a maximum of 4 bar, from where it is conducted around the insert (5) via the ring duct (4). The compressed air is blown onto the compressor wheel (C) through several inclined bores in the insert. On the one hand this provides additional air to the compressor, while on the other hand the compressor wheel is accelerated, thus increasing the charge air pressure.



Cleaning System – Dry Cleaning of the Turbine

Dry Cleaning of the Turbine

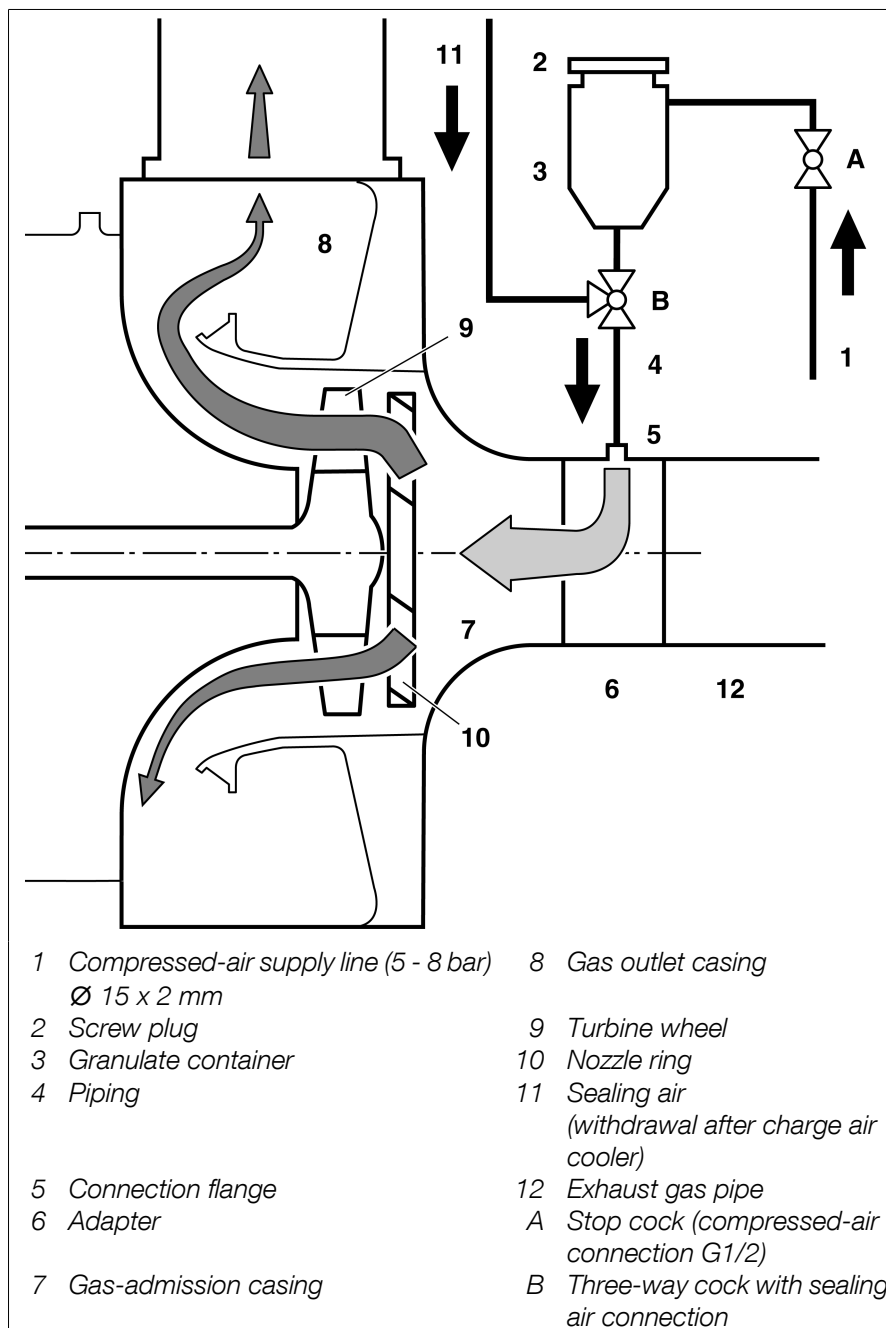


Figure 1: Diagram, dry cleaning of the turbine

NOTICE

Type and source of the hazard

The connection flange (5) can be attached either at the adapter (6) of the exhaust gas pipe (12) or directly at the gas-admission casing (7).

Description

Description
TCA

Functional Description

The dry cleaning of the turbine is performed during operation at normal load of the engine.

The granulate container (3) is equipped with a filling opening, a compressed-air supply line (1) and a pipe (4) leading to the gas-admission casing (7). The pipes for compressed air are fitted with stop cocks (A) and (B).

The granulate container is filled with cleaning granulates and shut tight. The stop cock (A) in the compressed-air supply line is opened and compressed air flows into the granulate container. Afterwards, the three-way cock in the pipe leading to the gas-admission casing is opened. The compressed air blows the granulates into the gas-admission casing. There, the exhaust gas flow transports the granulates to the turbine wheel. The granulates bounce against the nozzle ring and turbine wheel, thus removing deposits and contamination. The exhaust gas flow carries the granulates and contamination particles out of the system.

After the cleaning process has been carried out, the stop cock (A) and the three-way cock (B) are shut again. The sealing air (11) prevents clogging of the pipe during operation of the turbocharger. For this, a minor quantity of the air after the charge air cooler is used for continuous flow through the respective pipe system.

NOTICE

Cleaning the turbine with granulates on ship systems can lead to downfall on deck.

NOTICE

Type and source of the hazard

Granulates to be used for cleaning: See Chapter Technical Data [3.2.4]
Specified maintenance interval, see Maintenance Schedule [6.3.1]
For a detailed description on how to operate the cleaning device, see Work Card [500.22]

Assembling Instructions

The cleaning device is factory-mounted on a bracket at the gas outlet casing (8).. The pipings from stop cock (A) on are specifically adapted to the respective turbocharger, and included in the MAN Diesel delivery scope. For later equipping of the turbocharger with the cleaning device, the following points are to be observed:

- The granulate container (3) must be fastened at a suitable location. It may not be positioned lower than 1 m below the connection flange (5).
- The pipe (4) may not be longer than 6 m and must be supported against vibrations. Ensure unobstructed flow.
- Maximum operating temperature of the three-way cock (B) (exhaust gas): ≤ 150 °C.
- The connection flange (5) can be attached either at the adapter (6) of the exhaust gas pipe or directly at the gas-admission casing (7).

Cleaning System – Wet Cleaning of the Turbine (Optional delivery scope)

Wet Cleaning of the Turbine

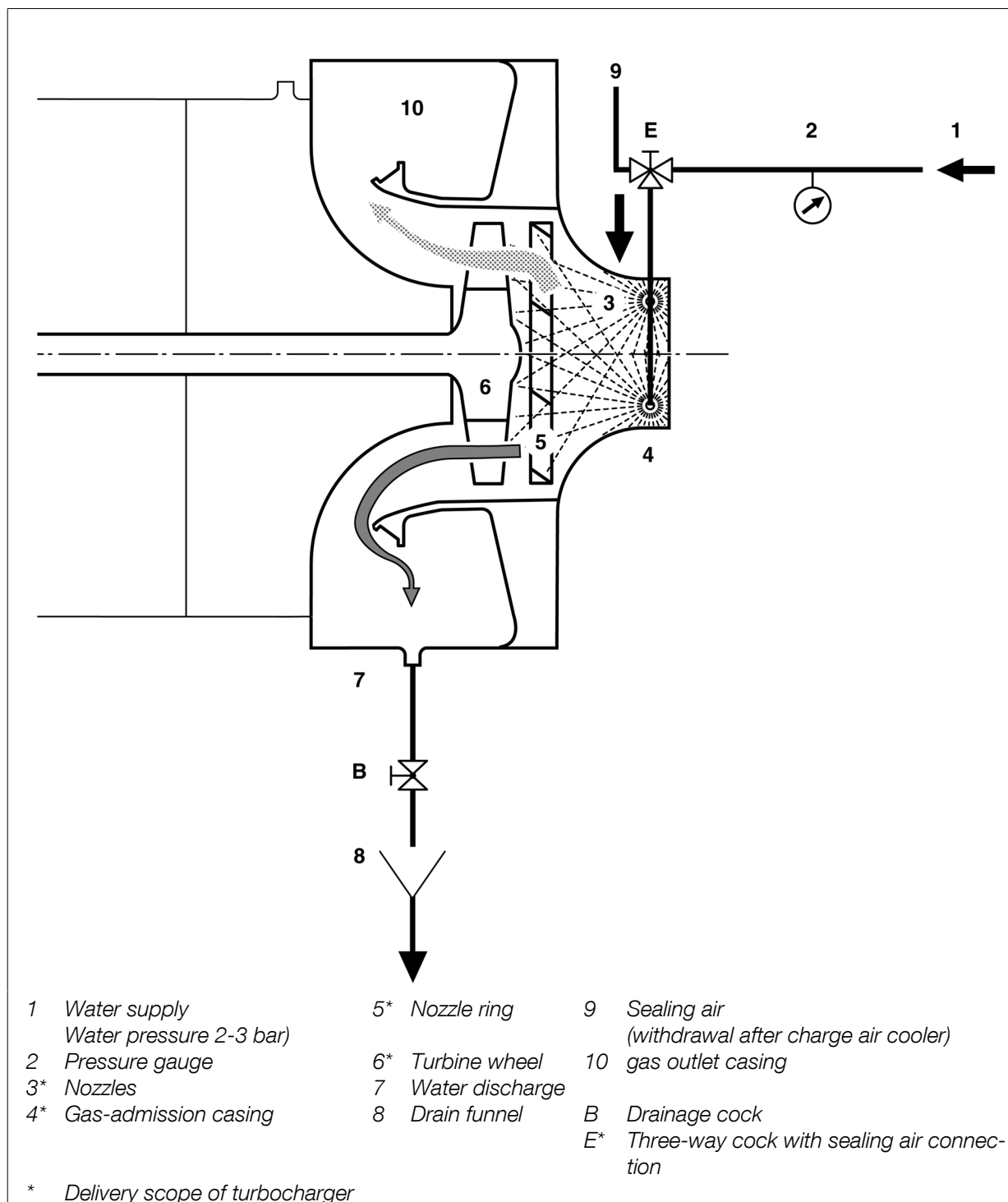


Figure 1: Diagram, wet cleaning of the turbine

Description

Description
TCA

Functional Description

Introductory Remarks	Depending on the quality of the fuel and the operating method of the engine, combustion residue can deposit in the turbine. For this reason, the turbine must be cleaned at regular intervals in accordance with the maintenance schedule. Cleaning of the turbine is carried out during operation of the turbocharger. In addition to cleaning the turbine with granulate(see Chapter Dry Cleaning [4.5]), the turbine can optionally also be cleaned with water (wet cleaning). In this, the advantages of both cleaning methods supplement each other.
Wet Cleaning of the Turbine	The wet cleaning is performed during operation with heavily reduced engine load, in order to avoid overload of the turbine blades (thermo shock). The washing water flows into the gas-admission casing (4) via three-way cock (E). The washing nozzles spray out the washing water into the gas-admission casing ahead of the turbine. The washing water droplets bounce against the nozzle ring (5) and the turbine (6), where they wear off contamination/deposits as follows: ▪ Mechanical cleaning from the impact of the water droplets ▪ Accumulated combustion residue are dissolved by the washing water ▪ The water vapor lightly reduces the temperature at the turbine blades. Due to the contraction of the turbine blade material caused by the lower temperature, the deposits on the turbine blades chip/crack off. The majority of the washing water is forced out into the open as steam via the flue, owing to the high flow rate in the gas outlet casing. A minor quantity of the washing water runs off through the washing water drain (7) and the drain valve (B). The washing water is conducted via a funnel (8) to a sediment tank, where it is collected. The sealing air (9) prevents clogging of the pipes during operation of the turbocharger. For this, a minor quantity of the air after the charge air cooler is used for continuous flow through the respective pipe system.

 **CAUTION**

The wet cleaning of the turbine may only be carried out at heavily reduced engine load (max. exhaust-gas temperature ahead of turbine ≤ 320° C)!

When the exhaust-gas temperature is too high, the washing water can trigger a thermoshock and cause damage to the turbine.

For operating parameters during the washing procedure, see Work Card [500.21]

NOTICE

Type and source of the hazard

For information on the requirements of the washing-water quality, see Technical Data [3.2.4]

Specified maintenance interval, see Maintenance Schedule [6.3.1]

For instructions concerning the washing procedure, see Work Card [500.21]



Cleaning System – Wet Cleaning, Compressor (Optional delivery scope)

Wet Cleaning System of the Compressor with Pressure Sprayer

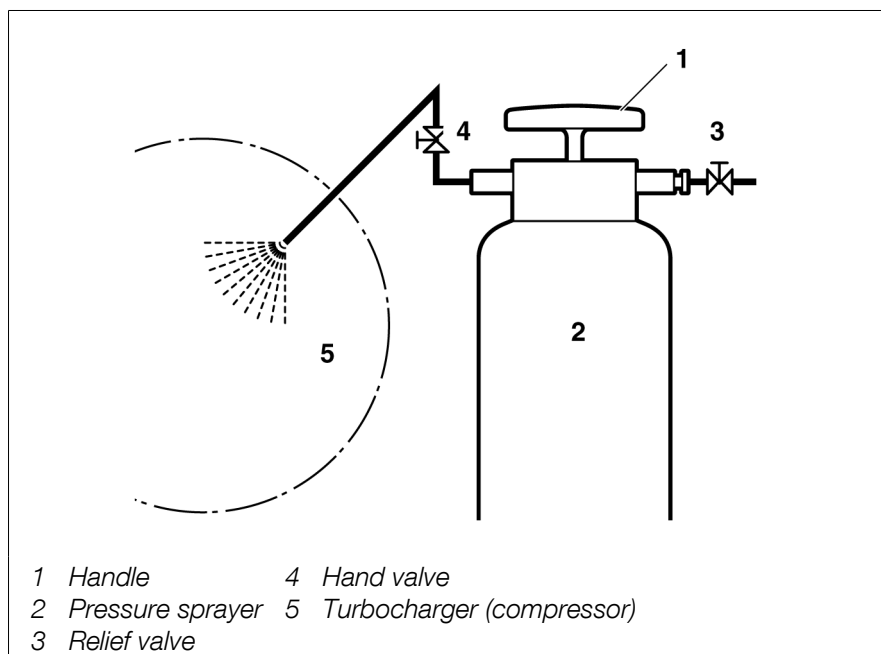


Figure 1: Wet cleaning system, compressor (pressure sprayer)

Functional Description

The wet cleaning of the compressor is performed during operation at full load.

The water container of the pressure sprayer (2) is filled with freshwater. An overpressure is created in the water container by means of the pump. By actuating the pushbutton on the hand valve (4), water is sprayed into the compressor casing (5) ahead of the compressor wheel. The water droplets bounce against the compressor wheel where they wear off contamination.

- 1 Safety**
- 2 Addresses**
- 3 Technical Data**
- 4 Description**
- 5 Operation**
- 6 Maintenance**
- 7 Work Cards**
- 8 Index**
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Safety during Operation

Containment Safety

The turbocharger is one of the highest loaded components in modern diesel engines. The high rotor speeds of a turbocharger result in high centrifugal force stresses at simultaneous high component temperatures. A consequence of additional extraordinary influences might be - in extremely rare cases - a failure of the components.

Precautions

MAN Diesel turbochargers correspond to the required containment safety according to the rules stipulated by the classification societies so that the state of the art prevents fragments from escaping.

In extreme cases, particularly if the turbocharger has not been sufficiently or properly maintained, there may be the risk of damages to persons or property in vicinity of the turbocharger.

To eliminate residual risks we kindly ask you to observe following rules:



Risk of injury!

Under no circumstances are persons permitted to be near the turbocharger without reason during operation.



- Turbine and compressor are to be cleaned at regular intervals as specified in the maintenance schedule.
- The maintenance of the turbocharger may only be carried out by trained personnel as well as in the maintenance intervals as stipulated in the operating manual.
- Inadmissible operating conditions, such as frequent load releases, compressor surging, excessive turbocharger speeds, must be avoided.
- Corrosive and erosive media in the intake air are to be avoided.

Technical Documentation

The technical documentation of the system, particularly the operating and maintenance manuals for the engine, turbocharger and the accessories required for engine operation, must be stored at a location where they are accessible for the operating and maintenance personnel at all times.

Regulations for Prevention of Accidents

When operating the system and for all maintenance and repair work, the relevant Regulations for Prevention of Accidents are to be observed. It is advisable to hang these regulations in the engine room and to repeatedly point out the dangers of accidents.

Safety Instructions Concerning Specific Danger Sources

Hearing Protection

WARNING**Sound levels exceeding 85 dB(A)!**

The danger of hearing impairment is given because sound levels are permanently above this value in the engine room when the engine is running.

- Always wear hearing protection in the engine room!

Exhaust Gases

In the engine room, exhaust gases with the harmful components NO_x, SO₂, CO, HC and soot can be set free.

WARNING**Health risk due to exhaust gases!**

The exhaust gases of the engine are harmful to one's health and detrimental to the environment when exceeding the limit values.

- Carry out maintenance in accordance with the maintenance schedule!
- Orient the operational management on behalf of the hazards!
- Observe operating results critically!

Danger of fire**WARNING****Potential danger of fire in the engine room!**

Fuel and lube oil are combustible and can easily ignite when contacting hot components in the engine room.

- Do not conduct fuel and oil pipes alongside hot engine components (exhaust pipe/turbocharger) that are not insulated!
- After maintenance and repair work on exhaust pipes and turbochargers, carefully and completely reattach all insulations and coverings!
- Regularly check the tightness of all fuel and oil pipes (maintenance schedule)!
- Eliminate any leakages immediately!

Fire-extinguishing Equipment

The engine room must be equipped with fire-extinguishing equipment. The proper function of this equipment is to be checked regularly.

DANGER**Danger to life through suffocation!**

When carbon dioxide fire-extinguishing systems are put into operation, danger of life is given due to suffocation.

- Do not put carbon dioxide fire-extinguishing systems into operation before it is ensured that no more persons are in the engine room!

Protective Covers

WARNING**Danger of burns and fire hazard from hot system components!**

Due to high operating temperatures the system components can become very hot. Persons can burn themselves on the hot components and/or inflammable operating fluids can ignite.

- Always have all coverings and insulations mounted to the turbocharger during operation of the engine!
- Regularly check the tightness of all fuel and oil pipes!

After assembly work, ensure that all coverings and protective covers from moving components as well as all insulations from hot components are reattached again. Engine operation with coverings removed is permitted only in special cases, such as for functional tests.

Alarm System

If important operating data are exceeded, an alarm system will set off alarm, reduction or stop signals, depending on the danger potential. The cause of the signals is to be thoroughly investigated and the source of the malfunction to be corrected. The signals may not be ignored or suppressed, except when ordered by management or in cases of increased danger.

WARNING**Ignoring or suppressing alarms and putting reduction and stop signals out of operation can lead to operational malfunctions which can result in serious personal injury and damage to property!**

Turbocharger and engine are fitted with an alarm system which sounds an alarm in case of technical problems or malfunctions. Therefore, an alarm always has a cause which must be rectified!

- Do not ignore or suppress any alarm, reduction or stop signals!
- Investigate the cause of every alarm, reduction or stop signal or report them!
- As far as possible, provide for swift rectification of the cause of alarms and signals!
- Ignore or suppress alarms, reduction and stop signals only in cases of increased danger or when ordered by management!

Personnel/ Qualification of Personnel

The engine and the systems required for its operation may be started, operated and stopped only by authorized persons. The personnel must be educated, trained and familiar with the system and the occurring dangers.

All persons operating the system or carrying out maintenance or repair work on the system must have read and understood all accompanying operating and maintenance instructions.

For ship systems:

Operational management through the chief engineer on board or through a technical officer.

Persons managing the operation must be in possession of a competence certificate that corresponds with the national regulations and international agreement (STCW). The number of required persons and their minimum qualification are usually stipulated through the national regulations or the international agreement (STCW).

For stationary systems (power plants):

The superintendent (engineer) must be available. Supervision/controlling of the engine and the accompanying supply system through a trained and specially briefed machinist or mechanic.

For managing persons in Germany, proof, among other things, must be furnished in accordance with the energy economy law (EnWG) that the technical operation management is guaranteed by qualified and sufficiently numbered personnel. In other countries, comparable laws/ guidelines are to be observed.

Safety during operation on the gas-powered engine

**DANGER**

Danger of fire caused by electrostatic charging

In the case of turbochargers on gas-powered engines, insulated mounting can cause electrostatic charging, resulting in ignition of the air/gas mixture in the air intake.

- If the turbocharger is mounted on the engine in such a way that it is electrically insulated, an earth connection must be installed for the equipotential bonding between the turbocharger and the engine.

Starting Operation/Putting Out of Operation

Initial Starting of Operation

1. Unpack the turbocharger.
2. Remove closing covers on gas-admission casing, gas outlet casing and compressor casing.
3. Remove preservation agents on the exhaust gas flanges of the turbine casing as well as on the pressure socket of the compressor casing with suitable scraper and cleaning agent.

WARNING



Danger of poisoning from toxic vapours!

The vapour-phase corrosion agent (increased corrosion protection) with which the openings of the turbocharger are treated sets toxic vapours free upon heating.

- Thoroughly blow out the turbocharger casing before assembling the turbocharger!
- Upon initial putting into operation, check if any corrosion protection agent is on the turbocharger!

NOTICE

For assembly of the turbocharger to the engine, see Work Card [500.31][500.31].

NOTICE

Upon delivery, the oil system of each turbocharger is factory-checked for cleanliness, and the oil supply and oil drainage are plugged. Therefore, when flushing the engine oil system, care must be taken that inadmissible contamination in the oil does not get into the turbocharger.

4. Flush the lube oil system. Plug the oil supply and oil drainage of the turbocharger and connect the supply and drain pipe on the engine side, or use an intake filter with low filter rating (20 µm recommended) in the lube oil supply. Measurement of the oil pressure after the intake filter.

NOTICE

For additional information on flushing the oil lubrication system, please refer to the engine manufacturers' manual.

Engine control

NOTICE

Indication of the pending alarm and the reaction of the engine control system must occur at the same time. Therefore, the engine control system must conform at least to category 3 in compliance with DIN EN 954-1.

Work Prior to Starting the Engine

1. Pre-lubricate engine and bearing points of the turbocharger (engine control system).
2. Set all shut-off devices of the engine system to the operating position.
3. Check the lube oil pressure in front of the turbocharger.

Starting Operation

Turbocharger

Start the engine. The engine exhaust gases drive the turbocharger and thus start it operating. If provided, the "Jet Assist" acceleration system is put into operation via the control system.

Checks during Operation

General

Despite automatic activation and control of the engine system, regular checks of the turbocharger are required so that possible malfunctions are detected as early as possible and the causes can be corrected in time.

NOTICE

The checks listed below are part of the operator's maintenance obligations. Results, observations and actions in connection with such monitoring measures are to be recorded in an engine log book. Reference values are to be defined in order to allow objective assessment of the results.

Continuous Checks on the Turbocharger

- Turbocharger speed
- Lube oil pressure in front of turbocharger
- Lube oil temperature in front of turbocharger
- Lube oil temperature behind turbocharger
- Exhaust gas pressure in front of turbine
- Exhaust gas temperature ahead of turbine
- Charge air pressure behind compressor
- Charge air temperature behind compressor
- Leakage of exhaust gas, charge air and oil pipes
- Contamination/saturation of air filter mat at silencer
- Smooth running of turbocharger

Smooth Running

Damage to the rotating element and bearings usually gives notice of appearance by unsmooth running, owing to imbalance or grazing of rotating parts. By auscultating the turbocharger (listening with e.g., a stethoscope), unsmooth running can often be detected at an early stage.

Speed

For clean air ducts of the compressor, a certain charge air pressure is assigned to each speed of the turbocharger. Thus, the charge pressure gives information about the speed and can be used for approximate speed control.

What to do in Case of Operating Faults

WARNING



Ignoring or suppressing alarms and putting reduction and stop signals out of operation can lead to operational malfunctions which can result in serious personal injury and damage to property!

Turbocharger and engine are fitted with an alarm system which sounds an alarm in case of technical problems or malfunctions. Therefore, an alarm always has a cause which must be rectified!

- Do not ignore or suppress any alarm, reduction or stop signals!
- Investigate the cause of every alarm, reduction or stop signal or report them!
- As far as possible, provide for swift rectification of the cause of alarms and signals!
- Ignore or suppress alarms, reduction and stop signals only in cases of increased danger or when ordered by management!

1. Either reduce the engine load or shut off engine, depending on the situation.

2. Determine and correct the cause of the malfunction (see Chapter [5.3]).

NOTICE

All malfunctions are to be corrected as soon as possible, as minor malfunctions can also lead to increased subsequent damage of the turbocharger or engine.

3. If the damage can not be corrected immediately, emergency operation is possible (see Chapter [5.2])

4. When the damage has been corrected, start the engine operation again as usual.

Putting Out of Operation

1. Shut off engine. The turbocharger gradually comes to a stop.

NOTICE

The run-down time of the turbocharger gives an indication of its mechanical condition. Sudden stopping indicates mechanical damage to the bearing, contact of the compressor or turbine wheel, trapped foreign objects or heavy fouling.

2. The lubrication oil runs out of the emergency and post-lubrication system to the turbocharger bearings so that these are still continuously lubricated for several minutes (post lubrication).

Emergency Operation on Failure of a Turbocharger

Introductory Remarks

If damage occurs to a turbocharger which can not be corrected immediately, emergency operation is still possible.

Available Appliances

Turbocharger (see Work Card [500.11] and [500.12])

- Securing fixture for blocking of the rotating element from the compressor side (intake cross-section remains open).
- End covers, for closing of the compressor and turbine back side when the rotating element is disassembled.

All appliances are designed in such a manner that continuous flow through the air and exhaust gas side of the turbocharger is possible.

Appliances for use on the engine:

- Screen/s (grid) for the side of the charge air pipe that faces away from the turbocharger. The screen/s is/are supposed to ease the suction operation of the engine.
- Blind flange/s for blocking of the partially mounted charge-air bypass pipe (if given).

Achievable Outputs/Speeds for Four-stroke Engines

The following criteria restrict the achievable engine load in engine-emergency operation:

- Maximum exhaust gas temperature after cylinder,
- Maximum exhaust gas temperature in front of turbocharger,
- Density of exhaust gas.

The following rating data apply only as reference values:

Turbocharger failed	L-Engine	V-Engine
Engine operation with variable speed	15%	15%
Engine operation with constant speed	20%	20%

Table 1: Emergency operation for failure of one turbocharger – Achievable outputs

Emergency Operation on Failure of a Turbocharger

WARNING



Despite compliance of the following measures, the danger of turbocharger destruction is still given!

For this case, there is imminent danger of personal injury or damage to property!

- Limit the duration of emergency operation to the absolute necessary minimum!
- Use the securing fixture (measure A) for emergency operation only when removal of the rotating element is not possible!
- Do not unnecessarily stay in the vicinity of the turbocharger during emergency operation!

NOTICE

Carry out measure A (see Table “Emergency operation in the event of turbocharger failure”) only when measure B cannot be performed for reasons of time. **Consequential damage possible.**

Emergency Measures	Supplementary Measures/ Precautions (see code number)
Engine may not be stopped for reasons of urgency Turbocharger remains unchanged	0-3
Engine may be stopped (short-term)	
Measure A: Block rotor from the compressor side using securing fixture (intake cross-section remains open). Afterwards, mount intake silencer or air intake casing again. (see Work Card [500.11])	0-5
Measure B: Disassemble rotating element including bearings; shut bearing casing on compressor and turbine side with end covers. Mount silencer/air intake casing and gas-admission casing again, as required. (see Work Card [500.12])	0-5

Table 2: Emergency operation for failure of a turbocharger

Legend

Code Number	Supplementary Measures/Precautions
0	Observe instructions of engine manufacturer!
1	Reduce engine power. The maximum exhaust gas temperatures behind cylinder and in front of turbocharger, and (for engines with two or more turbochargers) the maximum admissible turbocharger speed must not be exceeded. Pay attention to density of exhaust gas.
2	As a precaution, prepare fire-extinguishing measures!
3	When the rotating element is blocked or disassembled, stop lube oil supply because of danger of contamination or fire.
4	For suction operation (or with two or more turbochargers), the engine must be driven with reduced charge.
5	Engines with two or more turbochargers: Separate the exhaust gas intake side of the defective turbocharger from the gas flow of the second turbocharger with blind flange.

Malfunctions/Faults and their Causes

Trouble Shooting

NOTICE

In case of turbocharger irregularities upon or during engine operation, determine and correct the cause of the malfunction as soon as possible. Otherwise the danger is given that minor malfunctions can lead to increased subsequent damage on the turbocharger or engine.

No.	Malfunction	Possible Cause	Corrective Action	Work Card
1	Exhaust gas temperature in front of turbine too high	Silencer contaminated	Clean/replace	[500.24] [500.32]
		Compressor contaminated	Clean	[500.23]
		Turbine wheel and/or nozzle ring heavily contaminated	Clean	[500.21] [500.22]
		Rotating element grazes	Contact MAN Diesel Service	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged or eroded	Exchange defective components; contact MAN Diesel Service, if required	
		High air-intake temperature	Reduce output	Engine
		Charge air cooler contaminated	Clean, measure pressure differential	Engine
		Charge air pipe leaks	Seal charge air pipe	Engine
		Charge air temperature too high	Check charge air cooler	Engine
		Inlet/exhaust valves, valve slots of engine contaminated	Clean	Engine
		Exhaust gas pipe leaks	Seal exhaust gas pipe	Engine
		Exhaust-gas back-pressure behind turbine too high	Clean boiler	System
		Fuel injection at engine faulty	Repair fuel injection	Engine

No.	Malfunction	Possible Cause	Corrective Action	Work Card
2	Charge air pressure too low	Turbine speed too low	See malfunction No. 4	
		Silencer contaminated	Clean/replace	[500.24] [500.32]
		Compressor contaminated	Clean	[500.23]
		Turbine wheel and/or nozzle ring heavily contaminated	Clean	[500.21] [500.22]
		Defective bearings, imbalance of rotating element	Bearing inspection (noise and vibrations); contact MAN Diesel Service, if required	
		Rotating element grazes	Check components; contact MAN Diesel Service, if required	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged or eroded	Exchange defective components; contact MAN Diesel Service, if required	
		High air-intake temperature	Reduce output	Engine
		Charge air cooler contaminated	Clean	Engine
		Charge air pipe leaks	Seal charge air pipe	Engine
		Inlet/exhaust valves, valve slots of engine contaminated	Clean, measure pressure differential	Engine
		Exhaust gas pipe leaks	Seal exhaust gas pipe	Engine
		Exhaust-gas back-pressure behind turbine too high	Clean turbine; check for damage, if required	[500.21] [500.22]
		Fuel injection at engine faulty	Repair fuel injection	Engine
3	Charge air pressure too high	Nozzle ring lightly contaminated/restricted	Clean	
		Low air intake temperature	Reduce output	Engine

No.	Malfunction	Possible Cause	Corrective Action	Work Card
4	Speed too low	Silencer or air filter contaminated	Clean/replace	[500.24] [500.32]
		Compressor contaminated	Clean	[500.23]
		Turbine wheel and/or nozzle ring heavily contaminated	Clean	[500.21] [500.22]
		Defective bearings, imbalance of rotating element	Bearing inspection; Contact MAN Diesel Service	
		Rotating element grazes	Contact MAN Diesel Service	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged or eroded	Exchange defective components; contact MAN Diesel Service, if required	
		High air-intake temperature	Reduce output	Engine
		Charge air cooler contaminated	Clean	Engine
		Charge air pipe leaks	Seal charge air pipe	Engine
		Inlet/exhaust valves, valve slots of engine contaminated	Clean	Engine
		Exhaust gas pipe leaks	Seal exhaust gas pipe	Engine
		Exhaust-gas back-pressure behind turbine too high	Clean turbine; check for damage, if required	[500.21] [500.22]
		Fuel injection at engine faulty	Repair fuel injection	Engine
5	Speed too high	Nozzle ring lightly contaminated/restricted	Clean	
		Low air intake temperature	Reduce output	Engine
6	Lube oil pressure too low	Flow temperature of lube oil too high		Engine
		Lube oil filter contaminated	Replace lube oil filter	Engine
		Lube oil manometer defective	Replace lube oil manometer	Engine
7	Lube oil loss	Thrust ring or labyrinth ring damaged	Check components; contact MAN Diesel Service, if required	
		Defective labyrinth seal	Check components; contact MAN Diesel Service, if required	
		Seals damaged, connections untight	Check components; contact MAN Diesel Service, if required	
		Sealing air ineffective, oil coke behind turbine wheel	Check components; contact MAN Diesel Service, if required	
		Lube oil pressure too high	Reduce oil pressure	Engine/ system
		Overpressure in oil drain or in crankcase	Drain oil as described in Chapter [4.2][4.2]	

No.	Malfunction	Possible Cause	Corrective Action	Work Card
8	Hesitant starting, short run-down time	Compressor contaminated	Clean	[500.23]
		Turbine wheel and/or nozzle ring heavily contaminated	Clean	[500.21] [500.22]
		Thrust ring or labyrinth ring damaged	Check components; contact MAN Diesel Service, if required	
		Defective bearings, imbalance of rotating element	Bearing inspection; Contact MAN Diesel Service, if required	
		Rotating element grazes	Contact MAN Diesel Service	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged	Exchange defective components; contact MAN Diesel Service, if required	
		Sealing air ineffective, oil coke behind turbine wheel	Check components; contact MAN Diesel Service, if required	
9	Unusual heavy noise	Thrust ring or labyrinth ring damaged	Check components; contact MAN Diesel Service, if required	
		Defective bearings, imbalance of rotating element	Bearing inspection; Contact MAN Diesel Service, if required	
		Rotating element grazes	Check components; contact MAN Diesel Service, if required	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged	Exchange defective components; contact MAN Diesel Service, if required	
10	Turbocharger vibrates	Turbine wheel heavily contaminated	Clean	[500.21] [500.22]
		Thrust ring or labyrinth ring damaged	Check components; contact MAN Diesel Service, if required	
		Defective bearings, imbalance of rotating element	Bearing inspection; Contact MAN Diesel Service, if required	
		Rotating element grazes	Contact MAN Diesel Service	
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Turbine wheel, shroud ring, nozzle ring or compressor wheel damaged	Exchange defective components; contact MAN Diesel Service, if required	

No.	Malfunction	Possible Cause	Corrective Action	Work Card
11	Compressor surges	Turbine wheel and/or nozzle ring heavily contaminated	Clean	[500.21] [500.22]
		Foreign objects in front of/in turbine or compressor	Check turbine/compressor for damage; contact MAN Diesel Service, if required	
		Low air intake temperature	Reduce output	Engine
		Charge air cooler contaminated	Clean	Engine
		Charge air temperature too high	Reduce output	Engine
		Inlet/exhaust valves, valve slots of engine contaminated	Clean	Engine
		Exhaust-gas back-pressure behind turbine too high	Check exhaust gas pipe	Engine/ system
		Fuel injection at engine faulty	Repair fuel injection	Engine

Table 1: Trouble Shooting

- 1 Safety**
- 2 Addresses**
- 3 Technical Data**
- 4 Description**
- 5 Operation**
- 6 Maintenance**
- 7 Work Cards**
- 8 Index**
- 9 Spare Parts and Tools**
- 10 Spare Parts Catalog**

Safety during Maintenance/Repair

Qualification of Personnel

Performed by machinists or mechanical fitters and their assistants.

Briefing and supervision of difficult work tasks through a technical officer or the chief engineer.

Opening Pipes/Pressure Tanks



Danger of burns from hot components!

Many engine and turbocharger components become very hot during operation and require ample time for cooling down afterwards.

- Prior to taking any action, allow the engine and turbocharger to cool down sufficiently (1-2 h)!
- Wear protective clothing!



Danger of burning/scalding from running-out lube oil!

The lube oil is under pressure and very hot!

- Before opening pipes, flanges, screwed connections or gauges, free the system of pressure and check that it is pressure-free!
- Before opening oil pipes, wait until the lube oil has cooled down sufficiently (1-2 h)!
- Drain the system and check that the lube oil has been drained completely!
- Wear protective clothing!



On turbochargers with emergency and post-lubrication system, the lube oil will continue to drain for up to 1.5 h after shutting down the engine system.

Disassembly and Assembly of Pipes

During any work on the pipe systems, especially for fuel, lubrication oil and air, ensure utmost cleanness in order to avoid foreign objects or debris from entering the system. Therefore, all pipes subject to reassembly should be blocked with care upon disassembly.

New pipes being installed should be checked for cleanness and flushed, if required.

For extended storage, all individual parts are to be preserved.

Removal/Disassembly of Heavy Components

WARNING



Danger of injury from suspended loads!

Unsuitable or poorly maintained transport equipment as well as unsuitable set-down locations can lead to serious injury of people and to major damage.

- Use only transport equipment which is in perfect condition!
- Check if the transport equipment being used has the required carrying capacity!
- Choose only set-down locations that have sufficient loading capacity for the heavy component (this is not always the case for galleries, half-landings or grid-iron coverings)!

Danger of Crush Injuries from Rotating Components

WARNING



Danger of crush injuries or from blows!

When the engine is running during disassembly or assembly work, there is danger of becoming entangled in or coming against rotating components.

- Carry out all disassembly or assembly work while the turbocharger is shut down!
- During the work, secure the engine against starting!
- Set up danger signs!

Use of Cleaning Agents

When using cleaning agents, the manufacturer's notes for usage, possible dangers and disposal are to be observed.

WARNING



Cleaning agents can be caustic!

Cleaning agents on the skin, in the eyes or being inhaled can lead to serious damage to one's health or to allergies.

- Always observe the manufacturer's notes for cleaning agents!
- Wear protective clothing whenever required!
- Wear protective gloves whenever required!
- Wear safety goggles whenever required!
- Wear respiratory protection whenever required!
- Pay attention that other persons also wear the appropriate protective equipment whenever required or make these persons aware thereof!

Use of High-pressure Cleaners

When working with high-pressure cleaners ensure proper usage.

Air filters, shaft ends (also including such with lipped sealing rings), governors, splash-proof monitoring systems, cable sockets as well as sound and heat insulations under non-watertight coverings, must be covered or exempt from the high-pressure cleaning.

Spare Parts

During operation, the components of the turbocharger are subjected to very high forces. MAN Diesel has therefore developed special materials that can withstand these high forces. Only by using MAN Diesel original spare parts can you be sure that these high-quality materials have been employed.

**DANGER**

Bought-out spare parts can be manufactured from low quality materials that can not withstand the forces occurring during operation of the turbocharger!

This can result in serious damage to the turbocharger, engine room and engine, as well as danger to life for the operating personnel.

- Use only MAN Diesel original spare parts!
- Use bought-out spare parts only after these have been released by MAN Diesel!

MAN Diesel shall in no way be liable for any personal injury or material damage caused by use of spare parts from other manufacturers. Furthermore, any and all warranty claims relating to the turbocharger expire when using non-original spare parts.

General Remarks

Maintenance Work

Maintenance work belongs to the duties of the user, just as regular checks do. Their purpose is to retain operational readiness and operational safety of the system. They are to be carried out by competent personnel in accordance with the time schedule for maintenance.

Maintenance work supports the operating personnel in their effort to detect arising malfunctions/failures in time. They deliver hints towards necessary maintenance work or repairs to the persons responsible, and influence the planning of downtimes.

Maintenance and repair work can be carried out properly only when the required spare parts are available. In addition to these spare parts, it is advisable to stock a certain quantity of spares for unexpected outages.

Maintenance Schedule

The required tasks are Maintenance Schedule (Sheet [6.3.1][6.3.1][6.3.1]. It contains

- a brief description of the work,
- the repeat intervals,
- the personnel and time requirements,
- references to the respective work cards/working instructions.

Work Cards

The work cards describe

- the aim and object of the work

and contain information about

- the required tools/appliances and
- detailed descriptions and graphical depictions of the necessary work sequences and work steps.

see Work Card [7.1.1].

Spare Parts

Order numbers for spare parts can be found

- Turbocharger: in the spare parts catalogue of the technical documentation
- Tools: list of contents for tools
- On the respective work cards of the technical documentation.

Order numbers consist of the three-digit subassembly number and a three-digit item number, separated by a dot.

Special Services/Repair Work

MAN Diesel offers the following special services:

- Delivery of spare parts
- Consultation and support on questions concerning operation, maintenance and repairs
- Field-experienced training for operating personnel of MAN Diesel exhaust gas turbochargers
- Assessment and clarification of damage cases
- Dispatch of fitters and engineers

In case of requests please contact us at

- MAN Diesel SE, Augsburg Works
- MAN Diesel SE, Service Center Hamburg,
- Other Service Centers and authorized repair shops

Service Centers and Works of MAN Diesel SE can be found in Chapters [2.1] „MAN Diesel SE in Germany“ and „Worldwide Service Addresses“.

Tools/Special Tools

Inventory Tools

Commercially available tools that are usually included in the inventory of the engine system.

Inventory tools are not contained in the turbocharger's scope of supply.

Standard Tools

Included as standard in the turbocharger's scope of supply.

Normally, one set of standard tools is sufficient for a system with several turbochargers. The usual maintenance work can be carried out with these standard tools and the inventory tools.

The standard tool kit for the turbocharger(s) is combined in one or more boxes and supplied with a table of contents. The order numbers of the individual tools can also be taken from the table of contents.

Tools Requested by the Customer

Additional tools and appliances can be ordered at the MAN Diesel SE, as required.

Special Tools

Available only at the Service Centers; these tools are not required by the user of the turbocharger.

Some jobs, -more repair work than maintenance-, require special knowledge, experience and auxiliary equipment/appliances. For jobs like this, additional specialty tools are available to our Service Centers and possibly to some authorized repair shops. Therefore, we recommend that you always consult these Partners or commission work to them when your own time schedule, or the professional or personnel capacities are not sufficient.

Maintenance Schedule

Reliability and Adaptability

Validity of the Maintenance Schedule

The maintenance schedule contains a summary of the maintenance and inspection work, down to the major overhaul of the turbocharger.

A major overhaul of the turbocharger must be carried out latest after

- 24 000 - 30 000 operating hours

.

After each major overhaul, which best is carried out with a due engine service, the maintenance schedule starts from the beginning again.

A major overhaul includes the disassembly of the complete turbocharger for inspection of the actual condition, thorough cleaning and a check of all parts/components.

NOTICE

The maintenance must be scheduled well in advance, meaning that, for example, sailing timetables or power plant inspections must be taken into account. In order to ensure the operational safety of the turbocharger and the engine, maintenance work is to be performed as far as possible on schedule or beforehand.

Adaptation of the Maintenance Schedule

The maintenance schedule takes the following operating conditions and an annual operating period of approx. 6000 hours into account:

- Uniform loading within a range of 60 to 90% of the rated power,
- Observation of the specified operating media temperatures and pressures,
- Usage of the specified lube oil and fuel qualities,
- Attentive separation of fuel and lubrication oil.

The maintenance intervals must be shortened when the following operating conditions are given:

- Longer-term operation with peak loads or low loads, long idle periods, frequent heavy load cycles,
- Frequent engine starts and repeated run-up phases without sufficient warming-up,
- High stress on the engine before reaching the specified operating media temperatures,
- Too low lube oil, cooling water and charge air temperatures,
- Use of problematic fuel qualities and insufficient separation,
- Insufficient filtration of the intake air (particularly with stationary engines).

Maintenance Schedule

Turbocharger on the Two-stroke Engine

1, 2, 3					per	24	150	250	3000	12000	24000
Inspection (during operation)											
901	Check turbocharger for unusual noise and vibrations.	A	1	0,1	Turbo	X					
903	Check turbocharger and system pipes for leaks (charge air, exhaust gas, lubricating oil)	A	1	0,2	Turbo	X					
905	Check all the fixing screws, casing screws and pipe connections for tight fit.	7.2.1	1	1	Turbo			2	X		
Maintenance (during operation)											
911	Clean turbine - Dry clean, if available	500.22	1	0,3	Turbo	1					
913	Clean turbine - Wet clean, if available	500.21	1	1	Turbo			1			
915	Clean compressor (during operation)	500.23	1	0,3	Turbo		1				
917	Clean air filter (if available)	500.24	1	0,4	Turbo			1			
Maintenance (in common with an engine maintenance)											
931	Clean and check compressor casing, insert, diffuser and compressor wheel (visually check and clean, as required). Establish operational readiness of the turbocharger again.	500.41 500.42 500.43	2	4	Turbo					X	
941	Check thrust bearing, counter-thrust bearing, bearing disk	500.46	1	2	Turbo					X	
951	Major overhaul 24 000 ... 30 000 operating hours: Dismantle, clean and check all components of the turbocharger. Check gaps and clearances upon assembly	500.32 500.33 500.34 500.41 500.42 500.43 500.44 500.45 500.46 500.51 500.52 500.53 500.54 500.55 500.56 500.57 7.2.2	2	20	Turbo						X

Maintenance Schedule, Legend

24 ... 24000	Repetition intervals in operating hours
X	Maintenance work due
1	As required/depending on condition
2	Checking of new or overhauled parts required (once after the mentioned time)
A	No work card required/available

Remarks to the Maintenance Work

General Remarks

Disassembling the Turbocharger

The simplest method to go about the disassembly depends on the space conditions in the engine room. When performing maintenance and inspection work, it is sufficient in most cases to disassemble only subassemblies of the turbocharger.

Inspection Report

For a major overhaul of the turbocharger or for repair work concerning major components, it is recommended to record the condition of individual subassemblies and make entries into the operating log. Inspection report form, see Chapter [6.6].

NOTICE

In disassembly work, mark the position of the components relative to each other using a felt-tip marker pen. If required, document the disassembly/assembly with a camera.

Exchanging Components

Components with traces of wear or damage that influence especially the strength and the balancing precision and thus the smooth running of rotating parts, must be exchanged with original spare parts or repaired by an authorized repair facility or the manufacturer.

For dispatch, components must be packed and protected against corrosion in such a manner that they are not further damaged during transportation.

Screw Connections

NOTICE

Never attempt to loosen seized or tight screw connections with excessive force.

This can lead to screws being sheared off or parts being so badly damaged that they must be replaced. Sheared-off screws must be drilled out and the threads retapped.

Tightening Screw Connections

Apply plenty of diesel fuel or special dissolver (e. g. Caramba, Omnigliss, WD 40) to the screw connections and allow to penetrate for at least 1/2 hour, so that the dissolver can work its way through the threads. Observe the applicational tips of the dissolver manufacturer. Afterwards, apply light hammer blows to the screw/bolt head so that the threads run freely.

All screw connections on the turbocharger must be in perfect condition upon assembly or be renewed. When assembling lock washer pairs (1 pair consists of 2 equal washers), the long wedge surfaces (inside) must face each other, otherwise there will be no securing effect.

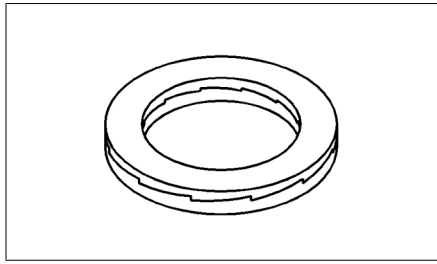


Figure 1: Assembly of wedge lock washers

NOTICE

Always tighten screws/bolts with the specified tightening torque!

See Technical Documentation, Work Cards.

Connections on the Turbine Side

Improperly tightened and secured screws/bolts and nuts can become loose due to vibrations and then be driven into the turbocharger by the air or exhaust gas flow, and cause severe damage.

Due to the high temperatures on the turbine side, screws/bolts, nuts and tab washers of non-scaling materials are used. These are marked on the face side (SM, VM or 4923, Z1 or 4828), to avoid them from being mistaken with regular parts. When disassembling, the fasteners marked in this manner are to be stored separately until being assembled.

All lock washer pairs on the turbocharger are made of non-scaling material and are not marked.

In order to avoid the screws/bolts and nuts on the turbine side from heating together, they must be treated with high-temperature lubricant (e. g. Molykote HSC) prior to each new assembly. Before treating with lubricant, the threads must be cleaned thoroughly. Screw-in threads from studs are not treated.

Inspection of Individual Components

General Remarks

The following remarks and questions are intended as a guide, e. g. for

- Assessment of the turbocharger condition,
- Operating log and inspection report,
- Inspection, reusing, repair and replacement of individual components.

Inspection Work

Gaps and Clearances	Admissible values, see Chapter [7.2.2].
Crack Test	Crack test on turbine blades (turbine rotor, compressor wheel) via acoustic testing or penetration testing with fluorescent color.
Streaks	Streaks all around the circumference of the turbine rotor, compressor wheel, insert: Check gaps and clearances,, see Chapter [6.6]. Streaks, locally on circumference of turbine rotor, compressor wheel, insert: Indication of imbalance.
Oil Coke behind Turbine Wheel	Sealing air not effective (e. g., pipes clogged), shaft seal on turbine side damaged or pre-lubrication pressure too high.
Erosion during HFO Operation	During HFO operation, small, hard particles (e.g. "catfines") are contained in the exhaust gas. These act particularly erosive where high gas velocities and high concentrations of such particles occur. The highest flow rates are effective at the outlet of the turbine nozzle ring. Here, the flow is strongly deflected towards the direction of the circumference. Besides this, the particles contained in the exhaust gas are hurled towards the outside. For this reason erosions can be noticed at the discharge of the nozzle ring, at the interior diameter of the shroud ring and at the exterior of the turbine blades.
"Catfines"	As a result of the refinery process and the achieved quality, "catfines", a fine grained powder with great hardness (recognizable through the Al and Si content) as well as other abrasive contaminants, are contained in the heavy fuel oil. Their amount can be reduced by means of well dimensioned, heated settling tanks and through sufficient separation.
Additions	Adding used engine lube oil is of particular disadvantage, recognizable through the increased amounts of Ca and Fe. Used engine oil not only contains residues of wear, but above all "Detergent dispersing additives", which bind fine particles and prevent "catfines" and other impurities from being eliminated through separation. For operation with "waste oil" added, a sharp increase of erosion damage is noticed. Combustion residue caused by incomplete combustion can also have erosive effects. Therefore, ensure clean combustion.
Silencer, if Available	How is the condition of the air filter mat? How is the condition of the absorption plates? Are cracks on any of the components?
Insert	Are there any traces of streaks?
Control Unit for Speed	Is the speed transmitter or the speed indicator defective?
Casing	Which casings are strongly contaminated from soot, oil or oil coke?

Compressor Wheel	Were the casings tight, particularly at the parting lines and the connection flanges? Do the casings show signs of cracks?
	Were screws/bolts torn? Were all screws/bolts and nuts firmly tightened and secured?
Turbine rotor	Was the seal air or were oil bores in the bearing casing clogged?
	Are all seals O.K.?
	How is the seat of the compressor wheel and the carrier?
	Are the seat faces bright or are there signs of fretting?
	Do the turbine blades show signs of cracks or heavy wear? Are there traces of streaks?
	Do the leading edges of the turbine blades show puncture notches caused by drawn-in foreign objects?
Turbine rotor	How is the run-out of the shaft? How is the condition of the bearing points?
	Are the labyrinth rings strongly worn?
	Are the seat faces bright or are there signs of fretting?
	Are the blades damaged or is heavy wear or erosion at the edges detectable? Do the blades have signs of cracks?
	Do the blades, disc or shaft show any traces of streaks?
	Are the fastenings at the foot of the blade O.K.?

NOTICE

In new condition, the blades are not fastened but inserted into the disc with clearance. At the blade end, they can be moved several millimeters both in axial and in radial direction. Due to deposits, this movability is limited after extended periods of operation.

Shroud Ring	Are there any traces of streaks from the turbine blades?
	The radial gap between shroud ring and turbine blade may not fall below the minimum value at any position, in order to avoid grazing during operation. The gap can become larger due to erosion. This considerably influences the efficiency. Therefore, the gap is measured at 4 locations around the circumference (every 90°) during checking. Admissible values, see work card .
Turbine Nozzle Ring	Are the blades bent? Are the blades heavily worn due to erosion?
	Are foreign objects obstructed in the ducts? Can traces be detected that point out to foreign objects?
Thrust ring	Can any cracks be detected at the blades?
	Are the seat faces bright or are there signs of fretting?
Bearings	Are there any signs of scoring or heavy wear?
	For checking of the bearings and bearing bushes, see Work Card and . Any wear across corners on both bearings is an indicator for imbalance of the rotating element.
	How is the seat of the bearing in the casing bore?
	Is the axial clearance of the rotating element and the bearing clearance within the admissible range?

NOTICE

When the axial bearing clearance is determined by axial displacement of the rotating element, then the rotating element is to be lifted lightly so that the labyrinth tips come free, out of the grooves in the cover seal.

Labyrinth Disk, End Cover Is the run-in pattern of the labyrinth tips normal?

Operating Life, Assemblies

Bearing Up to 50 000 operating hours

Turbine rotor Up to 100 000 operating hours

Compressor Wheel Up to 80 000 operating hours

The operating life of the compressor wheel depends on the intake air temperature, the charge pressure and the load profile of the engine. It can be considerably reduced by unfavorable conditions.

Casing Practically unlimited

These periods are empirical values and are reached with turbochargers that operate on engines powered with heavy fuel oil.

Continued use or replacement of components is decided upon at each major overhaul of the turbocharger. It is imperative that major overhauls are performed according to the periods specified in the maintenance schedule.

Inspection Report

Form

Accompanying the installer report	Date
Name of customer	Turbocharger
Address	Works No.
Site	Engine type
Name of vessel	Works No.
Turbocharger inspected/overhauled on by	
Reason	
Last inspection/Overhauled on by	
Operating hours since last inspection/overhaul h	
Operating hours since start-up h	

☒ Please mark where applicable ☒ checked ☐ Intake casing , if available .. ☐ contaminated ☐ Silencer , if available ☐ Air filter mat ☐ utilized ☐ replaced ☐ contaminated ☐ cleaned ☐ Speed transmitter , if available ☐ defective ☐ Insert ☐ utilized ☐ replaced ☐ contaminated ☐ traces of streaks ☐ axial ☐ radial ☐ all around the circumfer- ence ☐ locally on the circumfer- ence Location ☐ Compressor housing ☐ contaminated ☐ cracks ☐ damage from foreign ob- jects	☐ Diffuser ☐ utilized ☐ replaced ☐ contaminated. blades ☐ bent ☐ torn ☐ damage from foreign ob- jects ☐ Compressor wheel ☐ utilized ☐ replaced Reason ☐ contaminated. ☐ traces of streaks ☐ axial ☐ radial ☐ all around the circumfer- ence ☐ locally on the circumfer- ence Location ☐ blades ☐ bent ☐ torn ☐ damage from foreign ob- jects ☐ bore ☐ signs of fretting ☐ Thrust ring ☐ utilized ☐ replaced ☐ signs of scoring ☐ wear ☐ signs of fretting	☐ Turbine rotor ☐ utilized ☐ replaced reason ☐ balanced ☐ contaminated. ☐ traces of streaks ☐ axial ☐ radial ☐ all around the circumfer- ence ☐ locally on the circumfer- ence Location ☐ blades ☐ bent ☐ torn ☐ blade ☐ foot plate ☐ foot ☐ damage from foreign ob- jects ☐ erosion ☐ coverings ☐ light ☐ heavy ☐ one-sided ☐ Bearing points ☐ signs of scoring wear ☐ seatings ☐ signs of fretting Location ☐ labyrinth tips ☐ oil coke ☐ wear
---	---	---

- | | | |
|---|---|--|
| <p>☐ Bearing casing</p> <ul style="list-style-type: none"> ☐ Contaminated ☐ Cracks ☐ Sealing air bores <ul style="list-style-type: none"> ☐ clogged ☐ cleaned ☐ Oil bores <ul style="list-style-type: none"> ☐ clogged ☐ cleaned ☐ Seals <ul style="list-style-type: none"> ☐ replaced <p>☐ Sealing air valve, if available</p> <ul style="list-style-type: none"> ☐ Valve plater <ul style="list-style-type: none"> ☐ moves easily ☐ replaced <p>☐ Bearing disk</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ signs of scoring ☐ wear ☐ length (l) - compressor side
Taperfacemm ☐ length (l) - turbine side
Taperfacemm <p>☐ Counter-thrust bearing</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ signs of scoring ☐ wear ☐ length (l), taperfacemm | <p>☐ Labyrinth disk, compressor side(sealing cover)</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ abnormal run-in pattern ☐ wear ☐ traces of streaks <p>☐ End cover, turbine side</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ abnormal run-in pattern ☐ wear ☐ oil coke ☐ cleaned <p>☐ Labyrinth disk, turbine side</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ wear ☐ oil coke ☐ cleaned <p>☐ Turbine nozzle ring</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced ☐ blades <ul style="list-style-type: none"> ☐ bent ☐ torn ☐ damage from foreign objects ☐ puncture notches ☐ erosion ☐ coverings <ul style="list-style-type: none"> ☐ light ☐ heavy ☐ one-sided | <p>☐ Shroud ring</p> <ul style="list-style-type: none"> ☐ utilized ☐ replaced reason ☐ Contaminated ☐ Erosion ☐ Traces of streaks <ul style="list-style-type: none"> ☐ all around the circumference ☐ locally on the circumference Location <p>☐ Gas-admission casing</p> <ul style="list-style-type: none"> ☐ Contaminated ☐ Cracks ☐ Erosion ☐ Damage from foreign objects ☐ Screw connections <ul style="list-style-type: none"> ☐ loose ☐ torn off ☐ replaced <p>☐ Gas outlet casing</p> <ul style="list-style-type: none"> ☐ contaminated ☐ Cracks ☐ Damage from foreign objects ☐ Screw connections <ul style="list-style-type: none"> ☐ loose ☐ torn off ☐ replaced |
|---|---|--|

Gaps and clearances (admissible values, see C2 Chapter "Gaps and Clearances")

Pos. 6	Turbine blade/shroud ring	(Radial gap) mm
Pos. 10	Compressor wheel/insert	(Axial clearance) mm
Pos. 11	Compressor wheel/insert	(Radial gap) mm

Remarks:

Date

Signature

- 1 Safety**
- 2 Addresses**
- 3 Technical Data**
- 4 Description**
- 5 Operation**
- 6 Maintenance**
- 7 Work Cards**
- 8 Index**
- 9 Spare Parts and Tools**
- 10 Spare Parts Catalog**

7.1 Preface**7.2 Setting Data****7.3 Emergency Operation****7.4 Cleaning****7.5 Assembly of the Turbocharger and Auxiliary
Equipment**

Layout and Usage of the Work Cards

Numbering

The ordinal number of the work cards is located in the top right of the title. It consists of a 3-digit subassembly group number and a 2-digit counting number. The subassembly group number 500 hereby represents the collective subassembly group number for turbochargers.

Example - Turbocharger on engine: 500.31

- 500 = Subassembly group number (500 for turbocharger)
- 31 = Counting number (work card No. 31)

Cross references in work cards to corresponding work cards are in square brackets.

Example - Cross-reference to work card 500.43: [500.43]

In work card (labyrinth disk), there is a reference to work card 500.43 (compressor wheel).

Layout of the Work Cards

Work cards generally contain the following information:

- Required personnel and time,
- Affiliated work cards, meaning work cards that contain additional necessary or other useful information,
- The required tools and appliances,
- Possibly additional information/technical data and,
- The individual work sequences, always beginning with the starting condition, and followed by the individual work steps.

Parts and tools in the figures and text are each designated with their order number.

Example for the turbine rotor: 520.001

- 520 = Subassembly group number (520 für „Rotor, complete“)
- 001 = Position number

All assemblies and subassembly group numbers are listed in the spare parts catalog.

Order numbers for tools are (generally) identical with the tool number affixed on the tool.

Required Personnel and Time

The required time is mentioned in hours, both for the amount and qualification of the required personnel. The higher hour rate covers the time requirement for the complete scope of the respective work card. Personnel with lower hour rate is required only at times, e. g., for operation of a lifting tackle.

Tools / Appliances

Numbers mentioned under the Designation column represent spanner widths in mm, e. g. Socket wrench 22 = Socket wrench, size 22 mm.

Availability-codes under the Status column have the following meaning:

Standard	Standard tools (with tool number) If included in the scope of supply, turbochargers are equipped with a set of standard tools.
Option	Tools requested by the customer (with tool number)
Inventory	Commercially available tools/appliances These are not included in the supply scope of the turbocharger. It is assumed that these tools/appliances are included in the inventory of the engine system.
Special Tools	Special tools, e. g. for service centers

Tips and Pictographs

NOTICE

For notes concerning fault sources/handling errors; material damage can occur.
Additional notes, information and tips.

The instructions "Lubricate contact surfaces and threads", "Observe tightening torques" and "Use lock washer pairs" are additionally indicated by means of symbols in the marginal column:



Grease, lubricate and oil



Use
lock washers



Observe
tightening torques!

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**7.5 Assembly of the Turbocharger and Auxiliary
Equipment**

Screw Connections and Tightening Torques

Short Version

Tightening values for all important bolted connections/tightening torques for various friction coefficients μ .

Tightening Torques for Bolted Connections

Tightening of Bolted Connections by Torque

Whenever possible, bolted connections should be tightened with torque wrenches.

Particularly important bolted connections on the turbocharger are specified in "Special tightening torques at turbocharger - contact surfaces and threads lubricated with Molykote HSC)" and additionally on the respective work cards.

For all other bolted connections that are to be tightened with torque wrenches, the approximate values for the tightening torques are specified in the Table "Tightening torques for bolts, property class 8.8".

When tightening bolted connections observe as follows:

- Apart from the applied tightening torque, the tension force of a bolted connection depends on the lubricant being used, the surface condition of support and thread, as well as on the material combination. It is therefore very important that these conditions are completely observed.
- The tightening torques for the various threads, depending on their coefficient of friction (meaning depending on the selected lubricant), are listed in Table "Tightening torques for bolts, property class 8.8". The tightening torques apply for bolt materials of the 8.8 property class under utilization of approx. 70% of their yield strength. For other property classes, conversion factors are to be used and multiplied with the tightening torques from "Tightening torques for bolts, property class 8.8". The property class is impressed on the bolt head.

Property Class	5.6	6.8	8.8	10.9	12.9
Conversion Factor x	0.47	0.75	1.0	1.40	1.70

Table 1: Conversion factors for tightening torques depending on the property class of the bolts

Friction Coefficient μ

For the friction coefficient μ use approximately:

$\mu = 0.08$ for lubricants at temperatures

$\leq 200^\circ\text{C}$, e.g. Molykote paste D or Optimoly paste White-T

$\geq 200^\circ\text{C}$, e.g. Molykote paste HSC

$\mu = 0.14$ for lightly oiled surfaces without refinishing

Tightening Torques in General

Thread, Nominal Size	Tightening torque in Nm with Friction Coefficient μ	
	0.08	0.14
M 5	4	6
M 6	7	10

Thread, Nominal Size	Tightening torque in Nm with Friction Coefficient μ	
	0.08	0.14
M 8	17	25
M 10	34	50
M 12	59	84
M 14	95	135
M 14x1.5	100	145
M 16	141	205
M 16x1.5	150	220
M 18	200	280
M 18x1.5	215	320
M 18x2	205	300
M 20	275	400
M 20x1.5	295	450
M 20x2	285	422
M 22	370	540
M 22x1.5	385	595
M 22x2	380	565
M 24	475	690
M 24x2	500	750
M 27	700	1020
M 27x2	730	1100
M 30	950	1380
M 30x2	1015	1540
M 33	1270	1870
M 33x2	1350	2060
M 36	1640	2400
M 36x3	1710	2550
M 39	2115	3120
M 39x3	2190	3300
M 42	2630	3860
M 42x3	2760	4170
M 45	3260	4820
M 45x3	3415	5180
M 48	3950	5820
M 48x3	4185	6370

Table 2: Tightening torques for bolts, property class 8.8

Tightening Torques, Turbocharger

NOTICE

The tightening torques in the Table "Special tightening torques at turbocharger - contact surfaces and threads lubricated with Molykote HSC" apply to a friction coefficient $\mu=0.08$;

Work Card	Description	Tightening Torque in Nm
[500.11]	Undercut bolt (546.073)	75
	Cylindrical screw (596.023) for hub	34
[500.12]	Closing mechanism (emergency operation): Hexagon bolt (517.093)/ Hexagon nut (596.032)	60
[500.32]	Silencer at compressor casing: Hexagon nut (544.182)	140
[500.33]	Air intake casing at compressor casing: Hexagon nut (545.427)	140
[500.34]	Gravitation tank: Hexagon bolts for flanges/oil hoses	59
	Hexagon bolt (554.048)	59
	Hexagon bolt (554.056)	141
	Insert: Undercut bolt (546.073)	75
	Diffuser: Hexagon bolt (546.045)	60
	Compressor casing: Hexagon nut (506.104)	250
	Compressor wheel: Thrust screw for superbolt-tension nut (520.068)	34
	Gas-admission casing: Hexagon bolt (506.106)	165
	Nozzle ring/clamp ring: Hexagon bolt (513.013)	70

Table 3: Special tightening torques on the turbocharger - Contact surfaces and threads lubricated with Molykote HSC

Gaps and Clearances

Short Version

Gaps and clearances are to be measured and compared with the specified tolerances; the results are to be recorded and appropriate action is to be taken.

Required Tools/Appliances

Quantity	Designation	Number	Status
1	Feeler gauge	-	Inventory
1	Sliding caliper	-	Inventory
1	Dial gauge	-	Inventory

Introductory Remarks

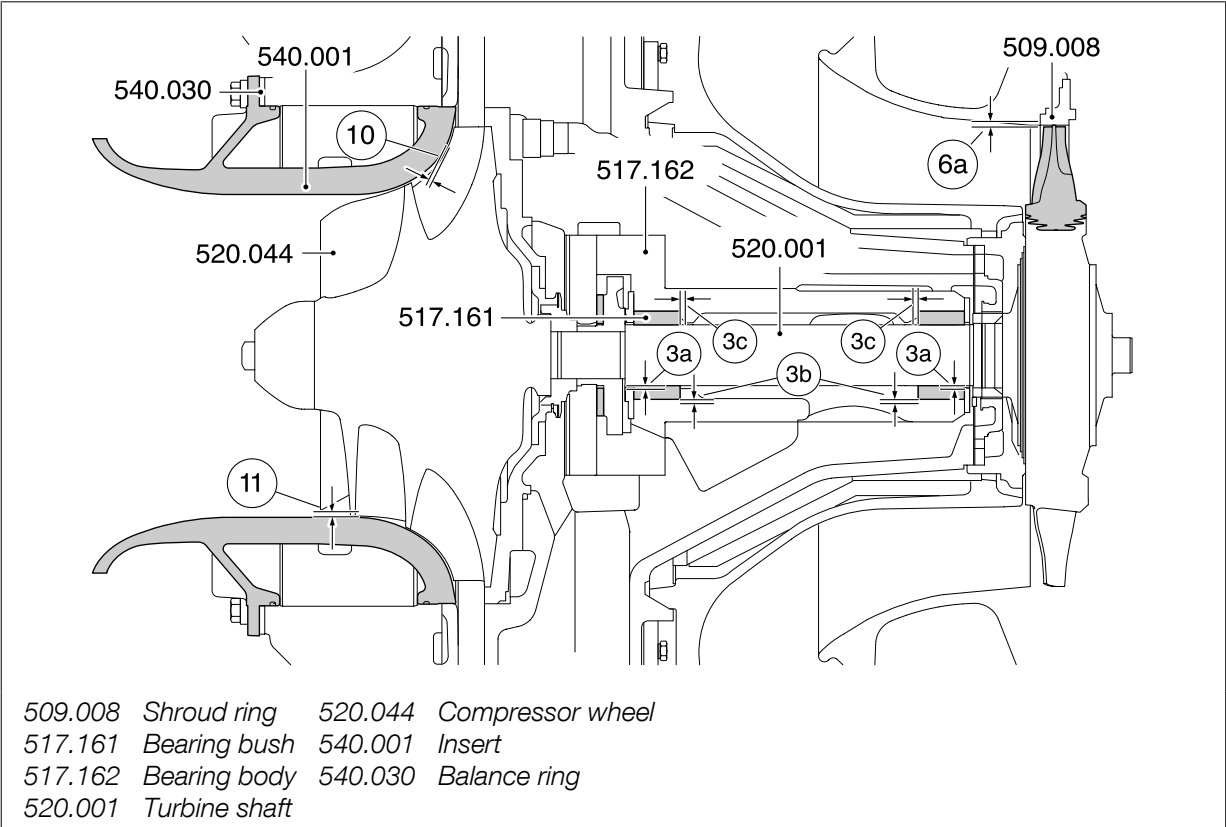
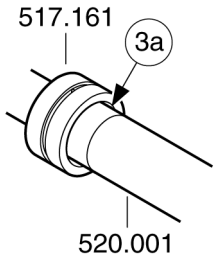


Figure 1: Overview, Gaps and Clearances

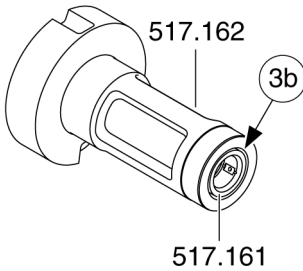
NOTICE

When tolerances are not met, the causes must be corrected, e. g., imbalance, malfunctions with the lube oil system (debris, pressure, volume, temperature, pre-lubrication and post lubrication).

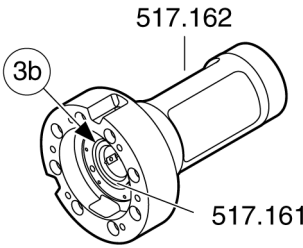
Radial Clearance Bearing Bush/Turbine Shaft

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	3a	–	0.05	0.09	–	0.10	

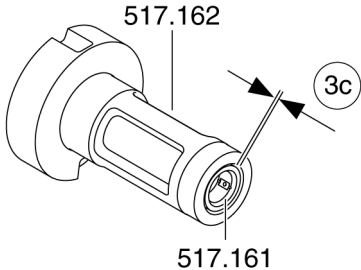
Radial Clearance Bearing Body/Bearing Bush: Turbine Side

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	3b	–	0.25	0.30	–	0.31	

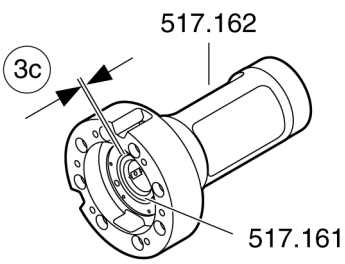
Radial Clearance Bearing Body/Bearing Bush: Compressor Side

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	3b	–	0.25	0.30	–	0.31	

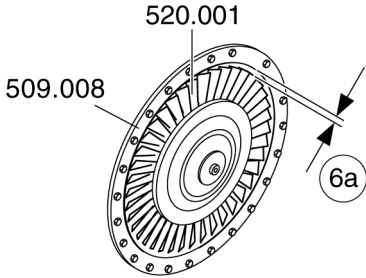
Axial Clearance Bearing Bush/Bearing Body: Turbine Side

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	3c	–	0.16	0.31	–	0.35	

Axial Clearance Bearing Bush/Bearing Body: Compressor Side

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	3c	–	0.16	0.31	–	0.35	

Radial Gap Turbine Blade/Shroud Ring

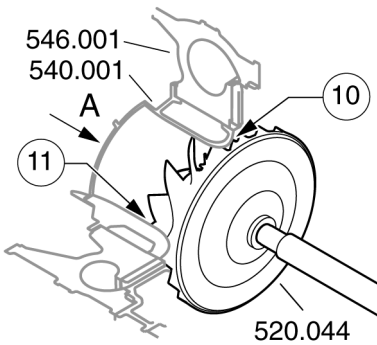
Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	6a ¹⁾	*	0.95	1.15	(0.80) ²⁾	1.40	

1) Setting specification for shroud ring, see Work card .

2) The minimum value may not be fallen short of at any location on the circumference.

Gaps and clearances marked with (*) are to be recorded in the inspection report.

Axial Gap (10) and Radial Gap (11) Compressor Wheel/Insert

Gaps and Clearances	Measuring Point		Condition as Manufactured		Replace Parts		Work Card
			Min. in mm	Max. in mm	Min. in mm	Max. in mm	
	10 ¹⁾	*	2.00	2.30	–	2.50	
	11 ²⁾	*	1.20	1.30	(1.00) ³⁾	1.50	
1) Set rotor towards system in direction (A). Measure the axial gap at 3 locations on the circumference via soft-metal marking, and calculate the mean value (determined value corresponds with the largest given (max.) axial gap).							
2) Take measurement at 4 locations of the circumference with a feeler gauge and calculate the mean value.							
3) The minimum value may not be fallen short of at any location on the circumference.							

NOTICE

Gaps and clearances marked with (*) are to be recorded in the inspection report.



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**7.5 Assembly of the Turbocharger and Auxiliary
Equipment**

Operation with Holding Device

Short Version

Enable emergency operation by blocking the rotating element.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	0,6
1	Assistant	0,6

Required Tools/Appliances

Quantity	Designation	Number	Status
1	Open-jaw wrench 55	–	Inventory
1	Open-jaw wrench 22	–	Inventory
1	Hex key 8	–	Inventory
1	Holding device	596.511	Standard
1	Arresting wrench	596.018	Standard
1	Hub	596.022	Standard
4	Cylindrical screw	596.023	Standard
4	Lock washer pair	596.024	Standard

Corresponding Work Cards

Work card	Work card	Work card
500.12	500.32	500.33
500.41		

Introductory Remarks

The damaged turbocharger can be shut down either with holding device [500.11] or with the closing mechanism [500.12].

Use the holding device (596.511) only when removal of the rotating element is not possible.

NOTICE

By using the closing mechanism (see work card [500.12] Operation without rotating element), further damage of the rotating element can be avoided.

Work Sequence – Assembly of the Holding Device

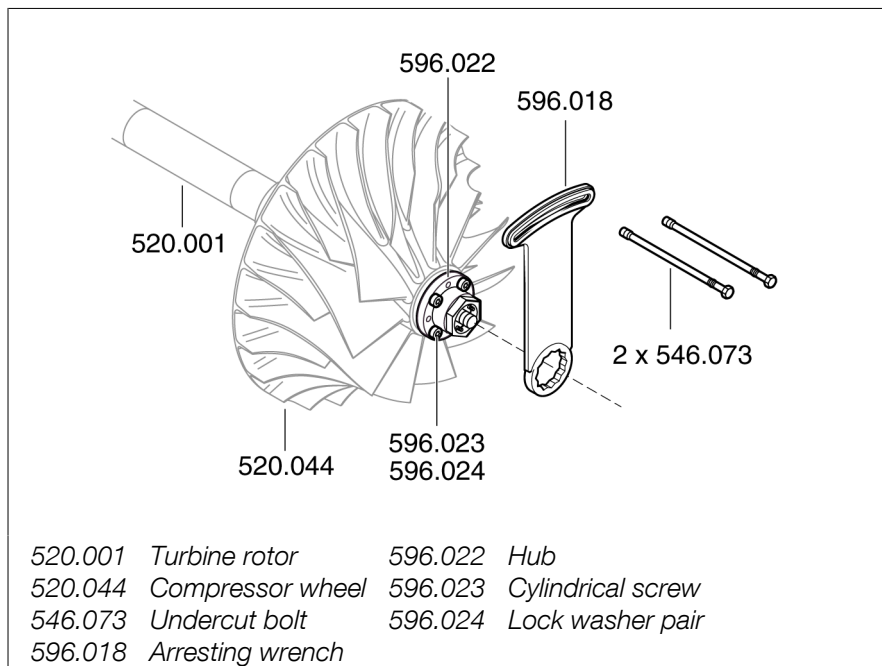


Figure 1: Holding device for emergency operation with rotating element

! DANGER**Danger to life from crush injuries or blows!**

When the engine starts during assembly of the holding device, there is danger of becoming entangled in or coming against rotating components and being seriously injured.

- Shut down the engine before installing the holding device in the turbocharger!
- During the work, secure the engine against starting!
- Set up danger signs!

Starting Condition

Silencer [500.32]/air intake casing [500.33] insert and balance ring disassembled.

Work Steps

1. Unscrew spinner from compressor wheel (520.044).
2. Fasten hub (596.022) on compressor wheel with four cylindrical screws (596.023) and lock washer pair (596.024).

NOTICE

The hub is formed accordingly so that the tension nut does not have to be loosened.

3. Attach arresting wrench (596.018) on hub and tighten with two undercut bolts (546.073) of the insert.

NOTICE

Use lock washer pairs (596.024)!



4. Mount silencer/air intake casing again.

NOTICE

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!

**WARNING****Danger of fire caused by lube oil!**

During emergency operation with the holding device, the danger of fire is given when lube oil gets into the turbocharger.

In emergency operation, always shut the oil supply pipe to the turbocharger!

**NOTICE**

Additional emergency operation notes, see Chapter [5.2].
Consequences

NOTICE**Observe tightening torques upon installation!**

Undercut bolt (546.073): 75 Nm

Cylindrical screw for hub (596.023): 34 Nm



Operation without rotating element

Short Version

Enable emergency operation by disassembling the rotating element and closing the bearing casing with closing covers.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	3,5
1	Assistant	3,5

Required Tools/Appliances

Quantity	Designation	Number	Status
1	Open-jaw wrench 19	–	Inventory
1	Closing mechanism	596.512	Standard
1	Closing cover (compressor side)	596.027	Standard
1	Closing cover (turbine side)	596.028	Standard
1	Gasket (compressor side)	596.029	Standard
1	Gasket (turbine side)	596.030	Standard
1	Stud screw	596.031	Standard
3	Hexagon nut	596.032	Standard
2	Hexagon bolt	596.033	Standard

Corresponding Work Cards

Work card	Work card	Work card
500.11	500.32	500.33
500.41	500.43	500.44
500.51	500.53	

Technical Data

Designation	Information
Closing mechanism	45 kg
Closing cover (compressor side)	31 kg
Closing cover (turbine side)	9 kg

Introductory Remarks

The damaged turbocharger can be shut down either with the holding device [500.11] or with the closing mechanism [500.12].

Use the holding device only when removal of the rotating element is not possible.

NOTICE

By using the closing mechanism (596.512 Operation without rotating element), further damage of the rotating element can be avoided.

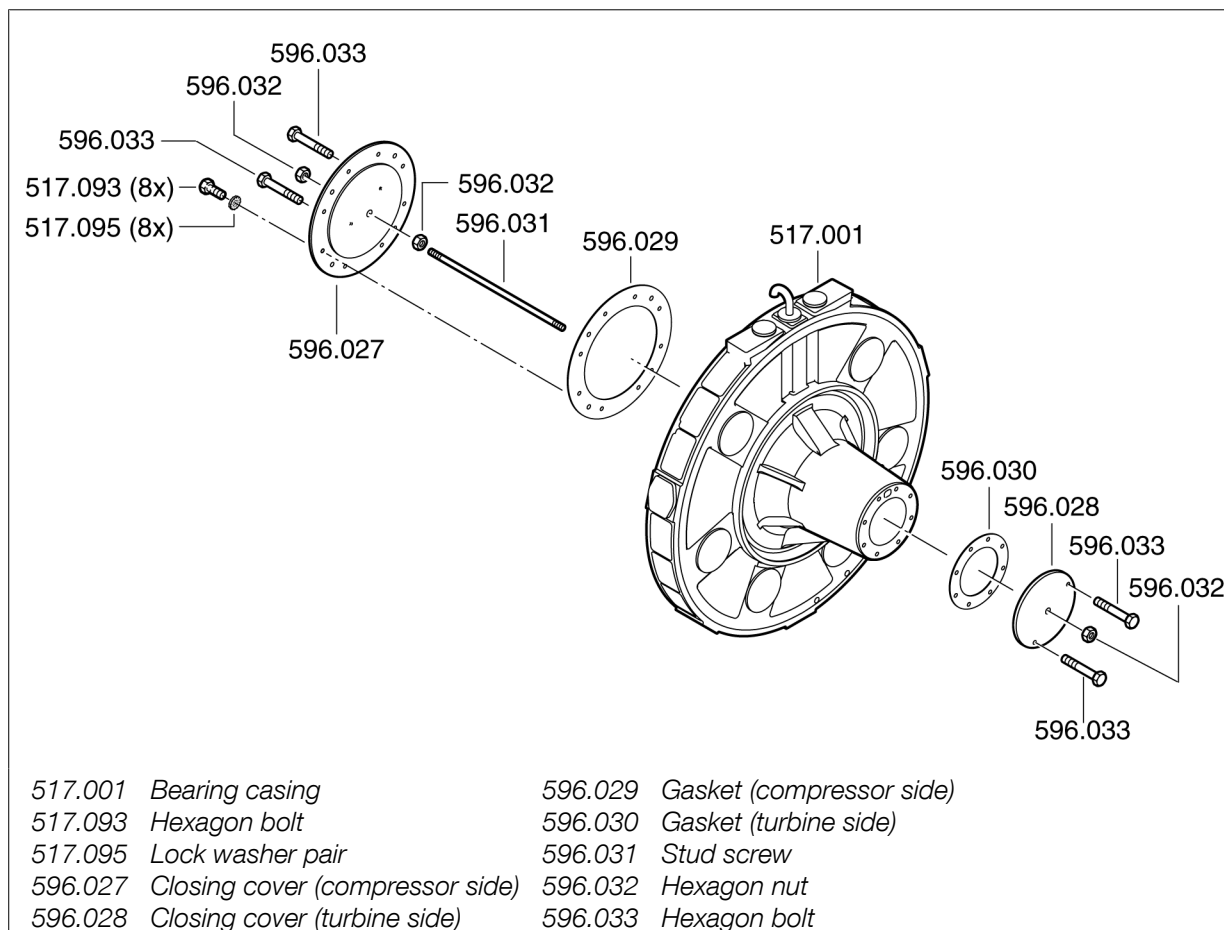
Work Sequence – Assembly of the Closing Covers

Figure 1: Closing mechanism at emergency operation without rotating element

! DANGER**Danger to life from crush injuries or blows!**

When the engine starts during removal of the rotating element or during assembly of the closing mechanism, there is danger of becoming entangled in or coming against rotating components.

- Shut down the engine before removal of the running gear as well as before installing the closing mechanism in the turbocharger!
- During the work, secure the engine against starting!
- Set up danger signs!

Starting Condition

Silencer [500.32]/air intake casing [500.33], insert and gas-admission casing disassembled.

Compressor wheel, labyrinth disk and turbine rotor disassembled.

NOTICE

Bearing bushes and counter-thrust bearings may remain installed.

Work Steps

1. Fasten stud screw (596.031) to closing cover (596.027) with two nuts (596.032).
2. Fasten closing cover and seal (596.029) to bearing casing (517.001) with eight hexagon bolts (517.093) and lock washer pairs (517.095) of the disassembled labyrinth disk.
3. Mount closing cover (596.028) and gasket (596.030) on turbine side and fasten to stud screw (596.031) with hexagon nut (596.032).

NOTICE

For easier installing, screw two hexagon bolts (596.033) each into the closing covers.

4. Mount gas-admission casing.
5. Mount silencer and air intake casing.

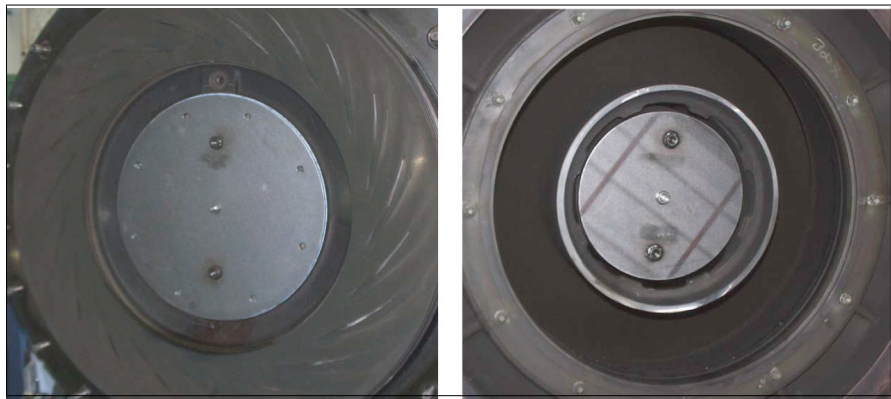


Figure 2: Closing covers mounted (compressor side and turbine side)

! WARNING**Danger of fire caused by lube oil!**

During emergency operation without rotating element, the danger of fire is given when lube oil gets into the turbocharger.

In emergency operation, always shut the oil supply pipe to the turbocharger!

**NOTICE**

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!



NOTICE

Additional emergency operation notes, see Chapter [5.2].

NOTICE

Observe tightening torques upon assembly!

Hexagon bolt (517.093): 60 Nm

Hexagon nut (596.032): 60 Nm



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**7.5 Assembly of the Turbocharger and Auxiliary
Equipment**

Cleaning the Turbine (Wet Cleaning)

Short Version

Carry out the wet cleaning in time according to the maintenance plan; clean components from debris/residue.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	0,6

Technical Data

Designation	Information
Washing nozzles	4
Water pressure	2 bar
Max. washing water-inlet temperature	50 °C
Exhaust gas temperature ahead of turbine	≤ 320 °C
Turbocharger speed	≤ 9500 min ⁻¹

Introductory Remarks

Cleaning Agent Use only fresh water for cleaning!

NOTICE

For information on the requirements of the washing water, see Technical Data - [3.2.4]

Cleaning Interval Every 250 operating hours (depending on the operating conditions the intervals can be shorter).

NOTICE

Cleaning instructions: See cleaning plate on the bracket at the gas outlet casing of the turbocharger.

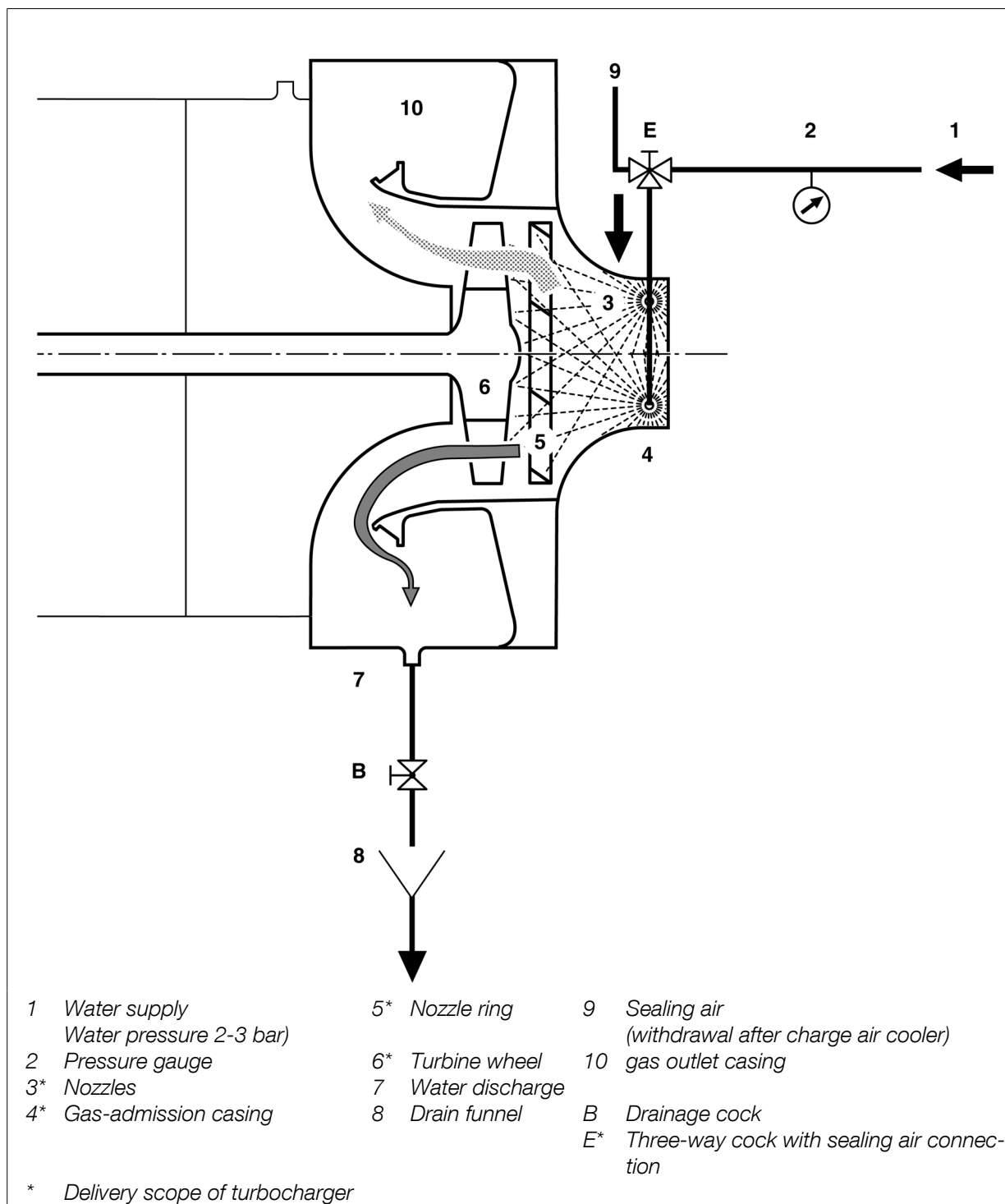


Figure 1: Diagram, wet cleaning of the turbine

NOTICE

Amount and layout of the washing nozzles (3) depends on the engine type and can vary from the illustration.

Work Sequence – Water Washing of Turbine (during Operation)

General

Water washing of the turbine is carried out at reduced engine power.

Washing is carried out in “intervals”, otherwise the turbine could graze.

The size of the nozzle slot and the water pressure are matched to the exhaust amount and temperature (quantity of heat) of the charger.

Work Steps

1. Run the engine at reduced engine power (1/4 load) for approx. 10 minutes (cooling down to the stabilizing condition).
2. Check the operating parameters for the washing of the turbine:
 - Exhaust gas temperature $\leq 320\text{ °C}$
 - Turbocharger speed $\leq 9500\text{ rpm}$
 - Water pressure: 2 bar

NOTICE

The exhaust gas temperature of 320 °C must not be exceeded under any circumstances, otherwise the turbine is liable to make contact during washing.

3. Open drainage cock (B). Check the outflow of exhaust gas at the opened cock; free opening, if required.
4. Open stop cock (E): Water is sprayed through the nozzles.
5. Washing duration 1 minute.

NOTICE

The water running out of the washing water outlet (7) gives information about the washing effect or about the contamination degree of the turbine. The water can be checked at the drain funnel (8).

6. Shut the stop cock (E).
7. Run the turbocharger dry for 1 minute at reduced engine capacity.
8. Repeat the washing procedure and the running dry (work steps 4-7) 3-5 times.
9. Shut drainage cock (B).
10. Run the turbocharger dry for 8 minutes at reduced engine power, then slowly increase the load.

NOTICE

If vibrations due to irregular heavy deposits of debris can not be corrected through the washing, the turbocharger must be dismantled and the respective parts must be cleaned mechanically.

NOTICE

In case vibrations of the turbocharger occur that were previously not existent, repeat the washing procedure.

Cleaning the Turbine (Dry Cleaning)

Short Version

Carry out the cleaning in time according to the maintenance plan; clean components from debris/residue.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	0,6

Technical Data

Designation	Information
Particle size of the granulate in mm (nutshells or soft activated charcoal)	1 mm (max. 1,5 mm)
Filling volume	1,5 l

Introductory Remarks

Cleaning Interval

At normal operating load of the engine, every 24 operating hours.

NOTICE

For turbine cleaning with this setup, do not use water, as in this case the cleaning is carried out at normal operating load of the engine.
Granulates to be used for cleaning: See Chapter Technical Data - [3.2.4]

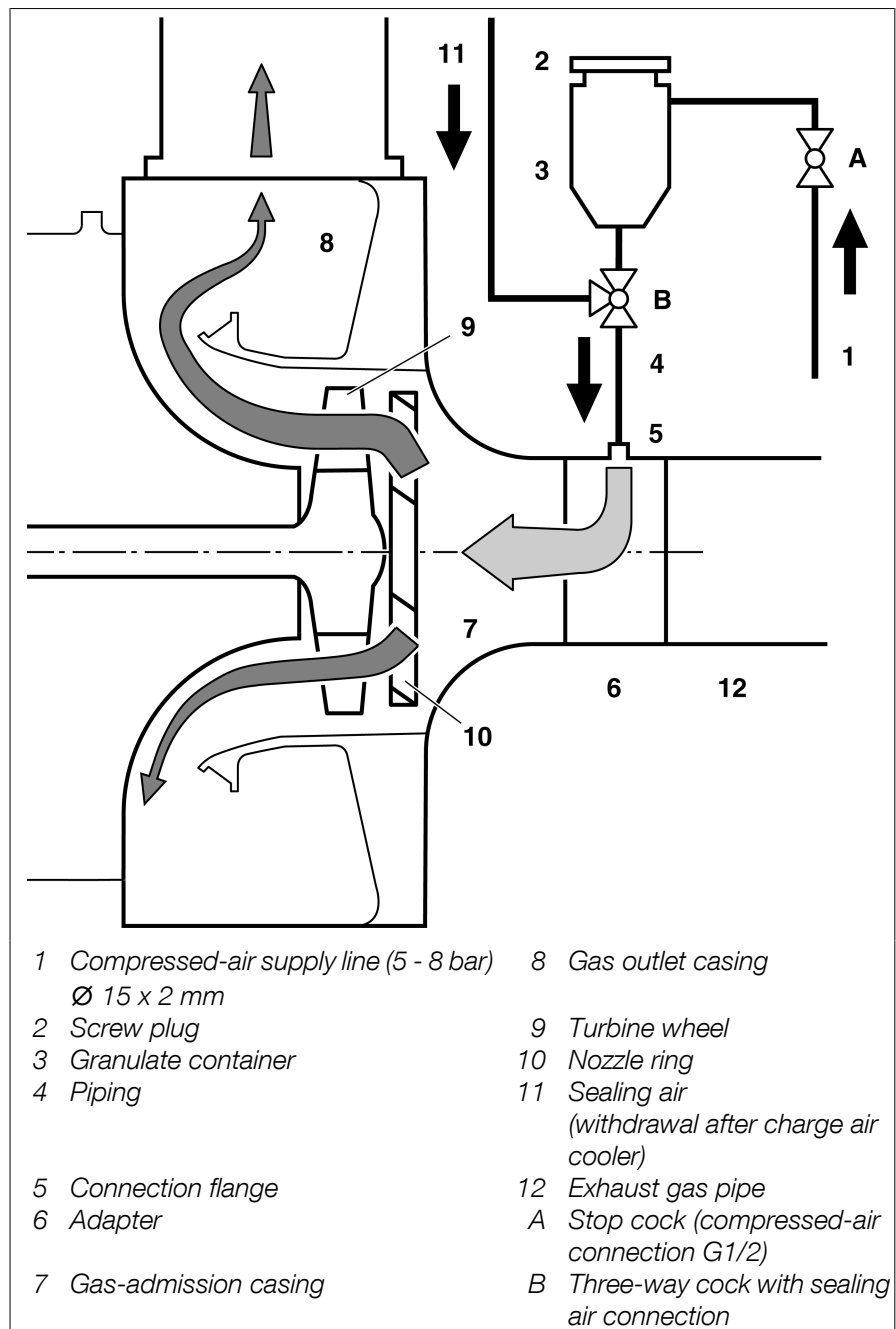


Figure 1: Diagram, dry cleaning of the turbine

General

The exact layout of the components depends on the engine type and can vary from the illustration.

NOTICE

Type and source of the hazard

The connection flange (5) can be attached either at the adapter (6) of the exhaust gas pipe (12) or directly at the gas-admission casing (7).

Work Sequence – Dry Cleaning of Turbine (during Operation)

Work Steps

1. Stop cock (A) must be shut.

2. Three-way cock (B) must be set to position „0“.
3. Unscrew screw plug (2), fill cleaning agent (granulates, active carbon) into the tank (3) and then shut again.
4. Set three-way cock (B) to position „I“
5. Open stop cock (A) (compressed-air connection) until a whistling sound indicates that the granulates are being blown in.
Blow-in duration approx. 30 seconds.
6. Shut the stop cock (A).
7. Set three-way cock (B) to position „0“ (sealing air). The sealing air prevents clogging of the pipe.

Wet Cleaning the Compressor (Pressure Sprayer)

Short Version

The cleaning of the compressor is carried out during operation. Carry out the work in time according to the maintenance schedule. Clean the components from debris/residue.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	0,3

Introductory Remarks

General

The layout of the components depends on the engine type and can vary from the illustration.

See plate with the cleaning specifications (see Fig. Wet cleaning system, compressor (pressure sprayer)) on pressure sprayer.

Cleaning Agent

Use only fresh water for cleaning.

Cleaning Interval

Every 150 operating hours.

The cleaning intervals depend on the contamination degree of the drawn-in air and range between 100 and 200 operating hours.

NOTICE

Carry out the cleaning only with the engine warmed up and in full load operation.

Do not shut down the engine immediately after cleaning, but continue to run it for several minutes until dry.

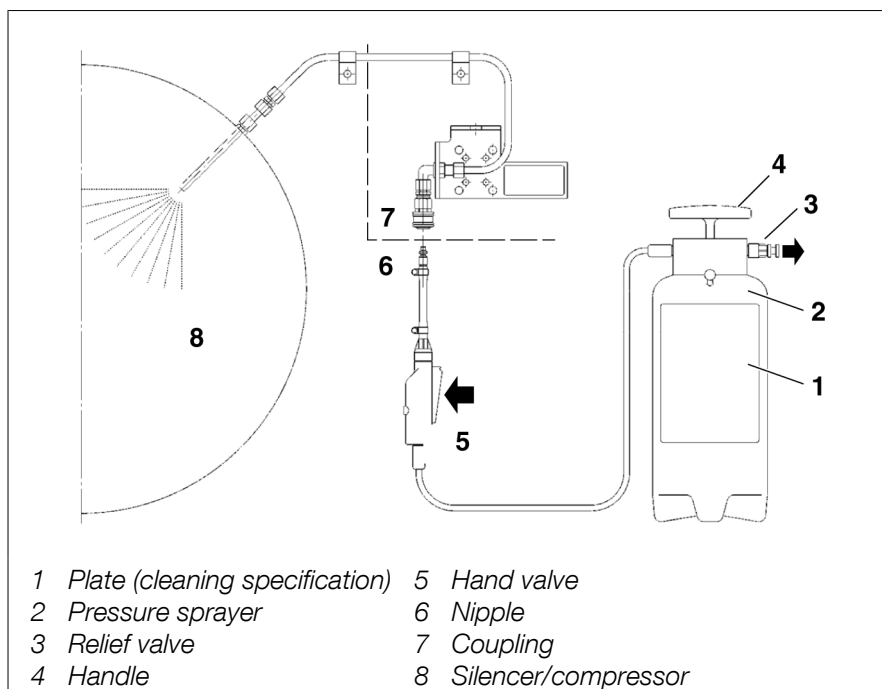


Figure 1: Wet cleaning system, compressor (pressure sprayer)

Work Sequence – Compressor Cleaning (during Operation)

Work Steps

1. Operate engine at full load.
2. Turn the handle (4) on the container of the pressure sprayer (2) to the left and take out the pump.
3. Fill a maximum of 2 l of freshwater into the container (to the 2 L mark on the container).
4. Screw in the pump and operate it until the relief valve (3) blows off.
5. Connect hose with nipple (6) to the coupling (7).
6. Operate the pushbutton (see large arrow in figure [Wet cleaning system, compressor \(pressure sprayer\)](#)) on the hand valve (5) until water is blown into the compressor (8) (approx. 30 seconds).
7. Release the plug connection and vent the container; for this, pull the spindle at the relief valve (3) outward (see small arrow in figure [Wet cleaning system, compressor \(pressure sprayer\)](#)).

Cleaning the Air Filter

Short Version

Clean the air filter regularly in accordance with the maintenance schedule. The work can be carried out during operation.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	0,3

Required Tools/Appliances

Quantity	Designation	Number	Status
–	Compressed air	–	Inventory
–	Cleaning agent (non-corrosive)	–	Inventory

Spare Parts

Amount	Designation	Number
1	Air filter mat	544.150
3	Lashing strap with ratchet	544.160
1	Cover strip	544.170

Introductory Remarks

General

The soiled air filter mat (544.150) can be cleaned several times, or replaced by a new mat.

Replacement is necessary when the air filter mat is strongly deformed, frayed out or perforated.

NOTICE

An increase in pressure loss, due to a contaminated air filter mat, should not be used as a reference for the cleaning interval.

DANGER



Danger to life from crush injuries or blows against rotating components!

When the inspection cover of the silencer is removed during operation, there is danger of becoming entangled in or coming against rotating components in the silencer or turbocharger and being seriously injured.

- Open the inspection cover for visual checks only when the engine is shut down!
- Change only the air filter mat during operation (see Work Sequence)!

Cleaning Interval

Every 250 operating hours.

The cleaning intervals depend on the contamination degree of the drawn-in air and therefore range between 250 and 500 operating hours.

NOTICE

In case of heavy contamination, it may be necessary to immerse the air filter mat into a cleaning agent for several hours. Therefore it is recommended to have a second air filter mat in stock at all times.



Figure 1: Inspection cover (1) and air filter mat with lashing strap on silencer

Work Sequence – Removal/Cleaning/Reinstallation of the Air Filter Mat

1. Loosen lashing straps and remove air filter mat (see Fig. [Inspection cover \(1\) and air filter mat with lashing strap on silencer](#)).
2. Clean the soiled air filter mat with warm water and a cleaning agent (non-corrosive).

NOTICE

Avoid excessive mechanical stress (e.g. wringing, sharp water jet). After cleaning, blow compressed air through the air filter mat.

3. Place the air filter mat over the perforated sheet of the silencer.
4. Place the cover strip over the butt joint of the air filter mat and then fasten the air filter mat with lashing straps.
5. Firmly tighten the ratchets of the lashing straps.

7.1 Preface

7.2 Setting Data

7.3 Emergency Operation

7.4 Cleaning

**7.5 Assembly of the Turbocharger and Auxiliary
Equipment**

Turbocharger to Engine

Short Version

Removing and refitting the turbocharger from and to the engine.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	6,0
1	Assistant	6,0

Required Tools/Appliances

Quantity	Designation	Number	Status
2	Shackle	-	Inventory
1	Torque wrench	-	Inventory
1	Open-jaw and ring wrenches (set)	-	Inventory
1	Socket wrench (set)	-	Inventory
1	Screwdriver	-	Inventory
1	Lifting tackle, ropes	-	Inventory
-	Wood (for placing underneath)	-	Inventory
1	High-temperature lubricant (e. g. Molykote HSC)	-	Inventory

Corresponding Work Cards

Work card	Work card	Work card
500.21	500.22	500.23
500.41		

Technical Data

Designation	Information
Turbocharger	5500 kg

Work Sequence - Deconserving

Condition at time of supply (as a rule)

- Packed in wooden box.
 - Accessories, spare parts and tool box are enclosed.
- For lists of spare parts and tools, see Chapter Spare Parts Catalog.
- Socket adjustments of the casings are factory set.
 - Connections are shut with sealing caps.
 - Interior components are conserved with corrosion prevention agents that are consistent with operating media.
 - Protective coating against corrosion on the outside.

Condition at time of supply with increased corrosion protection

In order to achieve increased corrosion protection (e. g., overseas, tropics, subtropics), the following measures are additionally carried out:

- Drying agent (bag, block) at the inside of closing caps/lids.
- Prior to closing the openings, a vapor-phase corrosion agent is sprayed into the gas intake, gas discharge and air-discharge connections.

Work Steps

1. Remove the protective coating against corrosion on the outside using diesel or kerosene.

2. Remove sealing caps/lids.

In case of increased corrosion protection:

1. Remove all drying agents, if applicable.
2. Blow compressed air through turbocharger casing, in order to remove vapor-phase corrosion agent.



Danger of poisoning from toxic vapours!

The vapour-phase corrosion agent (increased corrosion protection) with which the openings of the turbocharger are treated sets toxic vapours free upon heating.

- Thoroughly blow out the turbocharger casing before assembling the turbocharger!
- Upon initial putting into operation, check if any corrosion protection agent is on the turbocharger!

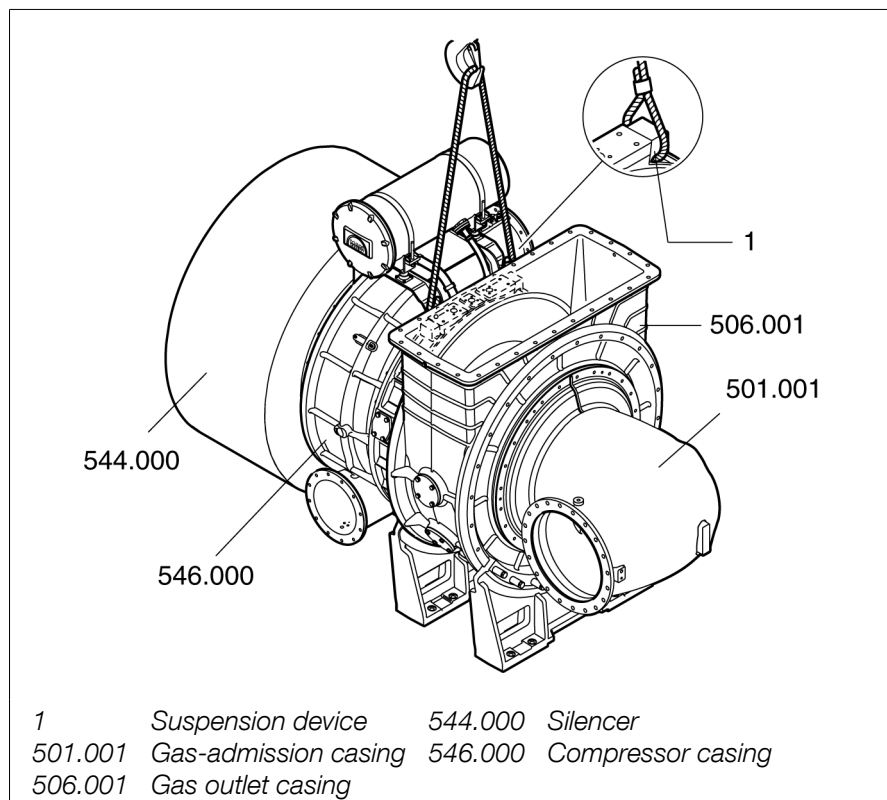
Work Sequence - Assembly to the Engine, Connection of the Charge Air and Exhaust Pipes

Figure 1: Suspension device for turbocharger with 90° gas-admission casing

NOTICE

Suspension device welded to turbocharger (condition of delivery).

Work Steps

1. Insert shackle and tackle line to suspension device (see Fig. [Suspension device for turbocharger with 90° gas-admission casing](#)).

WARNING**Danger of the turbocharger falling down!**

Improvised or incorrectly attached suspension devices can result in the falling down of the turbocharger and can seriously injure or kill persons located below.

- Lift the complete turbocharger only with the suspension device cast to the turbocharger!
- Never use other or self-designed suspension devices!
- Never use the lifting eye bolts to lift up the complete turbocharger, as these are intended only for the lifting individual subassemblies!

WARNING**Danger of injury from suspended loads!**

A rocking turbocharger is difficult to maneuver and can therefore seriously injure or kill people.

- Hang **turbochargers with axial gas-admission casing** onto the two lifting hooks between the compressor casing (546.000) and the gas outlet casing (506.001) and additionally onto the silencer (544.000) so that the turbocharger is in balance!
- Hang **turbochargers with the radial gas-admission casing** only onto the two lifting hooks between the compressor casing (546.000) and the gas outlet casing (506.001). The turbocharger is then in balance!
- Use tackle line/chains and adjust these to the appropriate lengths in order to guide the turbocharger!
- Move the turbocharger slowly in the engine room!

2. Lift turbocharger with lifting tackle and place onto bracket of engine.

3. Aligning the turbocharger:

The exhaust pipe must be in alignment with the gas-admission casing (501.001) and the compressor socket must be in alignment with the charge-air pipe of the engine.

NOTICE

The turbocharger must be aligned particularly accurate when a sliding sleeve is used between compressor casing (546.000) and air pipe instead of a compensator.

4. Lightly tighten the casing feet on the engine bracket. Use the fastening bolts and the resilient sleeves (delivery scope of engine) for this.
5. Connect the exhaust pipe of the engine and gas-admission casing (501.001) with the compensator (delivery scope of engine). Lightly stretch the compensator forward for this and use heat-resistant bolts.
6. Connect the pressure socket of the compressor casing (546.000) and the charge air pipe with the compensator or with the sliding sleeve (both in delivery scope of engine).
7. Tighten screw connection of the casing feet on the engine bracket with torque wrench.

NOTICE

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!

**NOTICE****Observe tightening torques during assembly!**

Screw connection of the casing feet to the engine bracket: see engine manual.



Work Sequence - Connection of the Lube Oil System

General

A backup in the lube oil outlet can lead to an oil-flood in the turbocharger. In the process, oil can seep out of the turbine side and cause a fire in the exhaust pipe.



The lube oil discharge must be installed sloped: Gradient > maximum possible system inclination + 5°. (See Chapter [4.2] [4.2])

Work Steps

1. Remove the closures at the bearing casing (517.001) and connect the lube oil supply and drain pipes to the turbocharger.

NOTICE

After fitting the lube oil pipes to size and prior to the final assembly, treat all pipes in acid bath and thoroughly clean them.

2. Mount and connect lube oil pipes, gauges and pressure controllers.

NOTICE

Use only seals that are in proper condition.

Work Sequence - Connecting the Speed Transmitter

Work Steps

Connect the read-out unit of the speed measuring device to the terminal box (compressor casing) according to the electrical wiring diagram.

Also see Work Card .

Work Sequence - Jet Assist Connection (Optional)

Work Steps

1. Check if the O-ring seals (540.015) are in the insert (540.001).
2. Fit air-supply pipe (delivery scope of engine) with pressure reducer, solenoid valve and non-return valve to the connection at the compressor casing and screw together.

Work Sequence - Cleaning System Connection (Optional)

- Connect the dry cleaning system for the turbine at the intended location on the engine/turbocharger [500.22].
- Mount the wet cleaning equipment for the turbine to the connections of the cleaning nozzles (gas-admission casing) and the dirty water drain [500.21].
- Connect the wet cleaning equipment for the compressor to the connection of the injection pipe in the compressor casing [500.23].

Work Sequence - Removal of the Turbocharger

Removal of the turbocharger is carried out in reverse order of assembly.

Due to restricted space conditions, it may be necessary to remove the turbocharger in partially disassembled condition (silencer, compressor and admission casing, and rotating element disassembled) (and).

 CAUTION

Danger of burns from hot components!

Many engine and turbocharger components become very hot during operation and require ample time for cooling down afterwards.

- Prior to taking any action, allow the engine and turbocharger to cool down sufficiently (1-2 h)!
- Wear protective clothing!

 CAUTION

Danger of burning/scalding from running-out lube oil!

The lube oil is under pressure and very hot!

- Before opening pipes, flanges, screwed connections or gauges, free the system of pressure and check that it is pressure-free!
- Before opening oil pipes, wait until the lube oil has cooled down sufficiently (1-2 h)!
- Drain the system and check that the lube oil has been drained completely!
- Wear protective clothing!

Silencer

Short Version

Removal and installation of the silencer is carried out in completed condition. Disassembly of the silencer is required for all work that must be performed from the compressor side.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	2,0
1	Assistant	2,0

Required Tools/Appliances

Quantity	Designation	Number	Status
1	Open-jaw wrench 24	-	Inventory
1	Torque wrench	-	Inventory
-	Lifting tackle, ropes	-	Inventory
-	Eye bolts, shackles	-	Inventory
-	Wood (for placing underneath)	-	Inventory

Spare Parts

Amount	Designation	Number
24	Lock washer pair	544.181

Corresponding Work Cards

Work card	Work card	Work card
500.24		

Technical Data

Designation	Information
Silencer	800 kg

Introductory Remarks

General

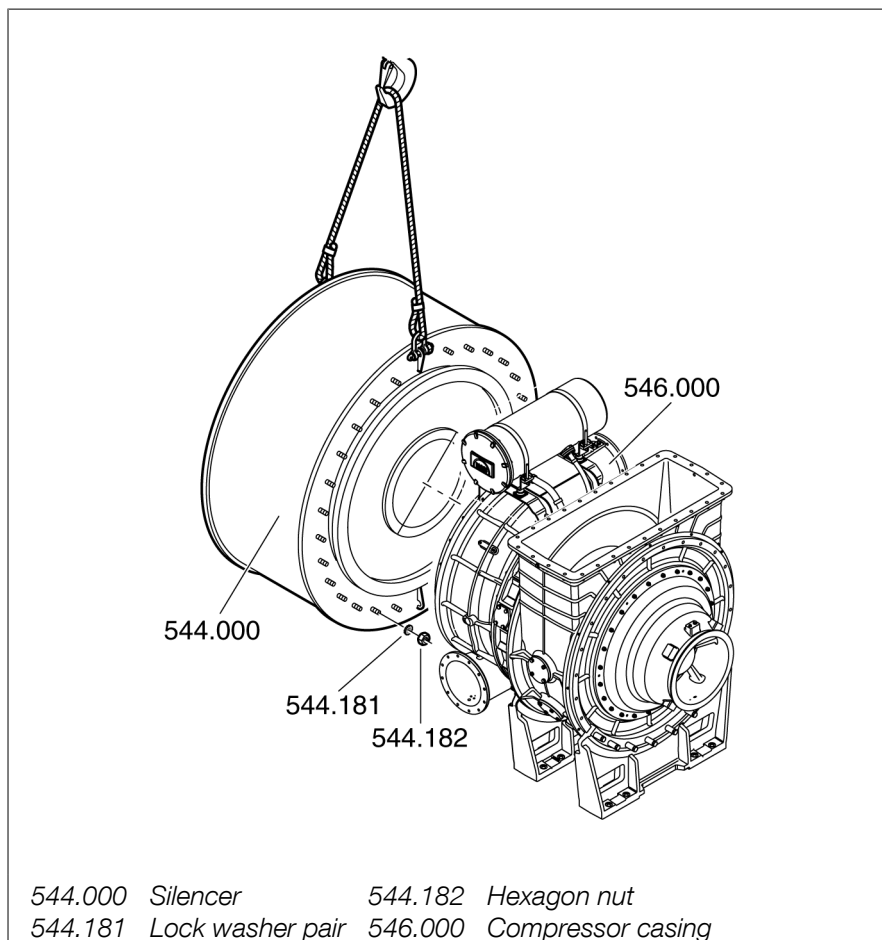


Figure 1: Installation/removal of the silencer at the compressor casing
For cleaning of air filter, see Work Card [500.24].

! DANGER



Danger to life from crush injuries or blows against rotating components!

Never remove the inspection cover (see Fig. in Work Card [500.24]) of the silencer during operation. There is danger of becoming entangled in or coming against rotating components of the silencer or turbocharger and being seriously injured.

- Shut down the engine before working on the silencer!
- During the work, secure the engine against starting!
- Set up danger signs!

WARNING**Danger of injury from suspended loads!**

A rocking silencer is difficult to maneuver and can therefore seriously injure or kill people.

- For attachment of the lifting equipment, use only the lifting hooks on the silencer which are intended for this purpose!
- Move the silencer slowly away from or towards the turbocharger!

Work Sequence - Removing the Complete Silencer

Starting Condition

Connection of the compressor washing equipment loosened (if available).

Work Steps

1. Fasten silencer (544.000) to lifting equipment using ropes.

NOTICE

The stud screws on the silencer casing are secured with Loctite 586. Therefore do not heat them upon disassembly.

2. Loosen hexagon nuts (544.182) at intermediate flange of compressor casing/silencer from the stud bolts of the silencer casing, preceding in circumferential direction.
3. Move silencer in axial direction away from the compressor casing and place it on wooden support with the flange surface down.

Work Sequence - Installation of the Complete Silencer

WorkSteps

Installation is carried out in reverse order of removal.

NOTICE

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!

**NOTICE**

Mount hexagon nuts each with lock washer pairs!

**NOTICE****Observe tightening torques upon installation!**

Hexagon nut (544.182): 140 Nm



Air intake casing

Short Version

Removal and refitting of the air intake casing is carried out in complete condition. Dismantling of the air intake casing is necessary for all jobs that must be performed from the compressor side.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	2,0
1	Assistant	2,0

Required Tools/Appliances

Quantity	Designation	Number	Status
2	Open-jaw wrench 24	-	Inventory
1	Torque wrench	-	Inventory
-	Lifting tackle, ropes	-	Inventory
-	Wood (for placing underneath)	-	Inventory
1	Shackle	596.046	Standard
1	Lifting eye bolt	596.051	Standard

Spare Parts

Amount	Designation	Number
48	Lock washer pair	545.428

Technical Data

Designation	Information
Air intake casing axial	237 kg
Air intake casing 90°	384 kg

Introductory Remarks

General

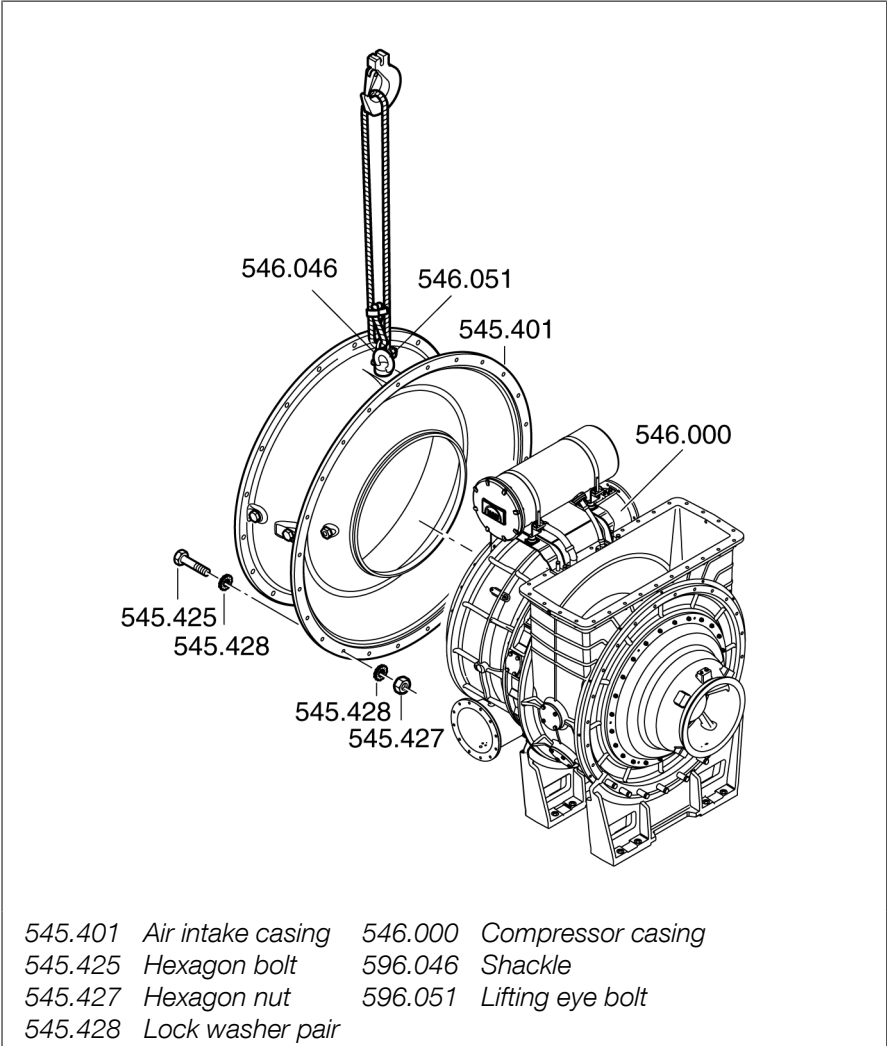


Figure 1: Installation/removal of the air intake casing at the compressor casing

Work Sequence - Removing

Starting Condition

Connection of the compressor washing equipment loosened (if available).

Work Steps

1. Screw eye bolt (596.051) into air intake casing (545.401).
2. Hang air intake casing via shackle (596.046) to lifting device; tension the rope.
3. Loosen hexagon nuts (545.427), hexagon bolts (545.425), lock washer pairs (545.428) in circumferential direction.
4. Move air intake casing away from compressor casing in axial direction and place down on wooden supports.

Work Sequence - Installation

Work Steps

Installation is carried out in reverse order of removal.

NOTICE

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!

**NOTICE**

Mount hexagon bolts and hexagon nuts each with lock washer pairs!

**NOTICE**

Observe tightening torques upon installation!
Hexagon nut (545.427): 140 Nm



Gravitation Tank

Short Version

The gravitation tank is removed only when the compressor casing is disassembled.

Required Personnel and Time

Number	Qualification	Duration in h
1	Technician	1,5

Required Tools/Appliances

Quantity	Designation	Number	Status
1	Open-jaw wrench 19	-	Inventory
1	Open-jaw wrench 24	-	Inventory
-	Lifting tackle, ropes	-	Inventory
-	Wood (for placing underneath)	-	Inventory

Spare Parts

Amount	Designation	Number
8	Lock washer pair	554.049
1	Sealing ring	554.061
1	Seal	554.062
1	Sealing ring	554.083

Technical Data

Designation	Information
Gravitation tank	140 kg

Work Sequence - Removal of the Gravitation Tank

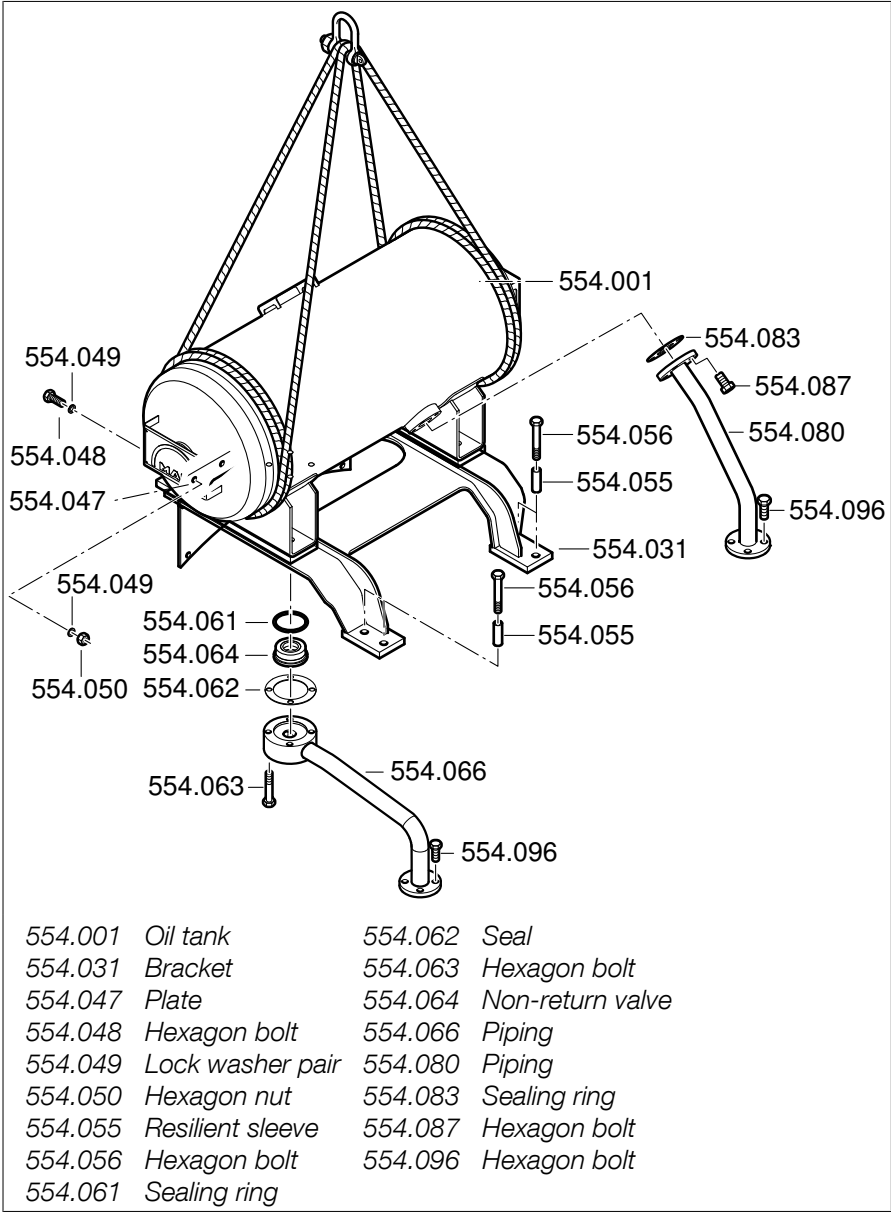


Figure 1: Removal of the gravitation tank (554.000)

Starting Condition

Pipes (554.066 and 554.080) have been released from bearing casing and oil tank, and oil drained. The openings at the gravitation tank and the bearing casing have been provisionally shut.

Work Steps

Check if the sealing rings (554.061, 554.062, 554.083) need to be replaced.

1. Fasten oil tank (554.001) to rope; use lifting device.
2. Loosen hexagon bolts (554.048) of plate (554.047).
3. Loosen hexagon bolts (554.056) of bracket (554.031).
4. Lift gravitation tank (554.000) with lifting device and set down on wooden support.

Work Sequence - Installation of the Gravitation Tank

Work Steps

Installation is carried out in reverse order of removal. Afterwards connect the pipings (554.066, 554.080). Use new sealing rings, as required.

NOTICE

Lubricate contact surfaces and threads (e. g., with Molykote HSC)!



NOTICE

Observe tightening torques upon installation!

Hexagon bolts flanges/ pipings: 59 Nm

Hexagon bolt (554.048): 59 Nm

Hexagon bolt (554.056): 141 Nm



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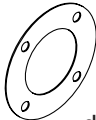
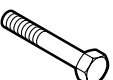
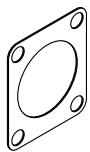
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9.1 Spare Parts

9.2 Tools

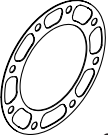
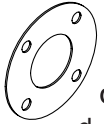
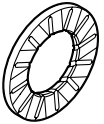
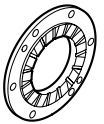
List of Spare Parts

595.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
506.027	1	Gasket	0.06	 $d_i = 93$ $d_a = 190$	509.012	4	Hexagon bolt	0.1	 M12 x 70 SM
506.066	2	Hexagon bolt	0.1	 M16 x 40 SM	509.014	12	Lock washer pair	0.01	 $d_i = 13$
506.106	6	Hexagon bolt	0.2	 M16 x 80 SM	513.013	10	Hexagon bolt	0.06	 M12 x 35 SM
506.107	27	Lock washer pair	0.01	 $d_i = 17$	513.014	21	Lock washer pair	0.01	 $d_i = 13$
506.108	1	Hexagon bolt	0.1	 M16 x 40 SM	517.026	1	Gasket	0.1	 $d_i = 91$

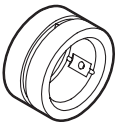
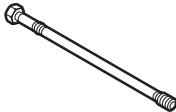
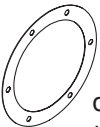
List of Spare Parts

595.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
517.063	1	O-ring seal	0.2	 $d_i = 1005$	517.102	2	Hexagon nut	0.02	 M12
517.070	1	Gasket, turbine side	0.1	 $d_i = 187$	517.109	1	Gasket	0.1	 $d_i = 53$ $d_a = 130$
517.075	1	Bearing disk	1.4		517.154	1	Counter-thrust bearing	2.0	
517.085	1	O-ring seal	0.1	 $d_i = 350$	517.156	12	Lock washer pair	0.01	 $d_i = 8.7$
517.095	14	Lock washer pair	0.01	 $d_i = 13$	517.157	3	Hexagon bolt	0.01	 M8 x 20

List of Spare Parts

595-01

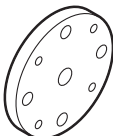
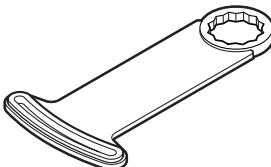
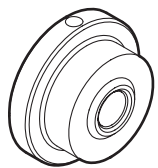
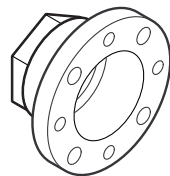
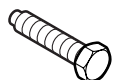
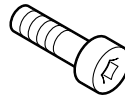
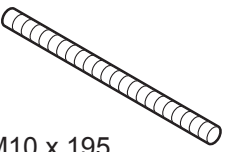
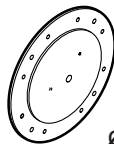
Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
517.161	2	Bearing bush	2.7		540.015	2	O-ring seal	0.15	 $d_i = 625$
517.172	2	Countersunk bolt	0.01	 M8 x 30	546.073	1	Undercut bolt	0.3	 M16 x 250
517.173	2	Lock nut	0.01	 VM8	554.013	1	O-ring seal	0.1	 $d_i = 388$
517.185	1	Gasket, compressor side	0.1	 $d_i = 178$ $d_a = 233$	544.150	1	Fleece mat	1.4	 5220 x 710 x 16
517.211	1	Seal ring	0.02	 $d_i = 45$ $d_a = 100$					

9.1 Spare Parts

9.2 Tools

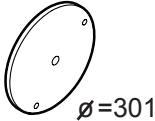
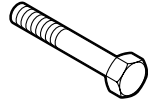
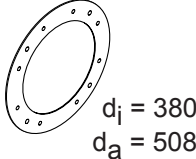
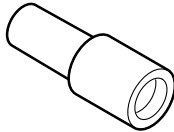
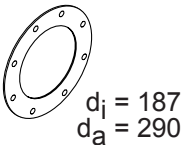
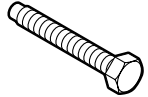
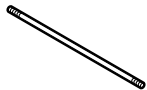
List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.001	1	Puller ring, compressor wheel	2		596.018	1	Locking key (emergency operation)	3.3	
596.002	1	Assembly ring, compressor wheel	2		596.022	1	Hub, (emergency operation)	1.6	
596.003	1	Jack bolt	0.3	 M20 x 2 x 100	596.023	4	Cylindrical screw	0.1	 M10 x 35
596.004	4	Threaded bar	0.1	 M10 x 195	596.024	4	Lock washer pair	0.01	 $d_i = 10,7$
596.005	4	Hexagon nut	0.01	 M10	596.027	1	Closing cover (compressor side), emergency operation	19.1	 $\varnothing = 508$

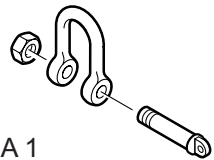
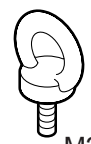
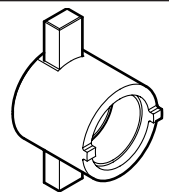
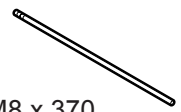
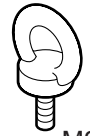
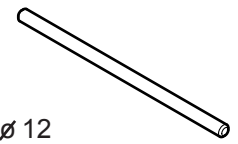
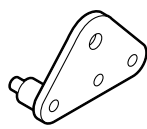
List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.028	1	Closing cover (turbine side), emergency operation	1.9		596.033	2	Hexagon bolt	0.1	 M12 x 120
596.029	1	Gasket (compressor side)	0.5		596.040	1	Extension shaft	0.4	
596.030	1	Gasket (turbine side)	0.2		596.041	3	Hexagon bolt	0.3	 M12 x 90
596.031	1	Stud screw	0.6	 M12 x 658	596.044	2	Guide bar	0.5	 M12 x 600
596.032	3	Hexagon nut	0.02	 M12	596.045	2	Guide bar	1.1	 M16 x 700

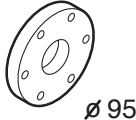
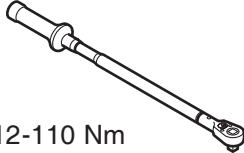
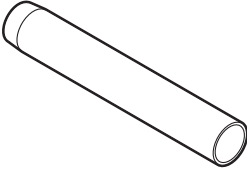
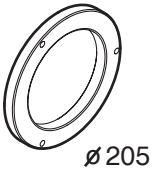
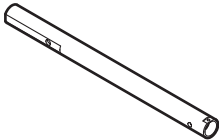
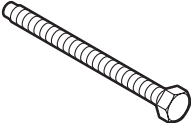
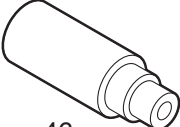
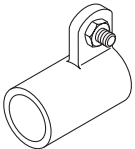
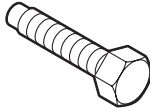
List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.046	1	Shackle	0.4	 A 1	596.058	2	Eye bolt	0.5	 M20 x 2
596.050	2	Guide bar	1.5	 M12 x 170	596.065	1	Spanner for ring nut	2.4	
596.051	2	Eye bolt	0.2	 M12	596.070	2	Threaded bar	0.1	 M8 x 370
596.052	1	Eye bolt	0,3	 M20 x 2	596.071	1	Turn spin	0.2	 Ø 12
596.055	1	Suspension plate, gas-admission casing	2.6		596.072	2	Hexagon nut	0.01	 M8

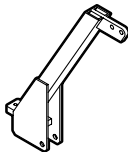
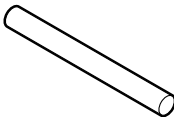
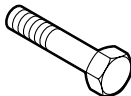
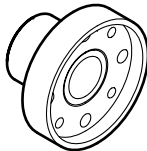
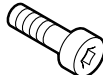
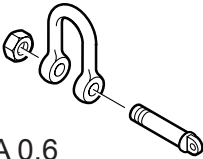
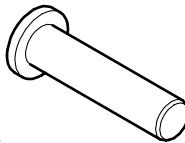
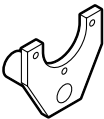
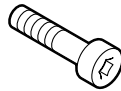
List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.073	1	Puller ring	1.1	 $\varnothing 95$	596.100	1	Torque spanner	1.5	 12-110 Nm
596.075	1	Protection sleeve, turbine rotor	6.6		596.101	1	Surface drive		 6 x 10
596.086	1	Assembly ring	2.2	 $\varnothing 205$	596.160	1	Pipe, suspension device for compressor wheel	5.3	
596.087	3	Hexagon bolt	0.1	 M10 x 130	596.167	1	Sleeve	1.2	 $d_a = 46$
596.504 • 596.092 • 596.124 • 596.125		Clamping sleeve • Clamping sleeve • Hexagon bolt • Hexagon nut	0.9		596.168	1	Hexagon bolt, suspension device for compressor wheel	0.1	 BM12 x 60

List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.170	1	Support	7.5		596.270	1	Pipe	1.8	 30 x 3 x 900
596.197	1	Hexagon bolt	0.2	 M16 x 80	596.271	1	Slide	3.5	
596.198	1	Hexagon nut	0.03	 M16	596.272	1	Cylindrical screw	0.03	 M10 x 35
596.208	1	Shackle	0.2	 A 0.6	596.273	1	Bolt	0.01	 ø 6
596.267	1	Lever carrier	1.6		596.274	3	Cylindrical screw	0.03	 M10 x 45

List of Tools

596.01

Order No.	Number	Designation	Weight [kg]	Design	Order No.	Number	Designation	Weight [kg]	Design
596.280	3	Hexagon bolt	0.1	 M16 x 50					

- 1 Safety**
- 2 Addresses**
- 3 Technical Data**
- 4 Description**
- 5 Operation**
- 6 Maintenance**
- 7 Work Cards**
- 8 Index**
- 9 Spare Parts and Tools**
- 10 Spare Parts Catalog**

Technical Documentation Exhaust Gas Turbocharger Spare Parts Catalogue

C3

Exhaust Gas Turbocharger. **TCA66**

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MAN Diesel SE • 86224 Augsburg, Germany • Phone +49 821 322 0 • Fax +49 821 322 3382



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3	Spare Parts	13
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501.05	Gas-admission Casing for VTA	
506.01	Gas Outlet Casing	
509.01	Gas Outlet Diffuser	
510.01	Variable Turbine Nozzle Ring (VTA)	
511.01	Adjusting Device for VTA	
513.02	Turbine Nozzle Ring	
517.03	Bearing Casing	
518.01	Casing Feet	
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542.01	Diffuser	
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579.03	Wet Cleaning of the Compressor (pressure sprayer)	
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591.02	Sound Insulation Compressor Casing (single outlet)	

Introduction

1

Preface

1.1

Spare Parts/Spare Parts Catalogue

Maintenance and repair work can be carried out properly only when the required spare parts are available.

They are to be kept in stock or to be ordered in good time. The order numbers required for ordering can be found in spare parts catalogues. The spare parts catalogue for the exhaust gas turbocharger is available.

The structural principle of this spare parts catalogue is explained in [Chapter 1.2](#). Order numbers for engine components, measurement and control equipment as well as for system accessories of the engine can be found in other modules of the technical documentation. See the enclosed Notes in [Chapter 1.3](#).

Sincerely yours

MAN Diesel SE

Validity

Validity	Spare parts catalogues are specially adapted to the respective turbocharger version. Therefore, the respective spare parts catalogue is valid only for the turbocharger or the turbocharger series with the works No. or factory-number group on the cover sheet. Important! <i>Spare parts catalogues may not be transferred from one turbocharger to another.</i> Spare parts catalogues which are handed out in advance for information purposes are marked with a red note sheet as "Informational Issue". Important! <i>Informational issues are not suitable for ordering spare parts, not suitable for the planning of inventory, not suitable to be taken over into electronic data processing systems!</i>
Replacement Sheets	In case of changes at a later point of time, the respective auxiliary or replacement sheets will be sent to you. Important! <i>Please ensure that such auxiliary or replacement sheets, as have been assigned to you, are properly filed into all corresponding spare parts catalogues.</i>
Changes/Alterations on the Turbocharger	If changes/alterations are carried out to the turbocharger over the course of the operating period for which no new drawings are drawn up and subsequently no new spare parts sheet is made out by the MAN Diesel SE, the operators of the engine themselves must enter a respective remark into the spare parts catalogue. This will help to avoid false orders.
Delivery of Changed Parts	Due to continuous development on the turbochargers, future spare parts for orders might be delivered that are changed in comparison to the original components, which nonetheless fulfill the same function. In such cases, the identifying card of the delivered part will contain an appropriate note.

Ordering Spare Parts

1.3

Overview

The turbocharger is part of a system from which the assemblies can be from various manufacturers. The following table gives an overview of the various spare parts catalogues/information aides from which the necessary data for spare parts orders can be taken.

Component/Assemblies	Information Aid	Note/ Link
Turbocharger	Spare parts catalogue, turbocharger	
	Data in work cards	
Tools (turbocharger)	List in the turbocharger tool box	
	Data in work cards	
Engine	Spare parts catalogue, engine	See technical documentation of engine and system
Measurement and control equipment	List of measurement and control equipment	
Engine and system accessories	Spare parts catalogue/Spare parts sheet depending on manufacturer and product	

Table 1. Spare parts catalogues and comparable information aides

Ordering of Spare Parts

To avoid questions when ordering spare parts, the following information is required:

Type of turbocharger. xxx xx
 Factory number of the turbocharger x xxx xxx
 (Type of engine) xxxxxx
 (Factory number of the engine). xxx
 Designation of the part xxxxxxxxx
 Order number xxx.xxx
 Quantity. xx
 Shipping address xxxxxxxxx
 Shipping method. xxxxxxxxx

Important!

If parts must be ordered that are not contained or can not be found in the spare parts catalogue, describe the part and the build-in location as precisely as possible (send photo along). For parts of subsuppliers which can not be found in their records, the information on the type plate must be quoted.

Address

Please direct your order to:

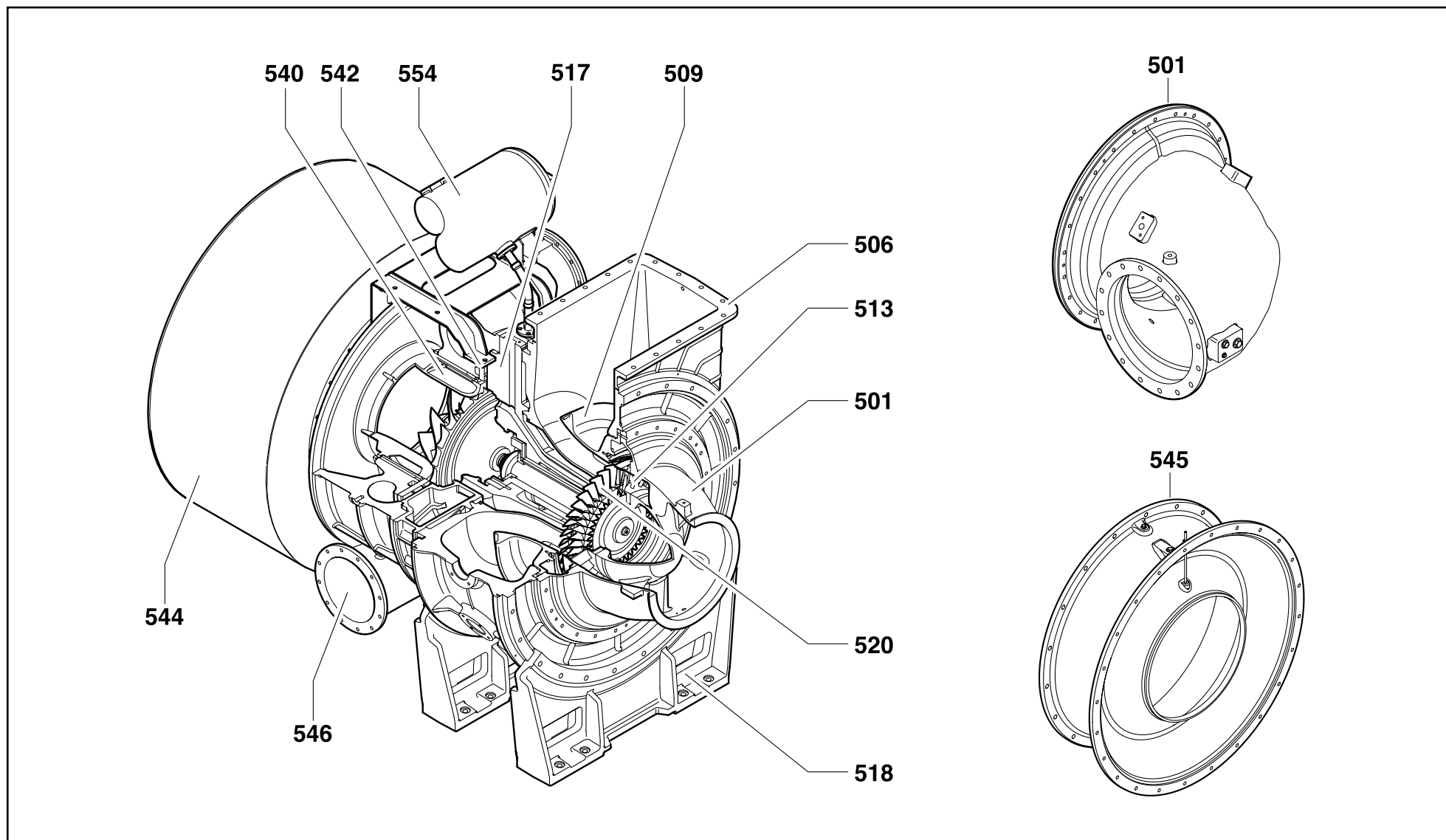
MAN Diesel SE
86224 Augsburg

Telephone: +49 821 322 3994
Telefax: +49 821 322 3998



List of Assemblies

2



List of Assemblies

Assembly	Designation
500	Exhaust gas turbocharger
501	Gas-admission casing
506	Gas outlet casing
509	Gas outlet diffuser
513	Turbine nozzle ring
517	Bearing casing
518	Casing feet
520	Rotor complete
540	Insert
542	Diffuser
(544)	Silencer
(545)	Air intake casing (axial/ 90°)
546	Compressor casing (single outlet, double outlet)
554	Gravitation tank
562 ¹⁾	Speed measuring device
578 ¹⁾	Cleaning device for turbine
579 ¹⁾	Cleaning device for compressor
591 ¹⁾	Covering and Insulation

1) Assemblies not shown in the figures
(...) if available



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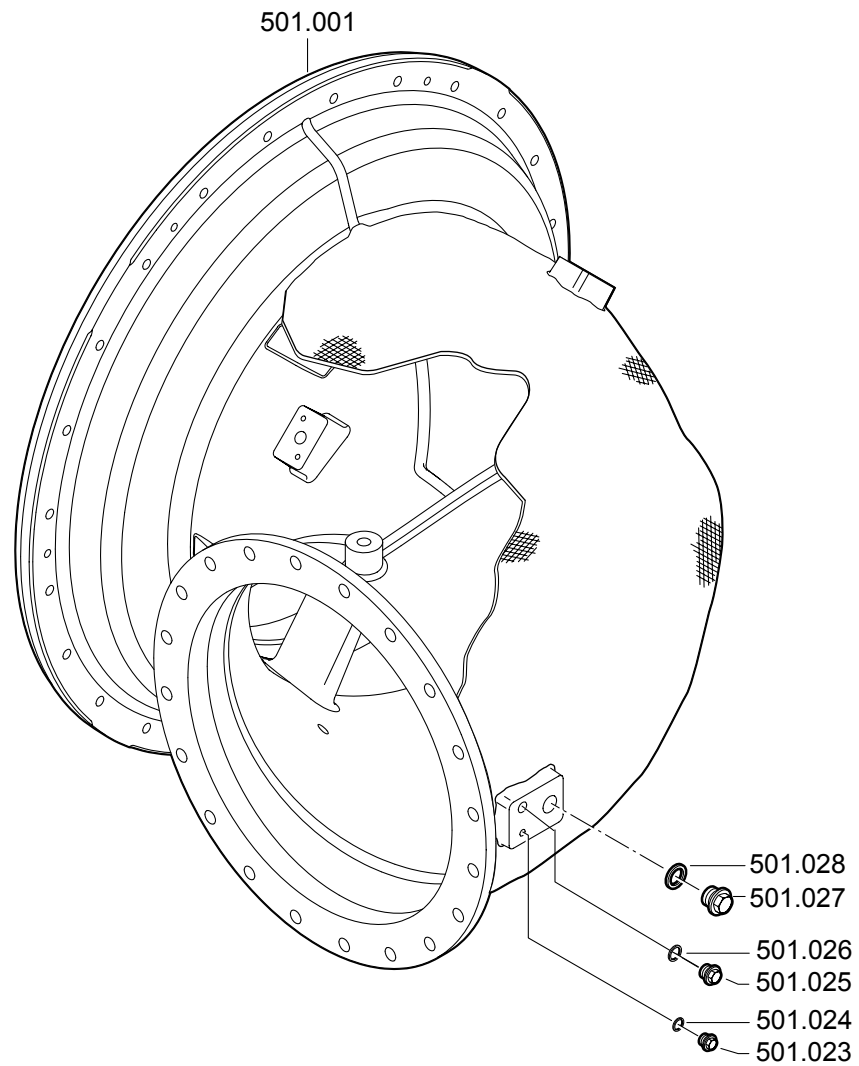
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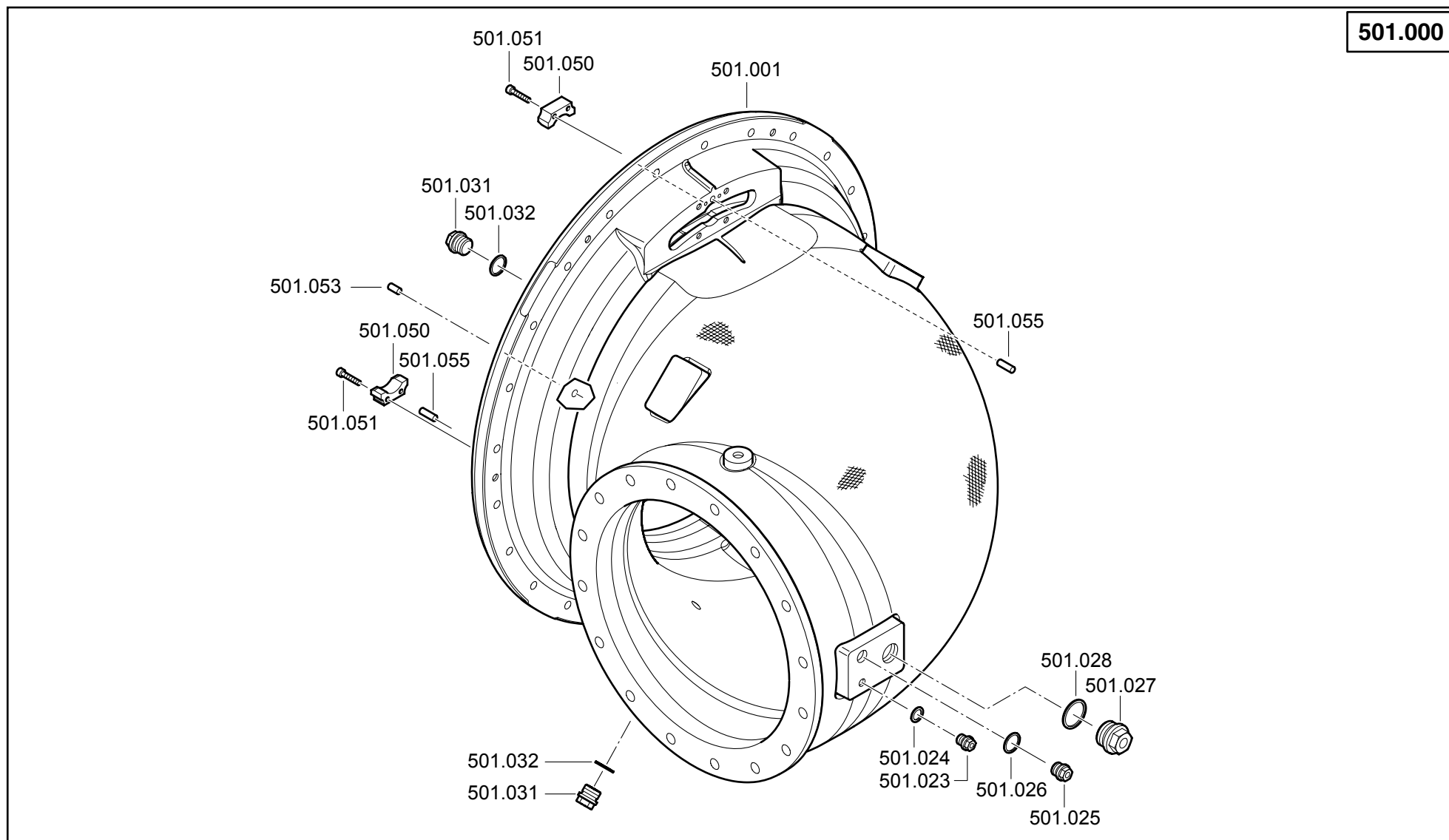


Gas-admission Casing

501.02

Order No.	Number	Designation	Order No.	Number	Designation
501.000	1	Gas-admission casing, compl.			
501.001	1	Gas-admission casing with insulation			
501.023	1	Screw plug			
501.024	1	Sealing ring			
501.025	1	Screw plug			
501.026	1	Sealing ring			
501.027	1	Sealing ring			
501.028	1	Screw plug			



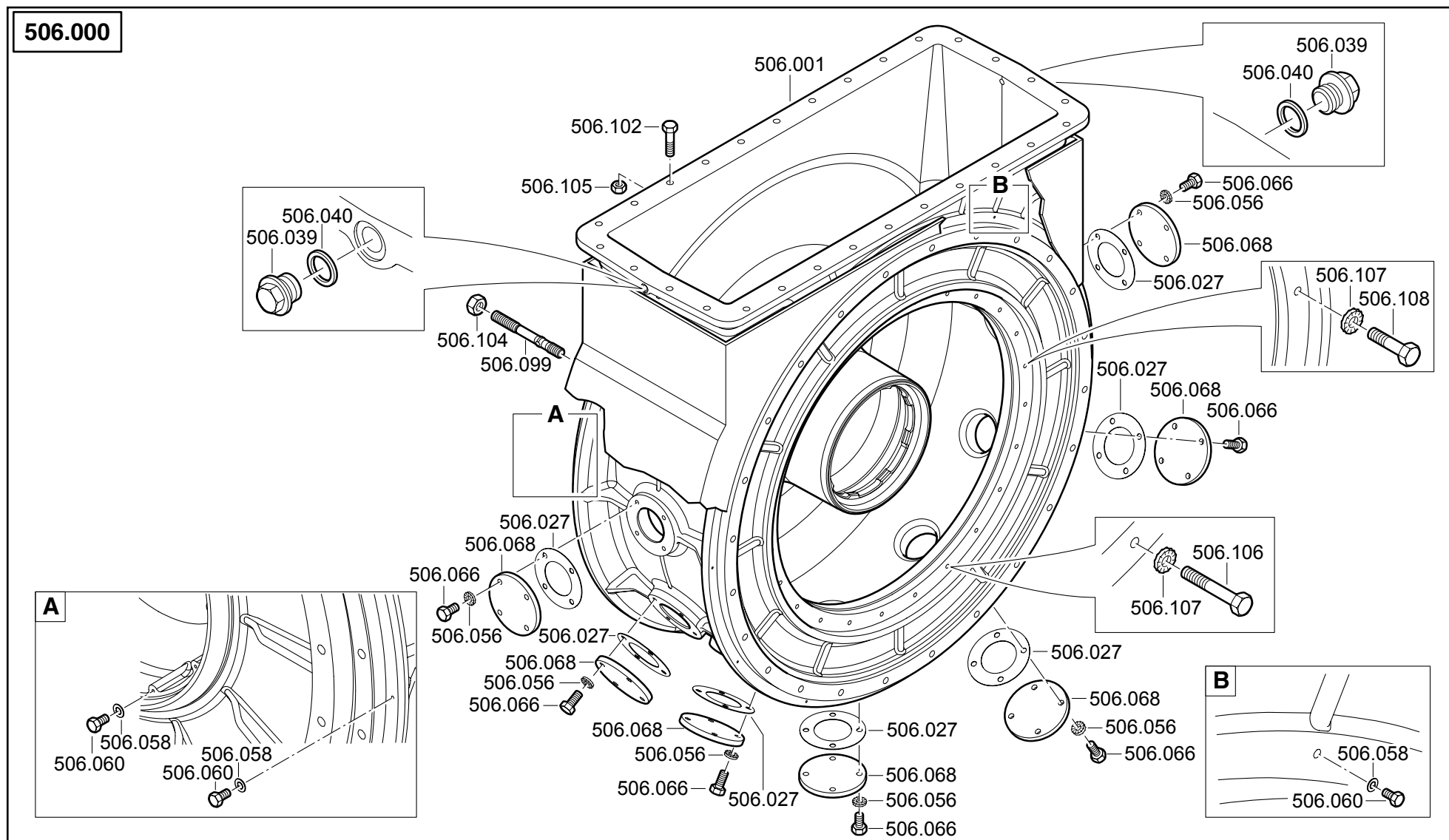


Gas-admission Casing for VTA

501.05

Order No.	Number	Designation	Order No.	Number	Designation
501.000	1	Gas-admission casing, compl.			
501.001	1	Gas-admission casing with insulation			
501.023	1	Screw plug			
501.024	1	Sealing ring			
501.025	1	Screw plug			
501.026	1	Sealing ring			
501.027	1	Screw plug			
501.028	1	Sealing ring			
501.031	2	Screw plug			
501.032	2	Sealing ring			
501.050	2	Guide piece			
501.051	4	Cylindrical screw			
501.053	1	Cylindrical pin			
501.055	2	Cylindrical pin			





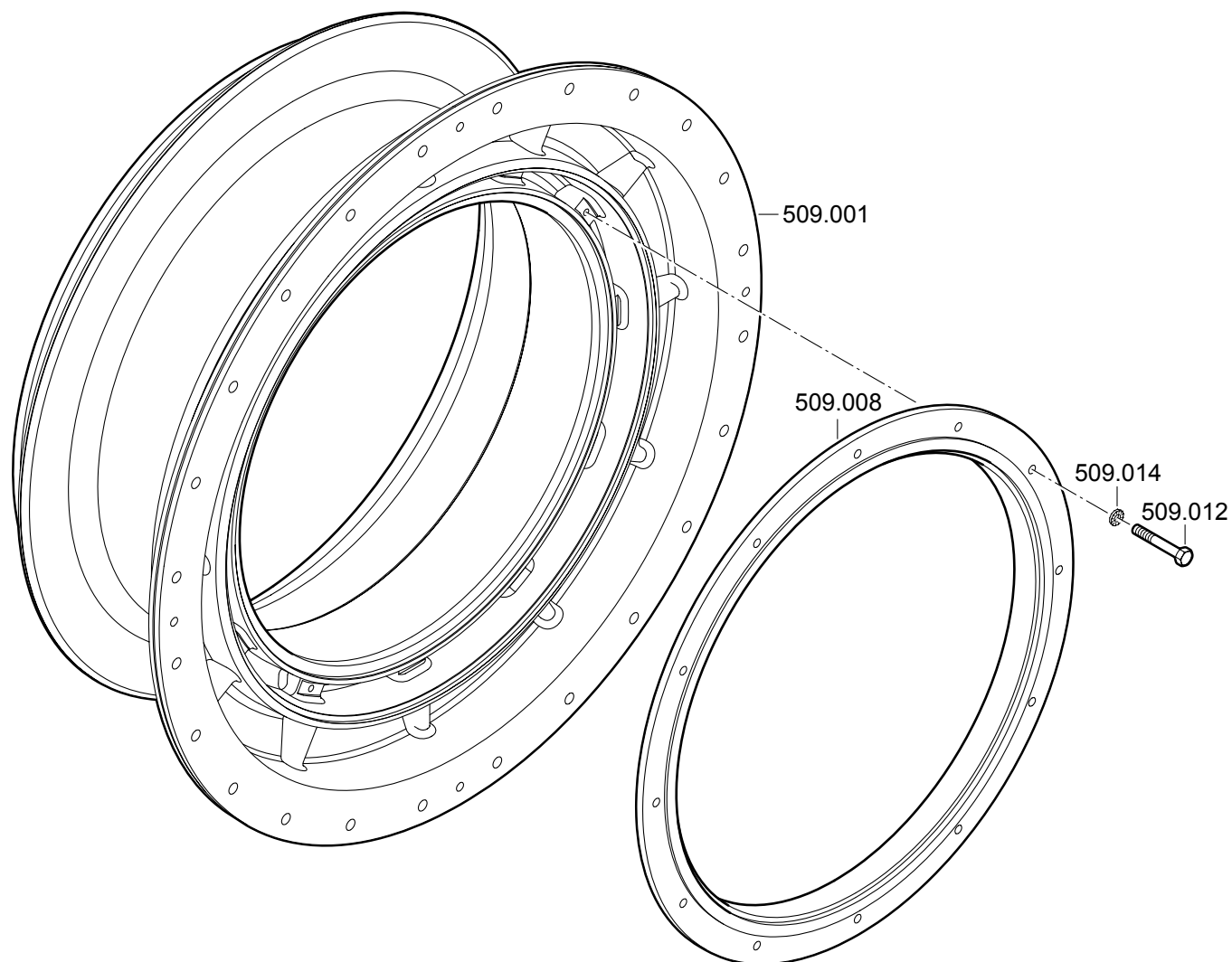
Gas Outlet Casing

506.01

Order No.	Number	Designation	Order No.	Number	Designation
506.000	1	Gas outlet casing, compl.			
506.001	1	Gas outlet casing with insulation			
506.027	7	Seal			
506.039	2	Screw plug			
506.040	2	Sealing ring			
506.056	28	Lock washer pair			
506.058	36	Lock washer pair			
506.060	36	Hexagon bolt			
506.066	28	Hexagon bolt			
506.068	7	Flange			
506.099	24	Undercut bolt			
506.102	26	Hexagon bolt			
506.104	28	Hexagon nut			
506.105	26	Hexagon nut			
506.106	24	Hexagon bolt			
506.107	27	Lock washer pair			
506.108	3	Hexagon bolt			



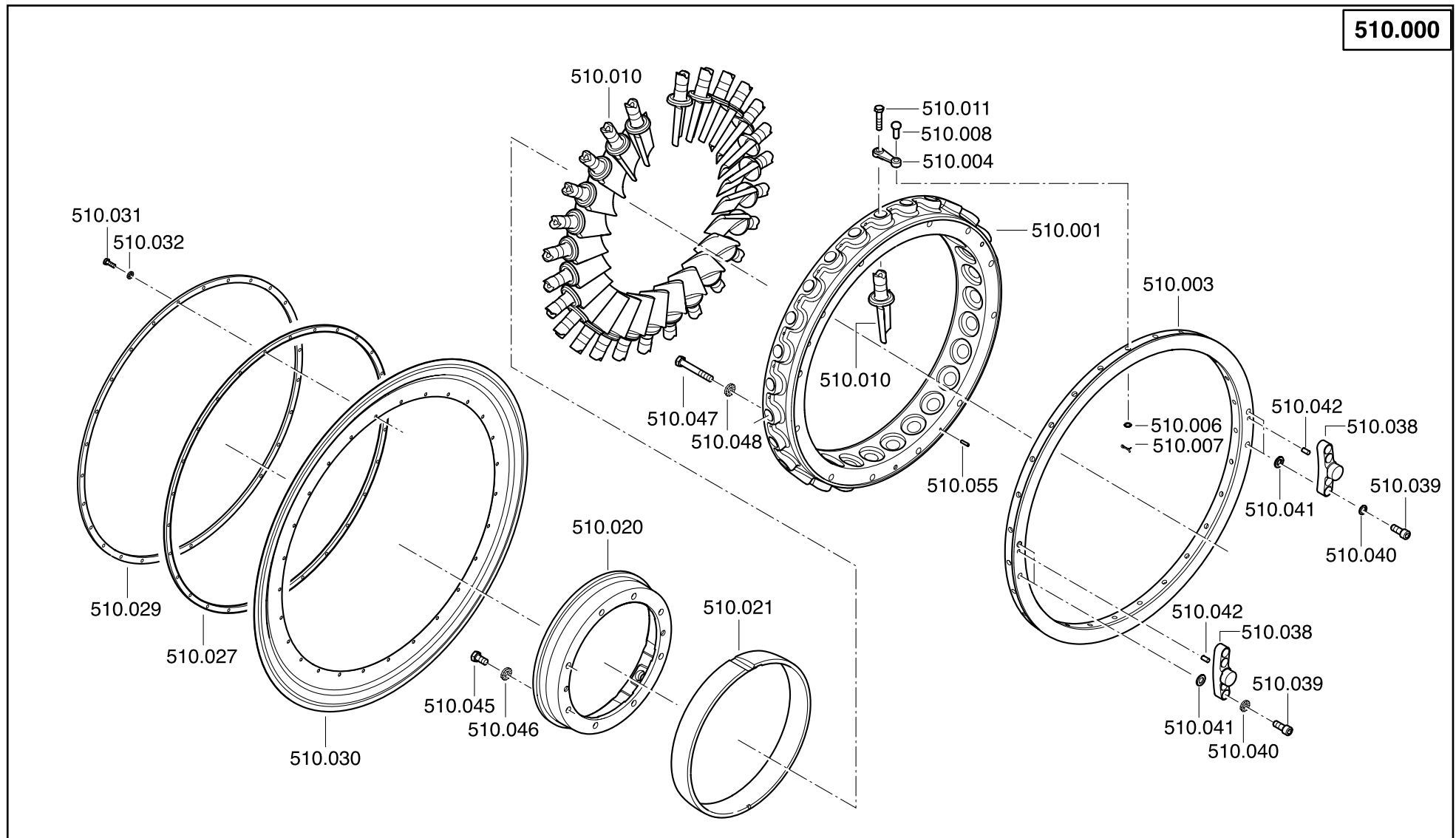
509.000



Gas Outlet Diffuser**509.01**

Order No.	Number	Designation	Order No.	Number	Designation
509.000	1	Diffuser, complete			
509.001	1	Diffuser outside wall			
509.008	1	Shroud ring			
509.012	12	Hexagon bolt			
509.014	12	Lock washer pair			

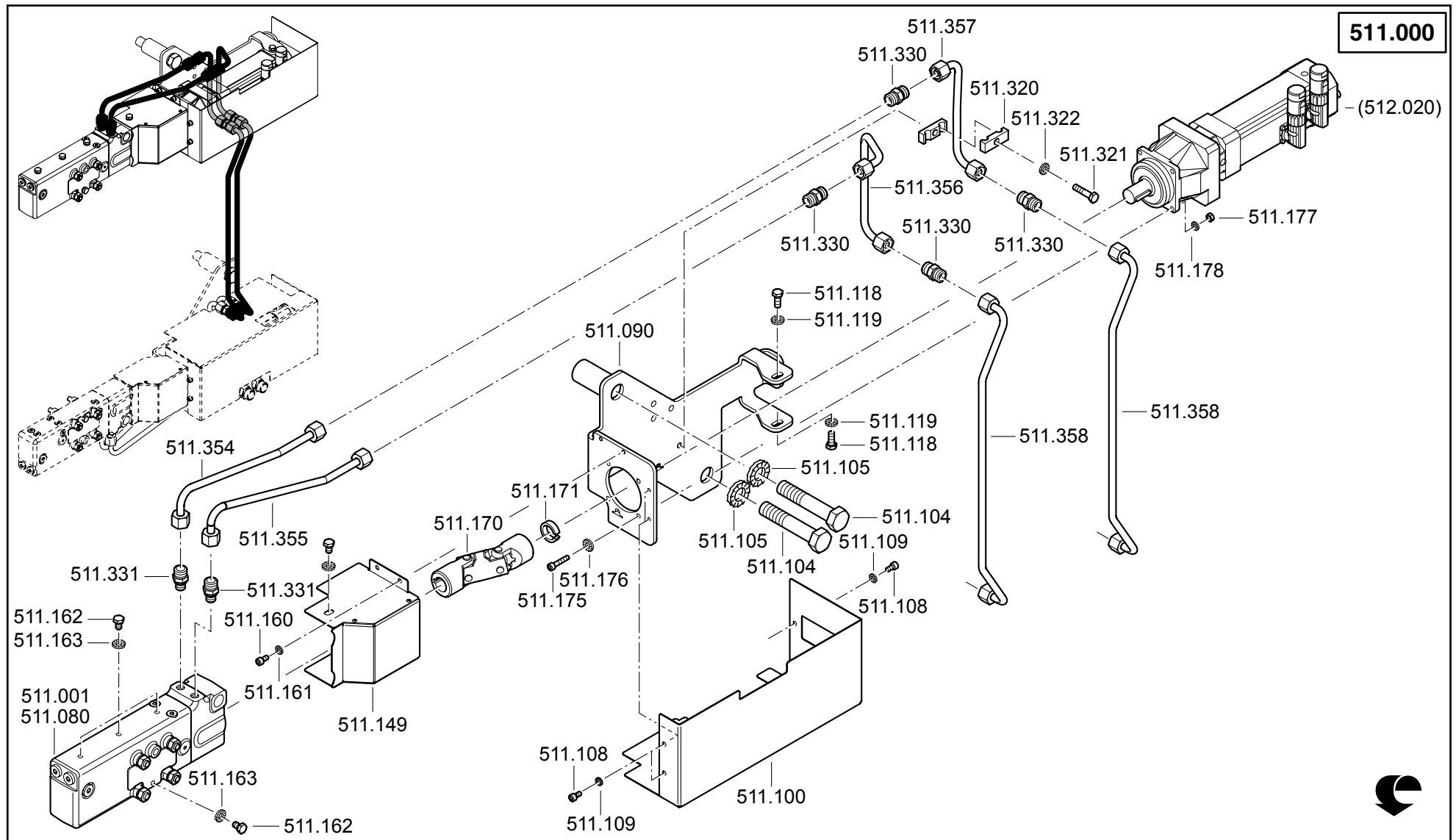




Variable Turbine Nozzle Ring (VTA)

510.01

Order No.	Number	Designation	Order No.	Number	Designation
510.000	1	Variable turbine nozzle ring, compl.	510.055	1	Clamping pin
510.001	1	Outer guide ring			
510.003	1	Setting ring			
510.004	27	Lever			
510.006	27	Disc			
510.007	27	Cotter pin			
510.008	27	Pin			
510.010	27	Turbine guide vane			
510.011	27	Hexagon bolt			
510.020	1	Holding ring			
510.021	1	Inner guide ring			
510.027	1	Intermediate ring			
510.029	1	Clamping ring			
510.030	1	Sealing cover			
510.031	27	Hexagon bolt			
510.032	27	Lock washer pair			
510.038	2	Carrier			
510.039	4	Cylindrical screw			
510.040	4	Lock washer pair			
510.041	4	Disc			
510.042	2	Cylindrical pin			
510.045	8	Hexagon bolt			
510.046	8	Lock washer pair			
510.047	12	Hexagon bolt			
510.048	12	Lock washer pair			

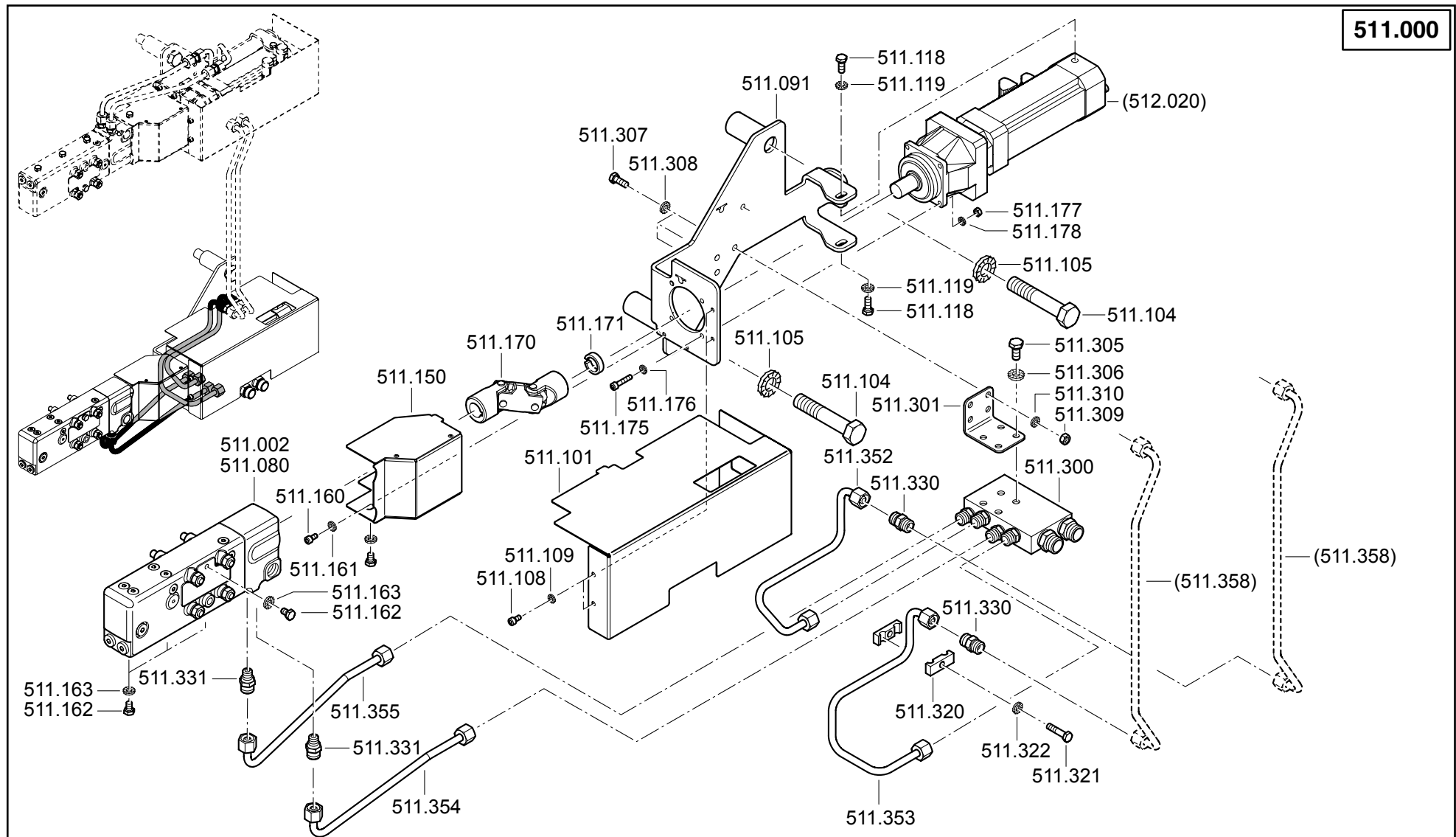


Adjusting Device for VTA

511.01

Order No.	Number	Designation	Order No.	Number	Designation
511.000	1	Adjusting device for VTA, compl.	511.330	4	Connecting socket
511.001	1	Spindle drive	511.331	2	Threaded socket
511.080	1	Lubrication grease	511.354	1	Cooling water pipe
511.090	1	Support	511.355	1	Cooling water pipe
511.100	1	Plate	511.356	1	Cooling water pipe
511.104	2	Hexagon bolt	511.357	1	Cooling water pipe
511.105	2	Lock washer pair	511.358	2	Cooling water pipe
511.108	3	Cylindrical screw	512.020	1	Servo motor
511.109	3	Lock washer pair			
511.118	2	Hexagon bolt			
511.119	2	Lock washer pair			
511.149	1	Plate			
511.160	2	Cylindrical screw			
511.161	2	Lock washer pair			
511.162	4	Hexagon bolt			
511.163	4	Lock washer pair			
511.170	1	Shaft joint			
511.171	1	Distance sleeve			
511.175	4	Cylindrical screw			
511.176	4	Lock washer pair			
511.177	4	Hexagon nut			
511.178	4	Lock washer pair			
511.320	1	Pipe clamp			
511.321	1	Hexagon bolt			
511.322	1	Lock washer pair			

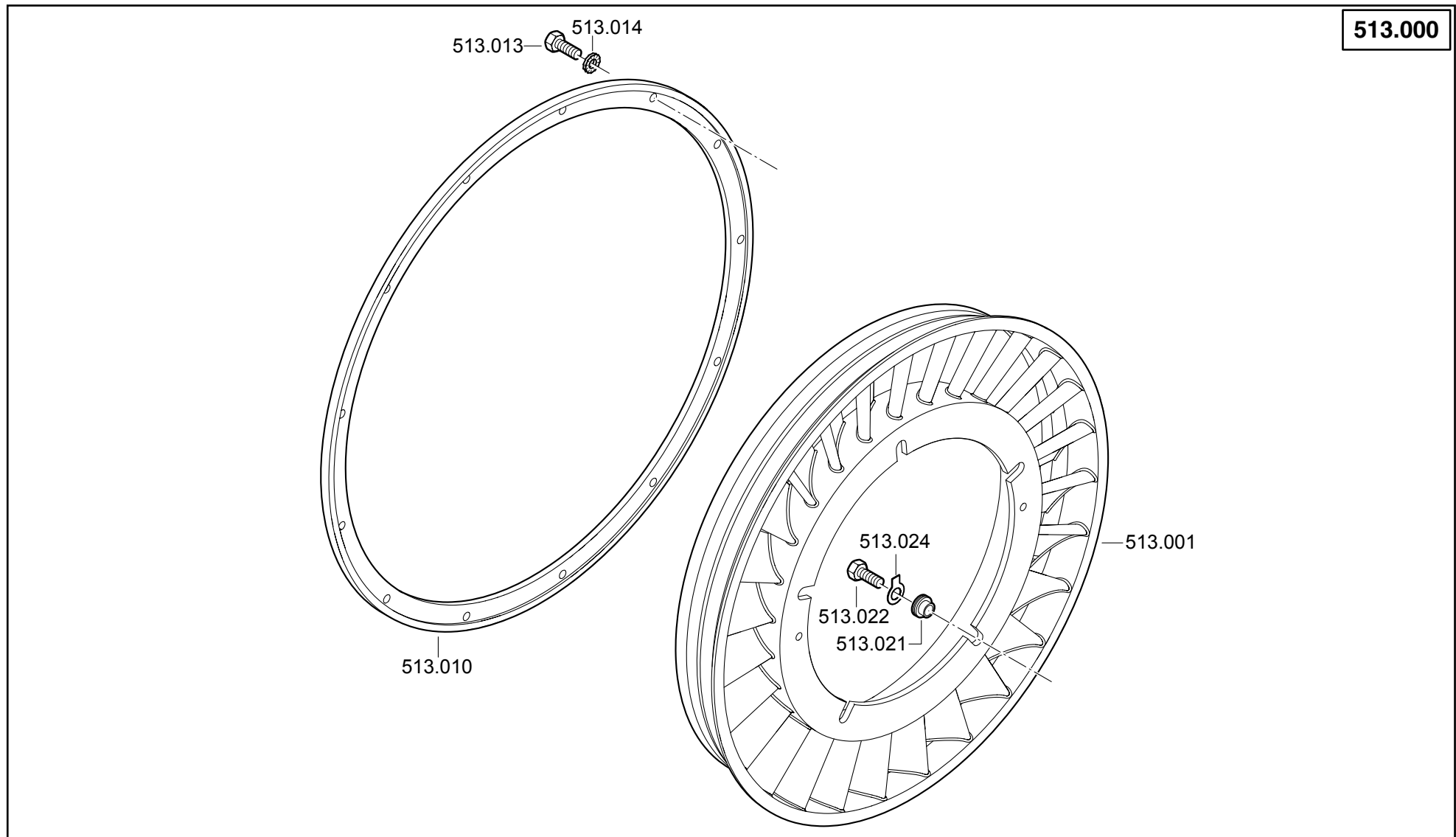




Adjusting Device for VTA

511.01

Order No.	Number	Designation	Order No.	Number	Designation
511.002	1	Spindle drive	511.307	3	Hexagon bolt
511.080	1	Lubrication grease	511.308	3	Lock washer pair
511.091	1	Support	511.309	3	Hexagon nut
511.101	1	Plate	511.310	3	Lock washer pair
511.104	2	Hexagon bolt	511.320	1	Pipe clamp
511.105	2	Lock washer pair	511.321	1	Hexagon bolt
511.108	3	Cylindrical screw	511.322	1	Lock washer pair
511.109	3	Lock washer pair	511.330	2	Connecting socket
511.118	2	Hexagon bolt	511.331	2	Threaded socket
511.119	2	Lock washer pair	511.352	1	Cooling water pipe
511.150	1	Plate	511.353	1	Cooling water pipe
511.160	2	Cylindrical screw	511.354	1	Cooling water pipe
511.161	2	Lock washer pair	511.355	1	Cooling water pipe
511.162	4	Hexagon bolt	512.020	1	Servo motor
511.163	4	Lock washer pair			
511.170	1	Shaft joint			
511.171	1	Distance sleeve			
511.175	4	Cylindrical screw			
511.176	4	Lock washer pair			
511.177	4	Hexagon bolt			
511.178	4	Lock washer pair			
511.300	1	Distributor casing			
511.301	1	Sheet-metal angle			
511.305	4	Hexagon bolt			
511.306	4	Lock washer pair			

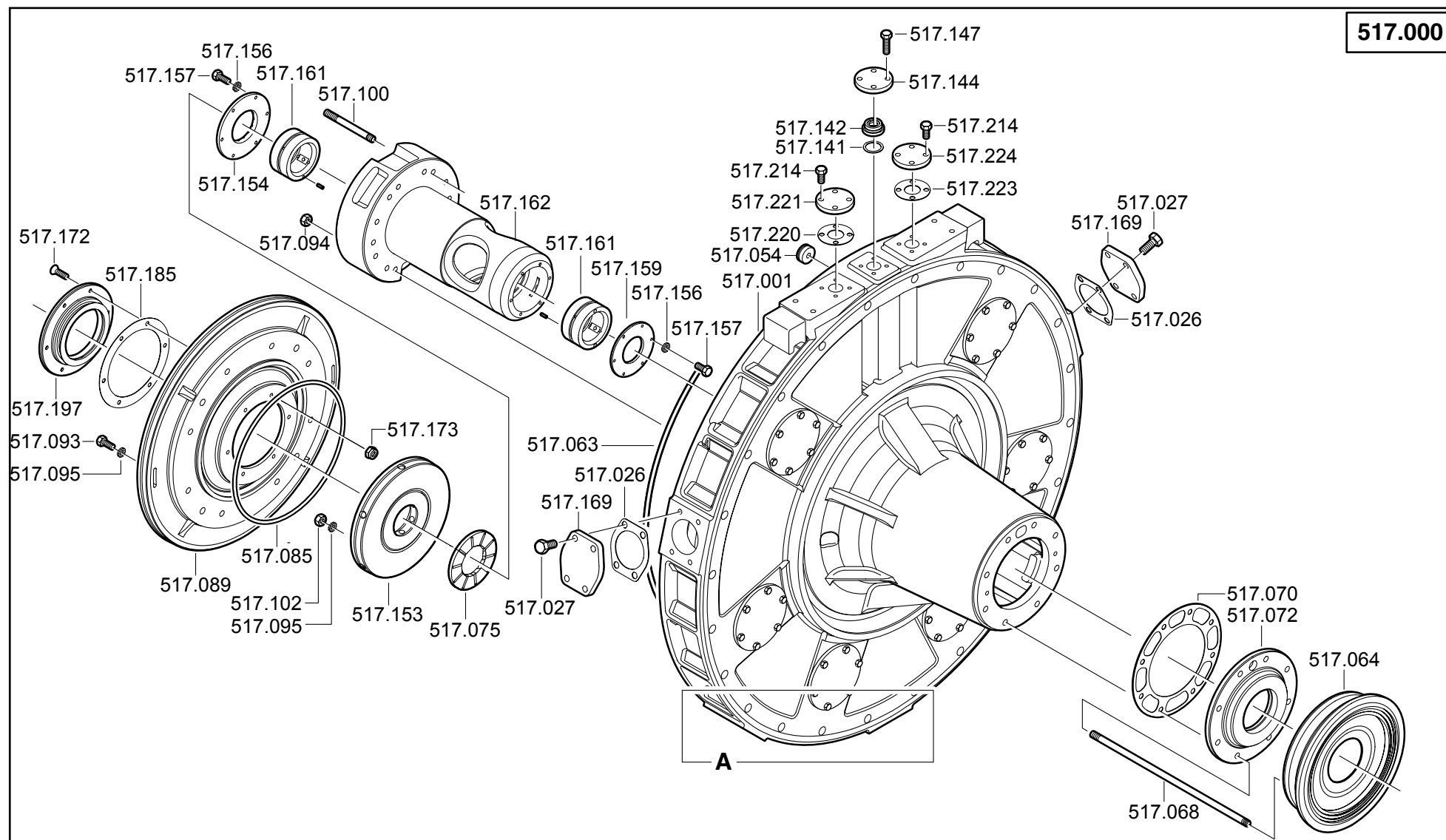


Turbine Nozzle Ring

513.02

Order No.	Number	Designation	Order No.	Number	Designation
513.000	1	Turbine nozzle ring, compl.			
513.001	1	Nozzle ring			
513.010	1	Clamping ring			
513.013	13	Hexagon bolt			
513.014	13	Lock washer pair			
513.021	5	Guide bolt			
513.022	5	Hexagon bolt			
513.024	5	Tab washer			





Bearing Casing

517.03

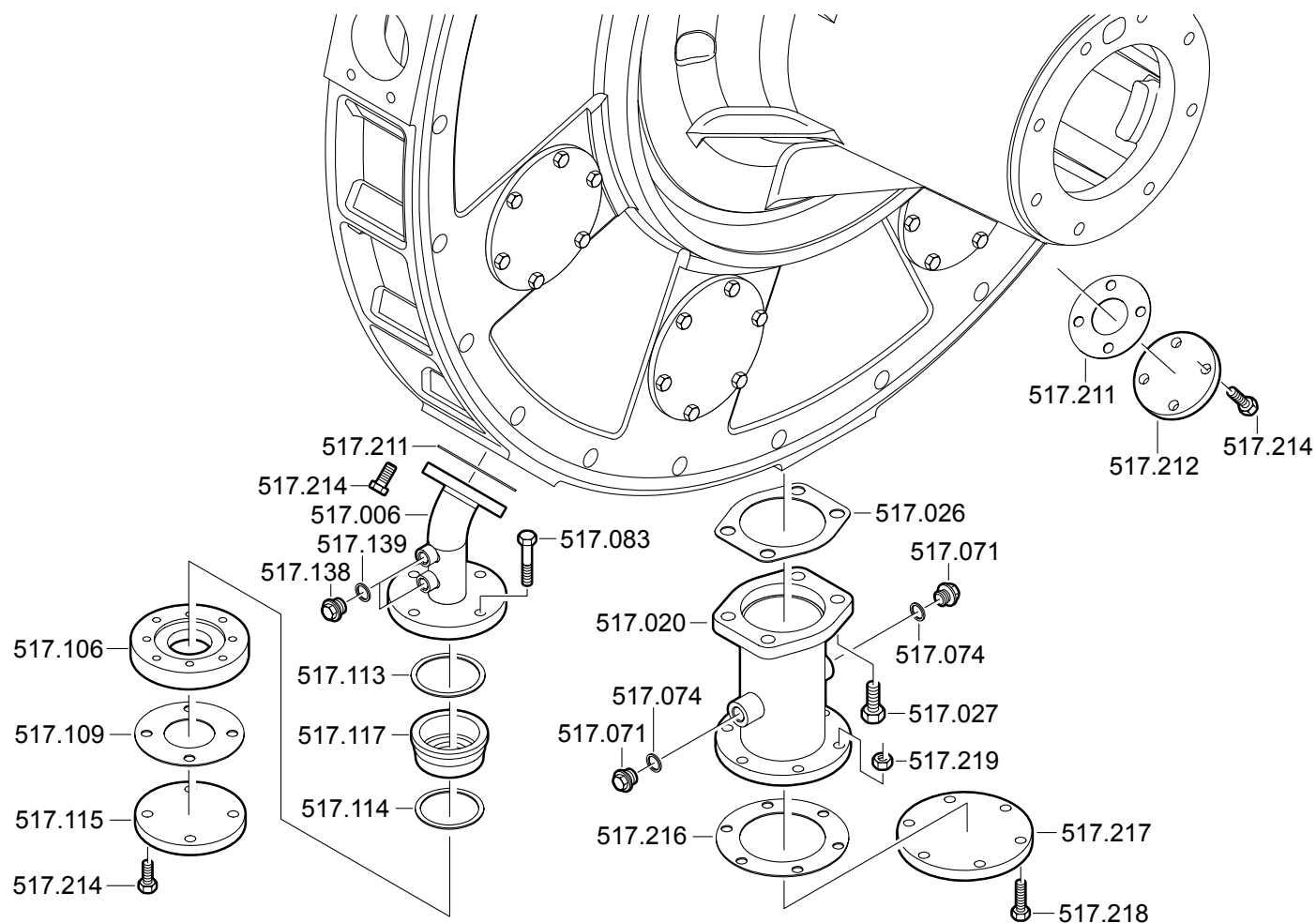
Order No.	Number	Designation	Order No.	Number	Designation
517.000	1	Bearing casing, compl.	517.154	1	Counter-thrust bearing
517.001	1	Bearing casing with items 517.100 and 517.162	517.156	12	Lock washer pair
517.026	2	Seal	517.157	12	Hexagon bolt
517.027	8	Hexagon bolt	517.159	1	Cover
517.054	1	O-ring seal	517.161	2	Bearing bush
517.063	1	O-ring seal	517.162	1	Bearing body
517.064	1	End cover, turbine side	517.169	2	Blind flange
517.068	8	Stud	517.172	6	Countersunk head screw
517.070	1	Seal, turbine side	517.173	6	Lock nut
517.072	1	Sealing cover, turbine side	517.185	1	Seal, compressor side
517.075	1	Bearing disk	517.197	1	Sealing cover, compressor side
517.085	1	O-ring seal	517.214	8	Hexagon bolt
517.089	1	Labyrinth disk, compressor side	517.220	1	Sealing ring
517.093	8	Hexagon bolt	517.221 ¹⁾	1	Flange
517.094	8	Lock nut	517.223	1	Sealing ring
517.095	14	Lock washer pair	517.224 ¹⁾	1	Flange
517.100	6	Stud			
517.102	6	Hexagon nut			
517.141	1	Sealing ring			
517.142	1	Non-return valve			
517.144	1	Flange			
517.147	4	Hexagon bolt			
517.153	1	Thrust bearing			

1) Only for version without gravitation tank



517.000

A



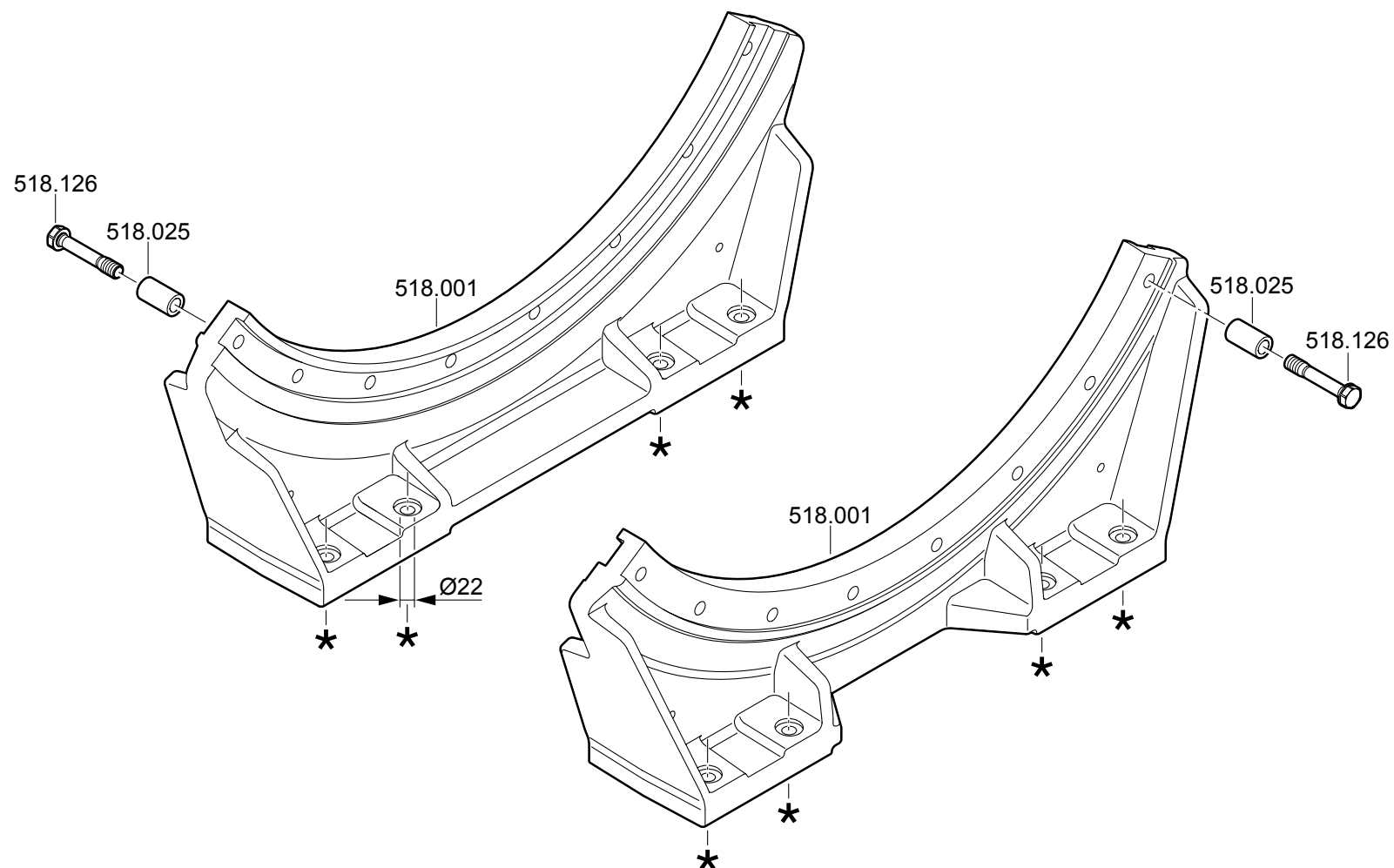
Bearing Casing (cut out A)

517.03

Order No.	Number	Designation	Order No.	Number	Designation
517.000	1	Bearing casing, compl.			
517.006	1	Pipe bend			
517.020	1	Pipe adapter			
517.026	1	Seal			
517.027	4	Hexagon bolt			
517.071	2	Screw plug			
517.074	2	Sealing ring			
517.083	4	Hexagon bolt			
517.106	1	Flange			
517.109	1	Seal			
517.113	1	Sealing ring			
517.114	1	Sealing ring			
517.115	1	Flange			
517.117	1	Non-return valve			
517.138	2	Screw plug			
517.139	2	Sealing ring			
517.211	2	Sealing ring			
517.212	1	Flange			
517.214	12	Hexagon bolt			
517.216	1	Sealing ring			
517.217	1	Flange			
517.218	6	Hexagon bolt			
517.219	6	Hexagon nut			



518.000



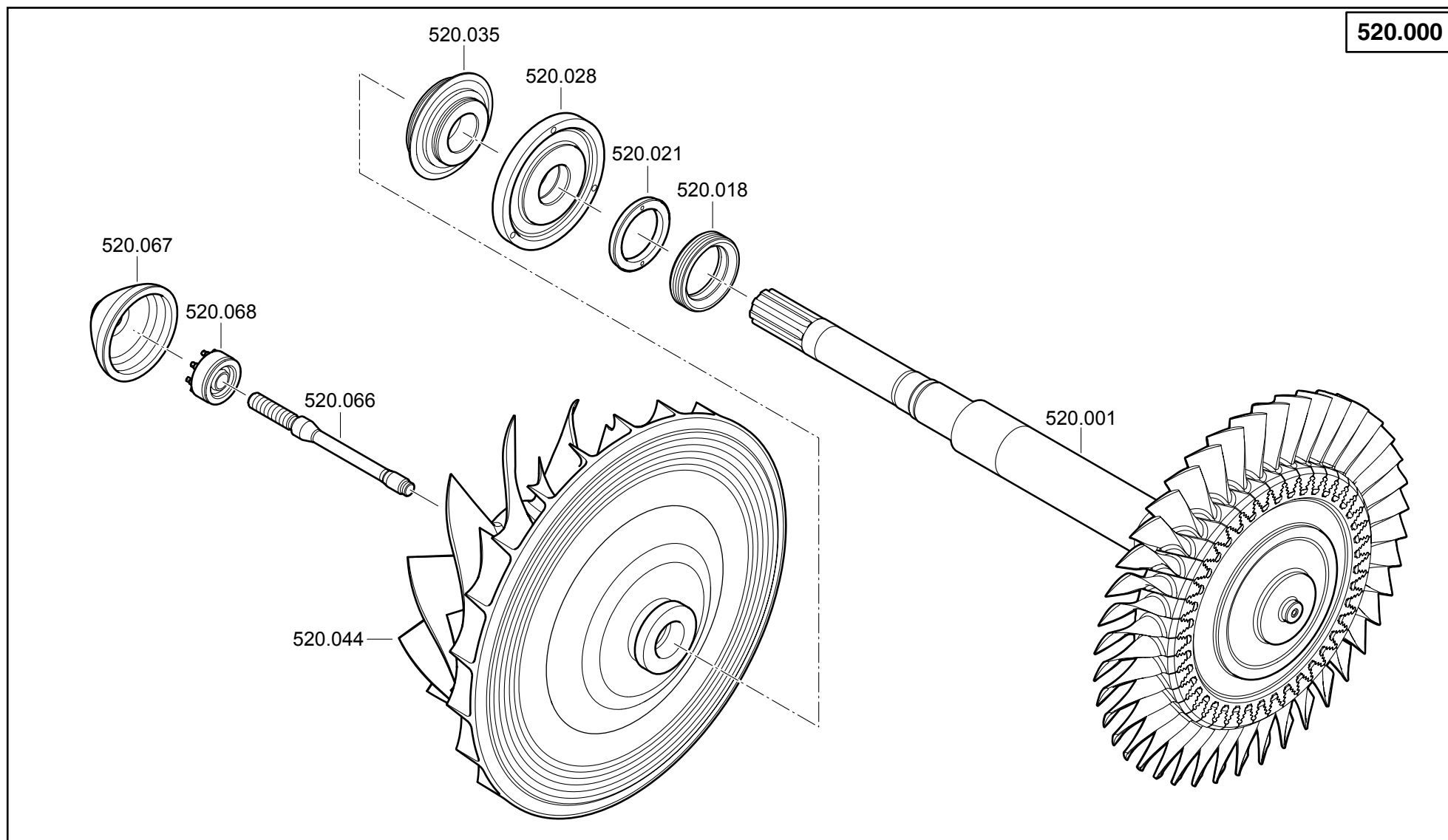
Casing Feet

518.01

Order No.	Number	Designation	Order No.	Number	Designation
518.000	1	Casing feet, compl.*)			
518.001	2	Casing foot			
518.025	16	Distance sleeve			
518.126	16	Undercut bolt			

*) Fixing elements see spare parts (engine)

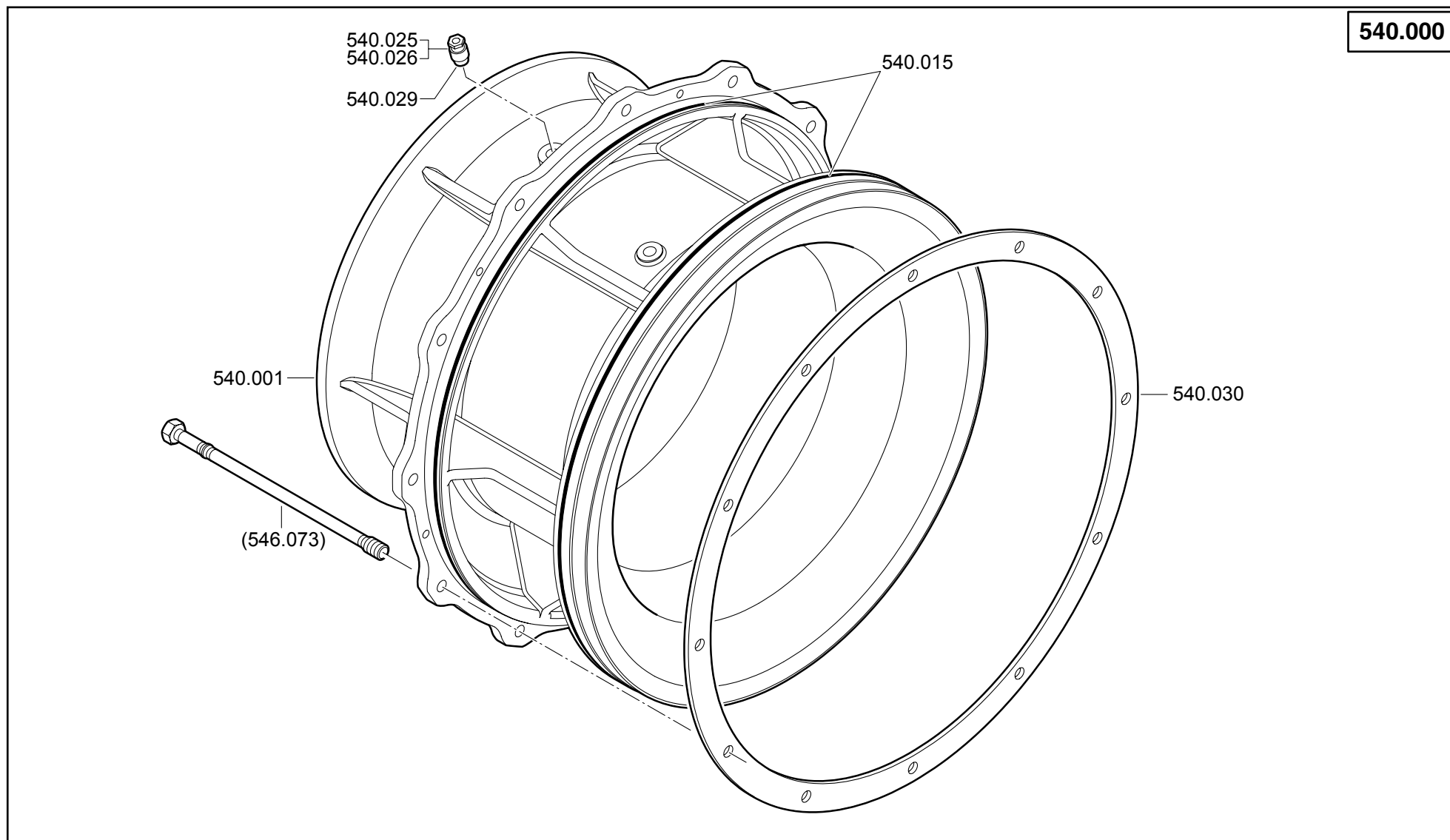




Rotor, complete**520.02**

Order No.	Number	Designation	Order No.	Number	Designation
520.000	1	Rotor, complete			
520.001	1	Turbine rotor with items 520.018 and 520.021			
520.018	1	Labyrinth ring, turbine side			
520.021	1	Ring nut			
520.028	1	Thrust ring			
520.035	1	Labyrinth ring, compressor side			
520.044	1	Compressor wheel			
520.066	1	Undercut bolt			
520.067	1	Spinner			
520.068	1	Tension nut			





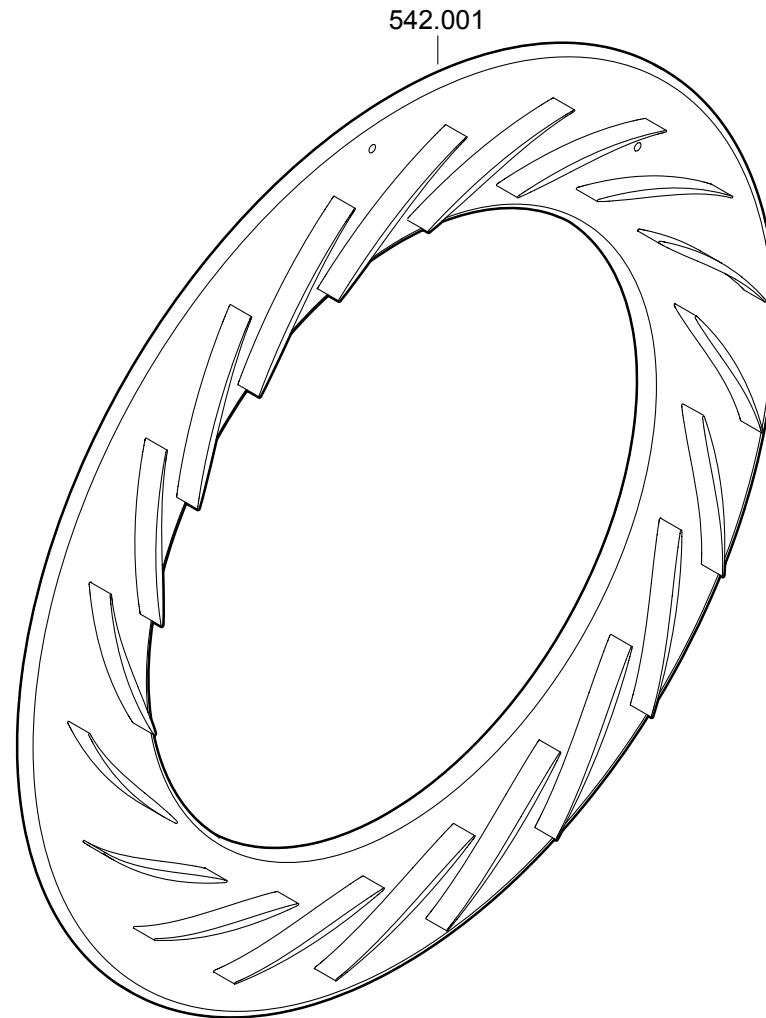
Insert, complete**540.01**

Order No.	Number	Designation	Order No.	Number	Designation
540.000	1	Insert, complete			
540.001	1	Insert			
540.015	2	O-ring seal ¹⁾			
540.025	1	Cable socket			
540.026	1	Seal disk			
540.029	1	Metal adhesive			
540.030	1	Balance ring			

1) Only for version with Jet-Assist



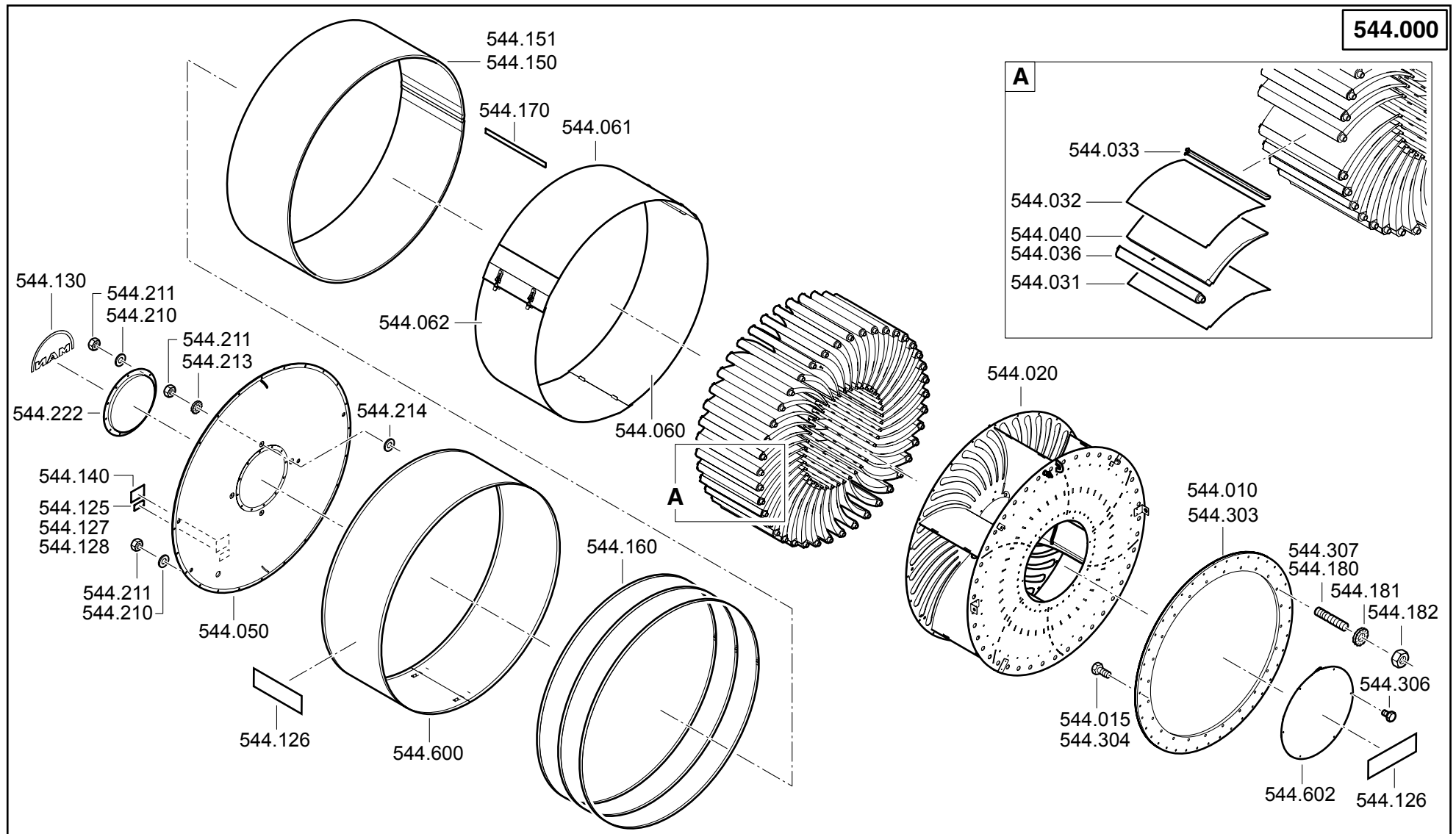
542.001



Diffuser**542.01**

Order No.	Number	Designation	Order No.	Number	Designation
542.001	1	Diffuser			



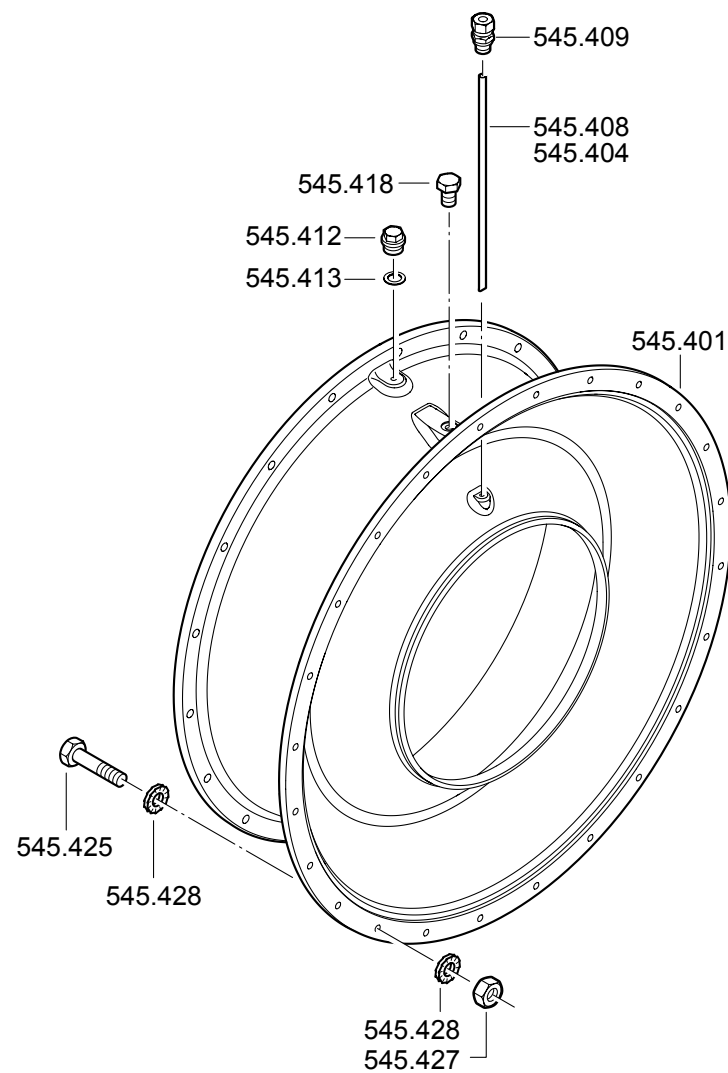


Silencer, complete**544.02**

Order No.	Number	Designation	Order No.	Number	Designation
544.000	1	Radial type silencer, complete	544.182	24	Hexagon nut
544.010	1	Flange	544.210	36	Washer
544.015	47	Hexagon bolt	544.211	40	Hexagon nut
544.020	1	Silencer casing	544.213	4	Lock washer pair
544.031	47	Inner perforated sheet	544.214	4	Washer
544.032	47	Outer perforated sheet	544.222	1	Cover
544.033	47	Inner guide rail	544.303	1	Adhesive
544.036	47	Outer guide rail	544.304	1	Adhesive
544.040	47	Damping pad	544.306	6	Hexagon bolt
544.050	1	Front wall	544.307	1	Loctite 586
544.060	1	Perforated sheet	544.600	1	Transport cover
544.061	1	Perforated sheet	544.602	1	Transport cover
544.062	1	Perforated sheet			
544.125	1	Plate			
544.126	2	Adhesive plate			
544.127	1	Sealant compound			
544.128	1	Activator			
544.130	1	Plate			
544.140	1	Sticker			
544.150	1	Fleece mat			
544.151	1	Fleece mat			
544.160	3	Lashing strap			
544.170	1	Cover strip			
544.180	24	Stud			
544.181	24	Lock washer pair			



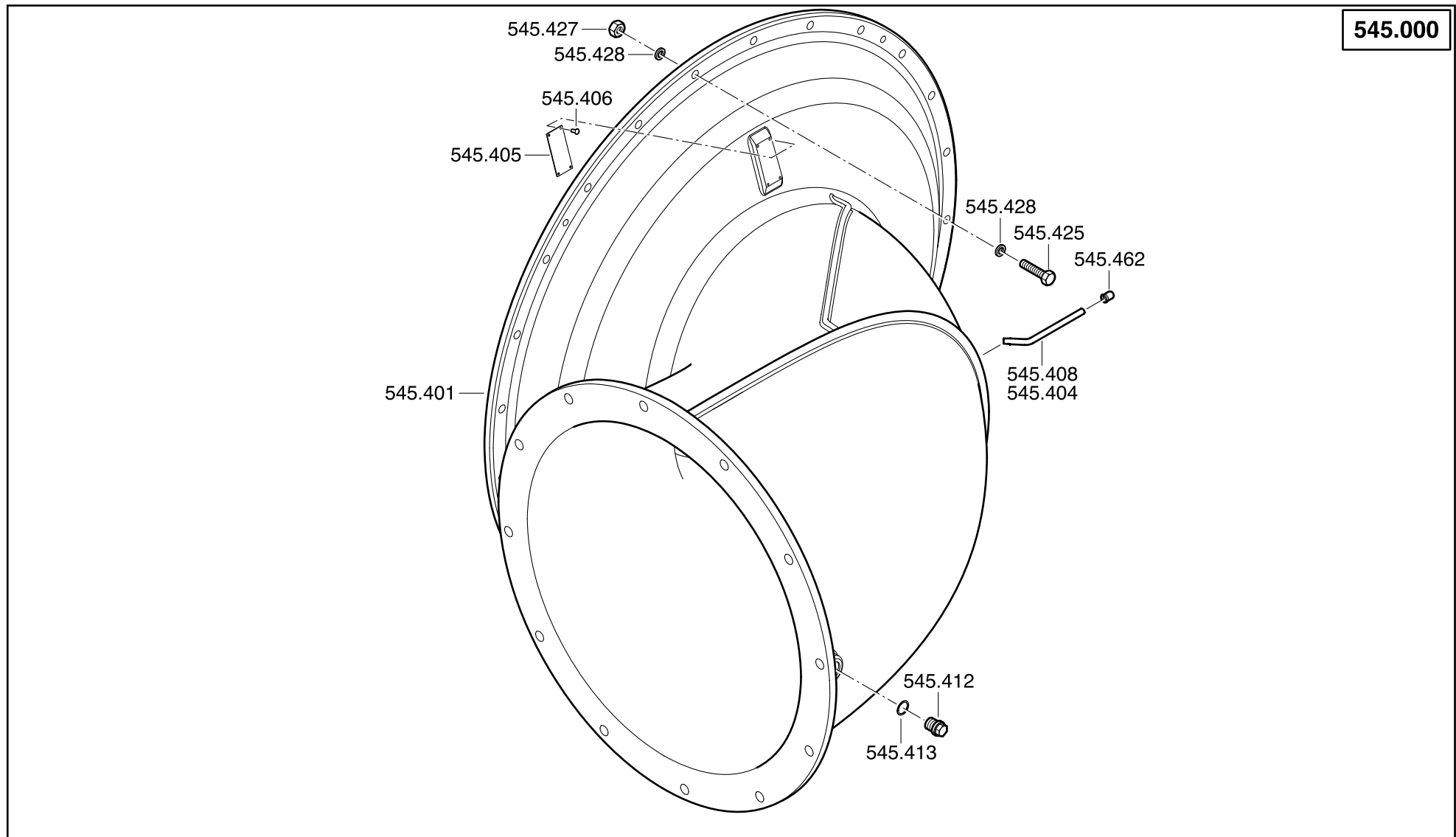
545.000



Air Intake Casing, axial**545.01**

Order No.	Number	Designation	Order No.	Number	Designation
545.000	1	Air intake casing, complete			
545.401	1	Air intake pipe			
545.404	1	Gasket			
545.408	1	Injection pipe			
545.409	1	Screw connection			
545.412	1	Screw plug			
545.413	1	Sealing ring			
545.418	1	Hexagon bolt			
545.425	24	Hexagon bolt			
545.427	24	Hexagon nut			
545.428	48	Lock washer pair			

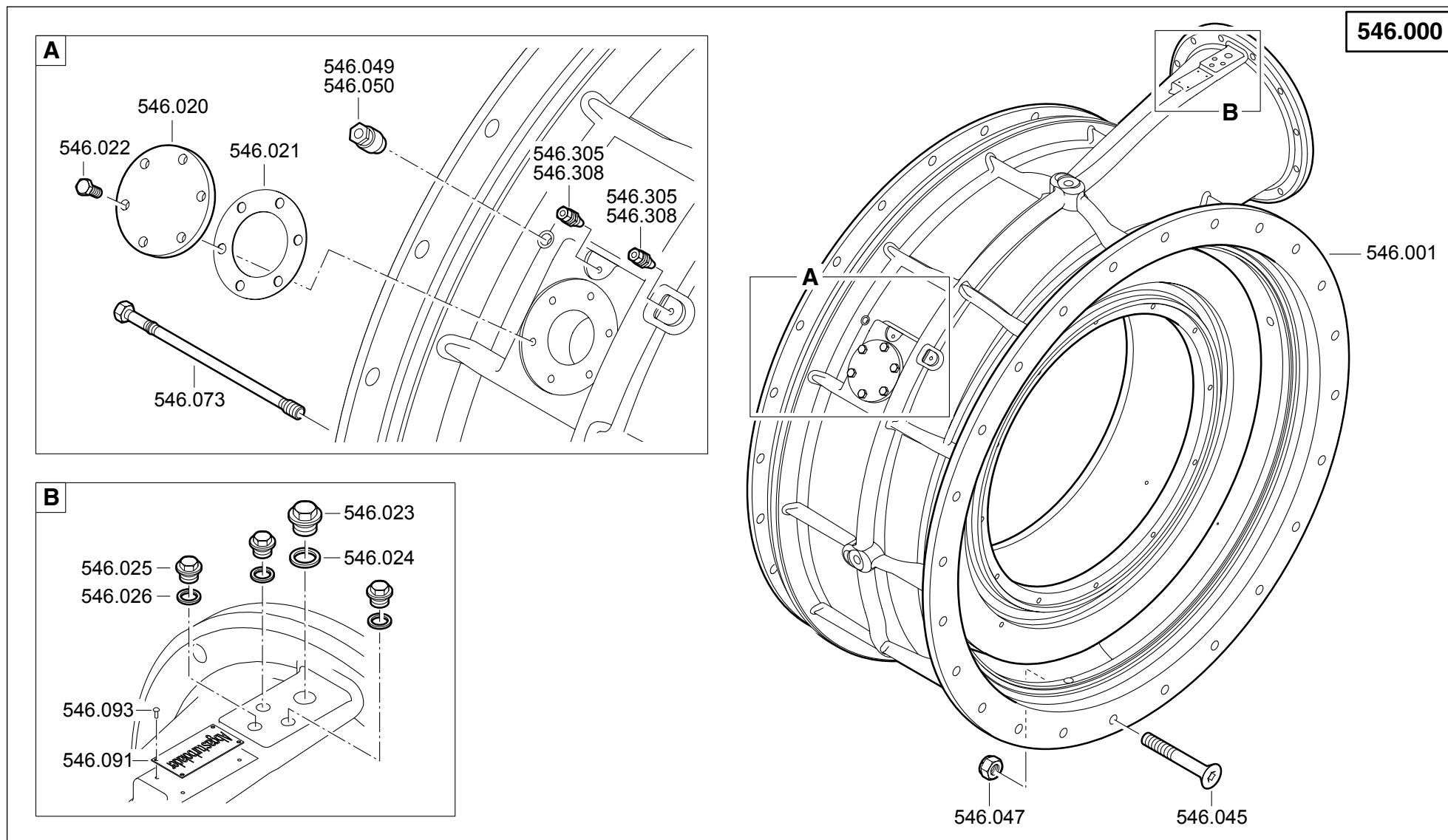




Air Intake Casing 90°**545.02**

Order No.	Number	Designation	Order No.	Number	Designation
545.000	1	Air intake casing, complete			
545.401	1	Air intake casing			
545.404	1	Metal adhesive			
545.405	1	Type plate			
545.406	4	Notched nail			
545.408	1	Injection pipe			
545.412	1	Screw plug			
545.413	1	Sealing ring			
545.418	1	Hexagon bolt			
545.425	24	Hexagon bolt			
545.427	24	Hexagon nut			
545.428	48	Lock washer pair			
545.462	1	Protecting cap			



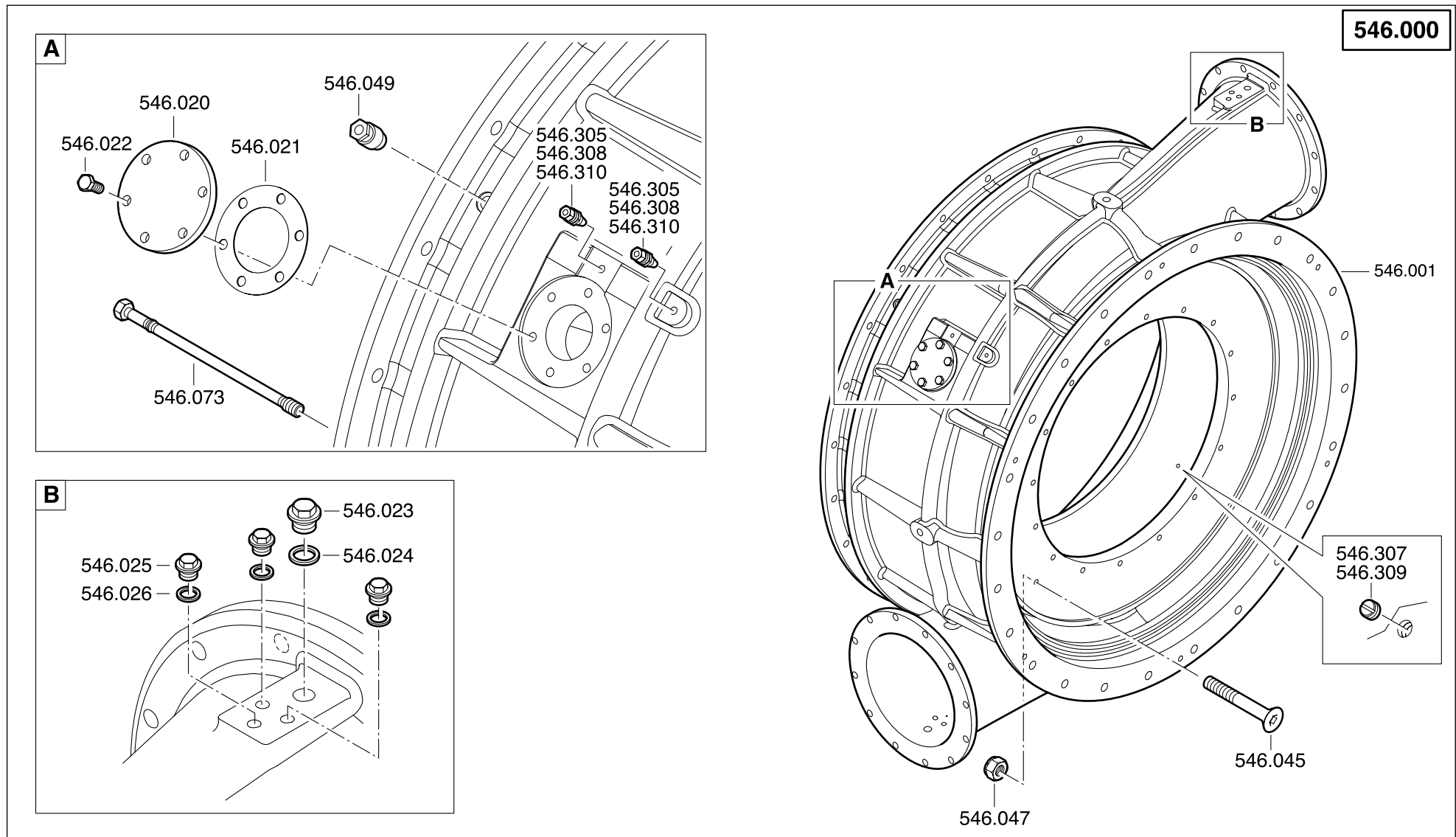


Compressor Casing (single outlet)**546.01**

Order No.	Number	Designation	Order No.	Number	Designation
546.000	1	Compressor casing, compl.			
546.001	1	Volute casing			
546.020	1	Flange			
546.021	1	Sealing ring			
546.022	6	Hexagon bolt			
546.023	1	Screw plug			
546.024	1	Sealing ring			
546.025	3	Screw plug			
546.026	3	Sealing ring			
546.045	1	Countersunk bolt			
546.047	1	Lock nut			
546.049	1	Cable socket			
546.050	1	Seal disk			
546.073	12	Undercut bolt			
546.091	1	Type plate			
546.093	4	Notched nail			
546.305	2	Screw connection			
546.307	2	Screw plug			
546.308	2	Closure plug			
546.309		Metal adhesive			
546.310	1	Taper bush			

Compressor Casing (double outlet)

546.02



6671 C3 546.02 E

03.09

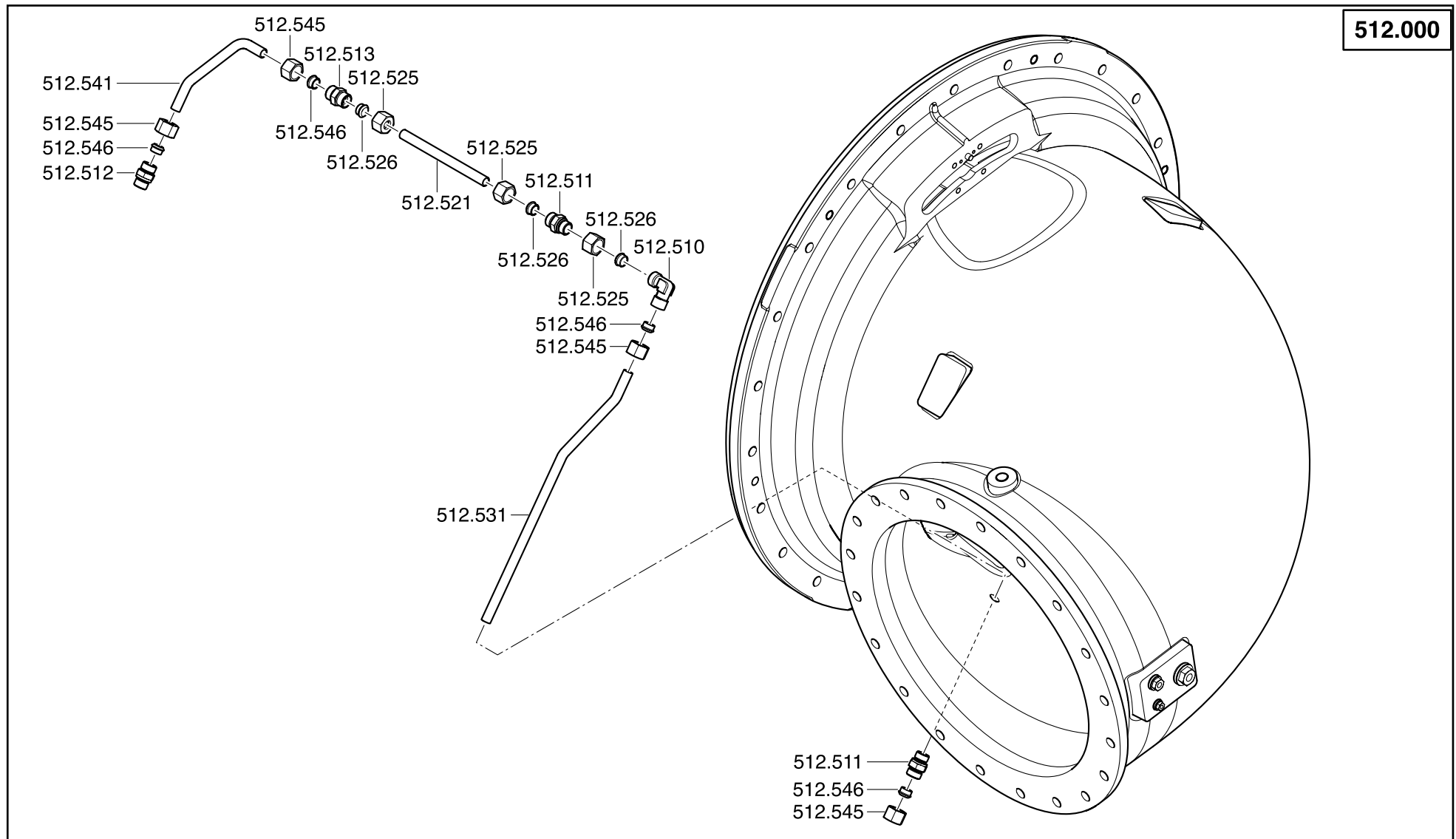
TCA66

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Compressor Casing (double outlet)**546.02**

Order No.	Number	Designation	Order No.	Number	Designation
546.000	1	Compressor casing, compl.			
546.001	1	Volute casing			
546.020	1	Flange			
546.021	1	Sealing ring			
546.022	6	Hexagon bolt			
546.023	2	Screw plug			
546.024	2	Sealing ring			
546.025	6	Screw plug			
546.026	6	Sealing ring			
546.045	1	Countersunk bolt			
546.047	1	Lock nut			
546.049	1	Cable socket			
546.073	12	Undercut bolt			
546.305	4	Screw connection			
546.307	4	Screw plug			
546.308	4	Closure plug			
546.309	1	Metal adhesive			
546.310	4	Taper bush			



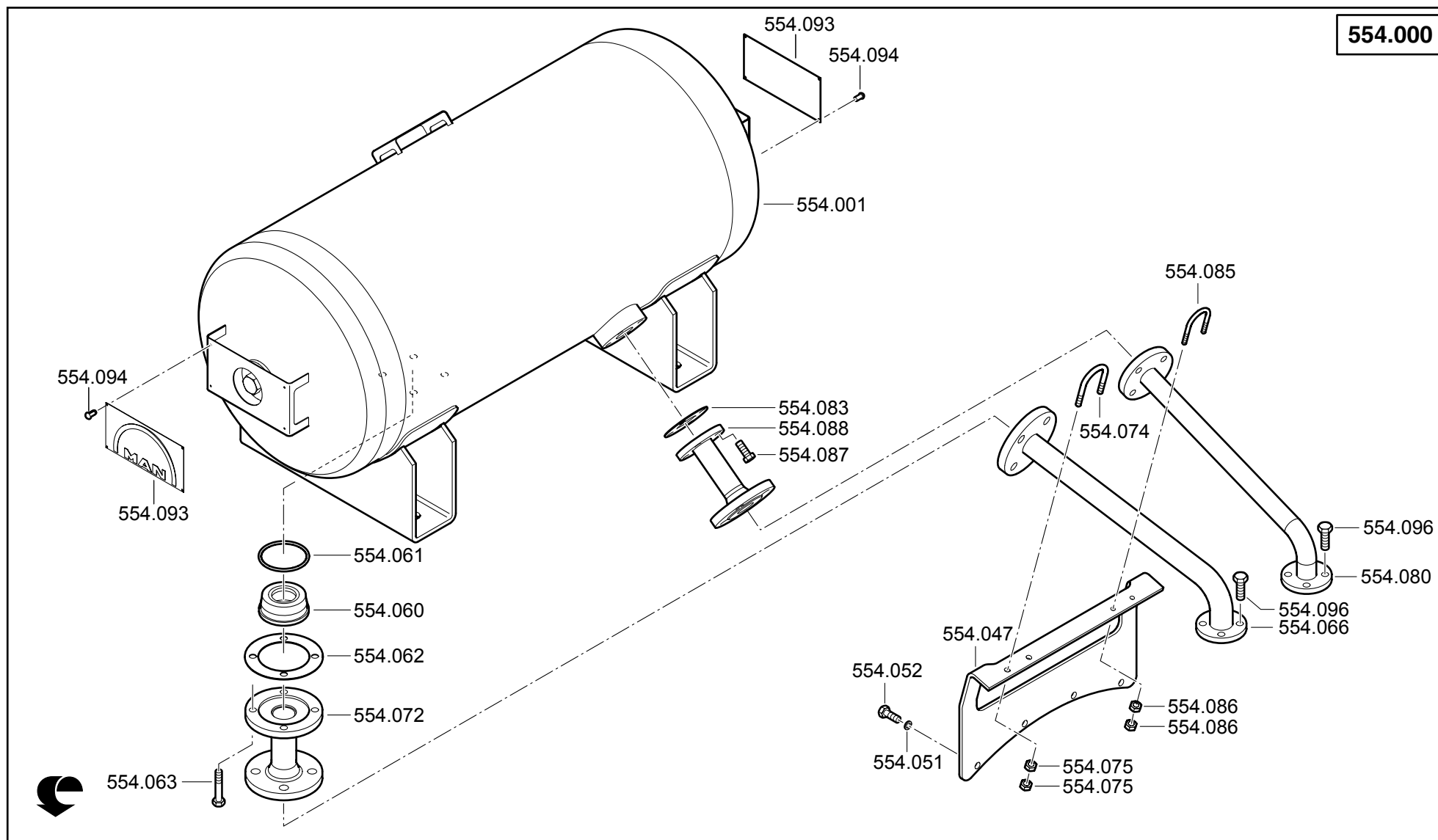


Sealing Air Pipe for VTA

549.04

Order No.	Number	Designation	Order No.	Number	Designation
512.000	1	Sealing air pipe, complete			
512.010	1	Socket			
512.011	2	Threaded socket			
512.012	1	Threaded socket			
512.013	1	Connection socket			
512.021	1	Air pipe			
512.031	1	Air pipe			
512.041	1	Air pipe			

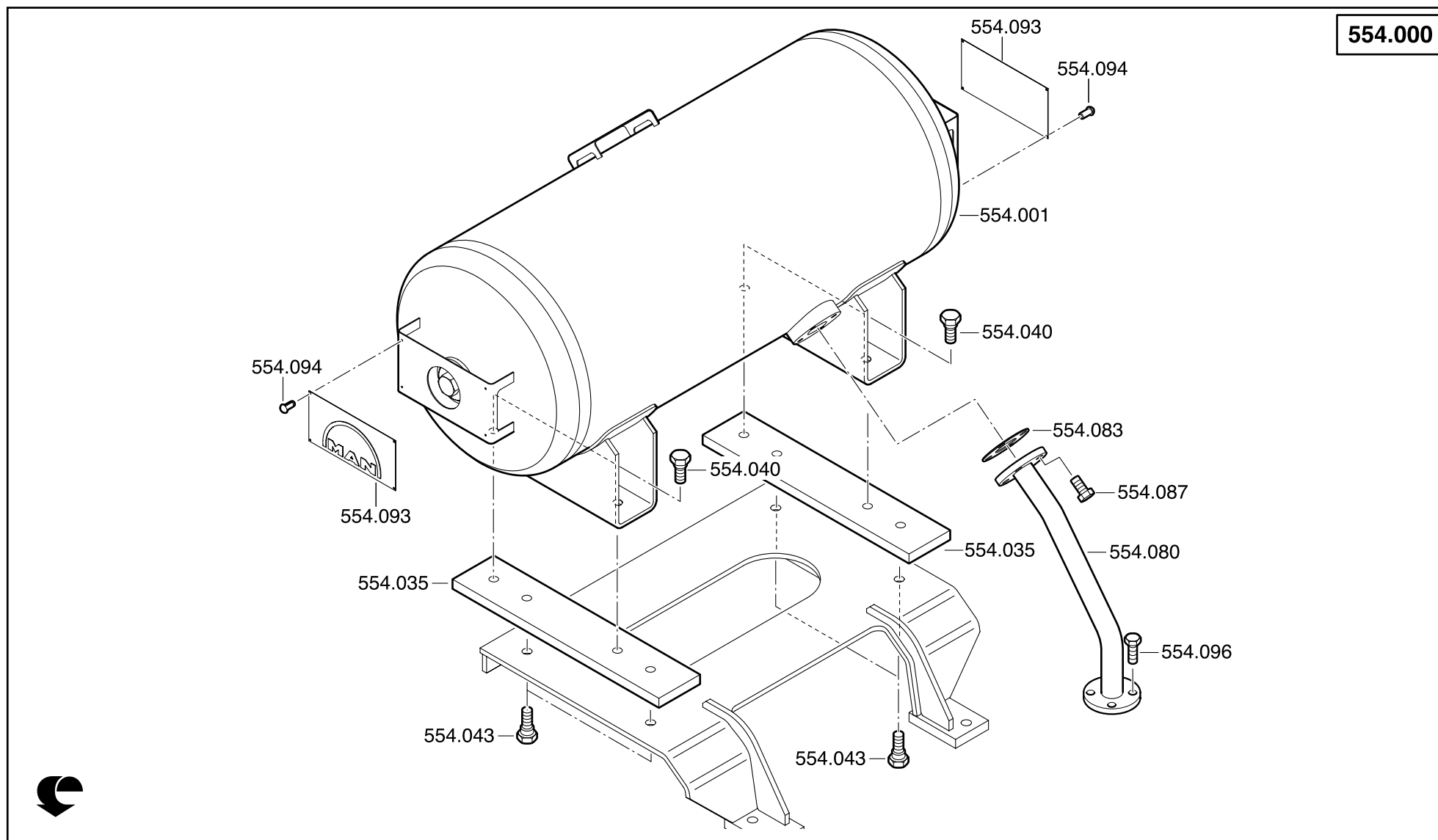




Gravitation Tank, removed**554.04**

Order No.	Number	Designation	Order No.	Number	Designation
554.000	1	Gravitation tank, complete			
554.001	1	Oil tank			
554.047	1	Plate			
554.051	4	Lock washer pair			
554.052	4	Hexagon bolt			
554.060	1	Non-return valve			
554.061	1	Sealing ring			
554.062	1	Seal			
554.063	4	Hexagon bolt			
554.066	1	Adapter			
554.072	1	Adapter			
554.074	1	U-bolt			
554.075	4	Hexagon nut			
554.080	1	Adapter			
554.083	1	Sealing ring			
554.085	1	U-bolt			
554.086	4	Hexagon nut			
554.087	4	Hexagon bolt			
554.088	1	Adapter			
554.093	2	Plate			
554.094	8	Blind rivet			
554.096	8	Hexagon bolt			



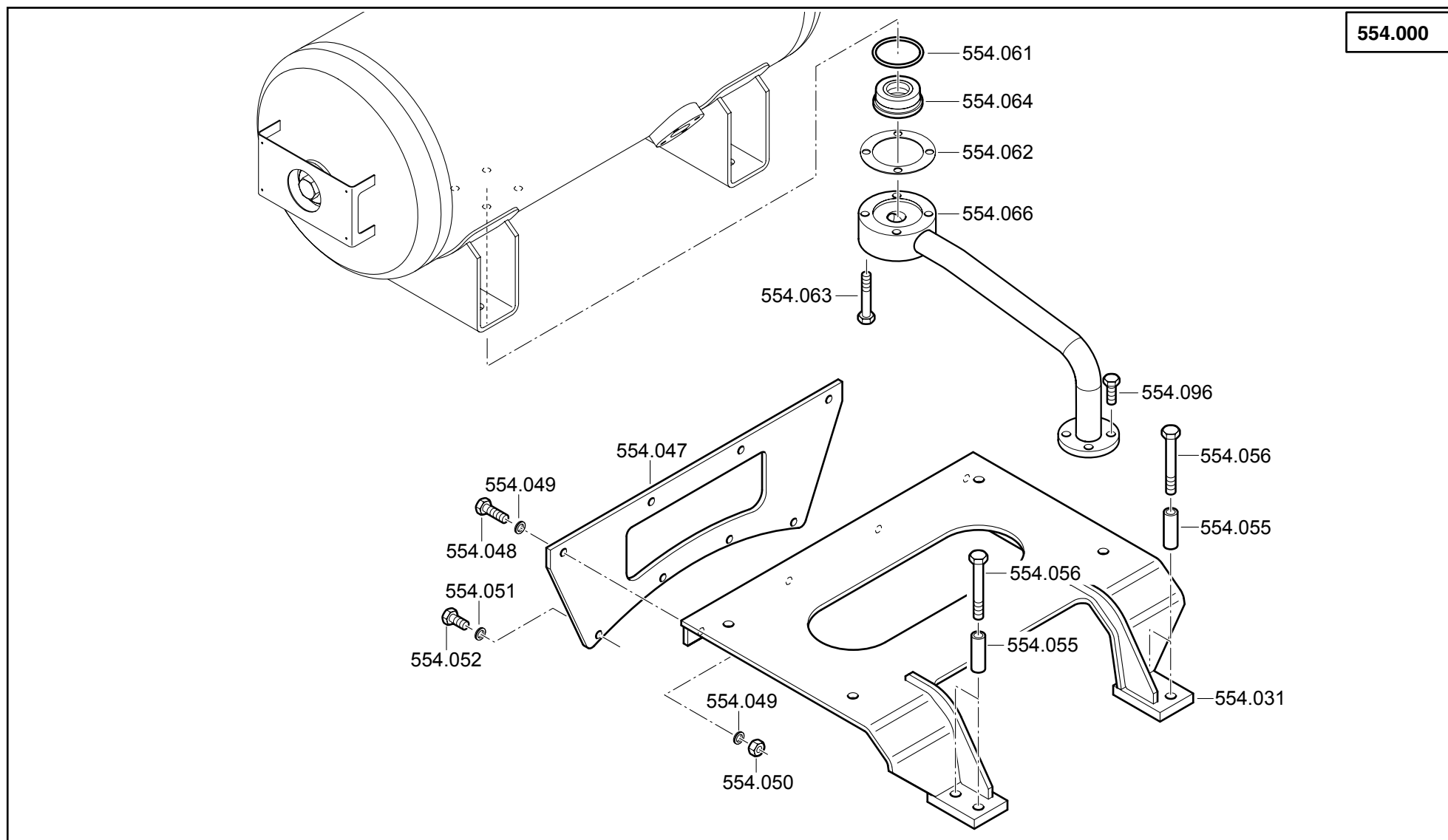


Gravitation Tank

554.05

Order No.	Number	Designation	Order No.	Number	Designation
554.000	1	Gravitation tank, complete			
554.001	1	Oil tank			
554.035	1	Plate			
554.040	4	Hexagon bolt			
554.043	4	Hexagon bolt			
554.080	1	Piping			
554.083	1	Sealing ring			
554.087	4	Hexagon bolt			
554.093	2	Plate			
554.094	8	Blind rivet			
554.096	4	Hexagon bolt			



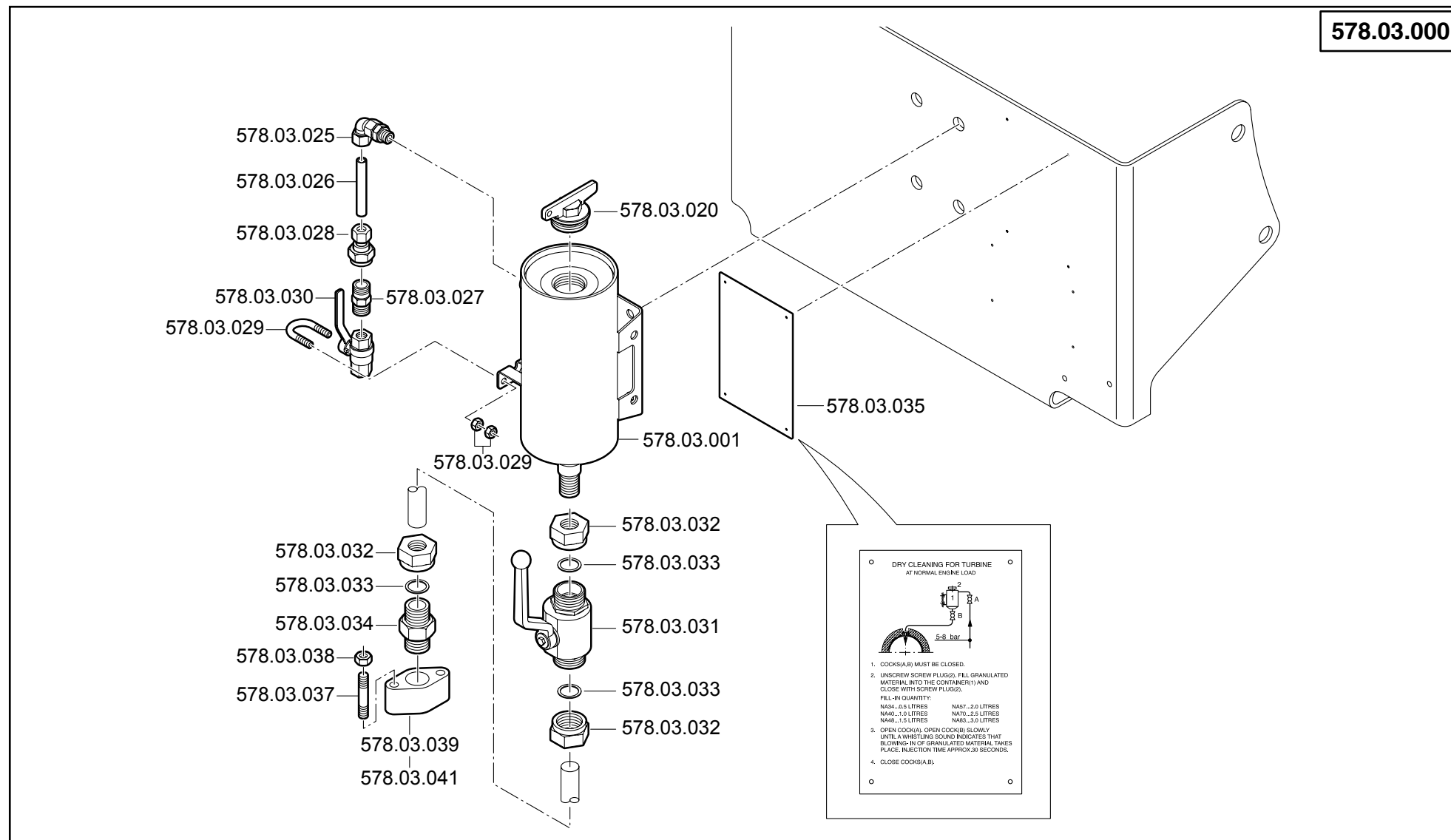


Gravitation Tank

554.05

Order No.	Number	Designation	Order No.	Number	Designation
554.031	1	Bracket			
554.047	1	Plate			
554.048	4	Hexagon bolt			
554.049	8	Lock washer pair			
554.050	4	Hexagon nut			
554.051	4	Lock washer pair			
554.052	4	Hexagon bolt			
554.055	4	Resilient sleeve			
554.056	4	Hexagon bolt			
554.061	1	Sealing ring			
554.062	1	Seal			
554.063	4	Hexagon bolt			
554.064	1	Non-return valve			
554.066	1	Piping			
554.096	4	Hexagon bolt			





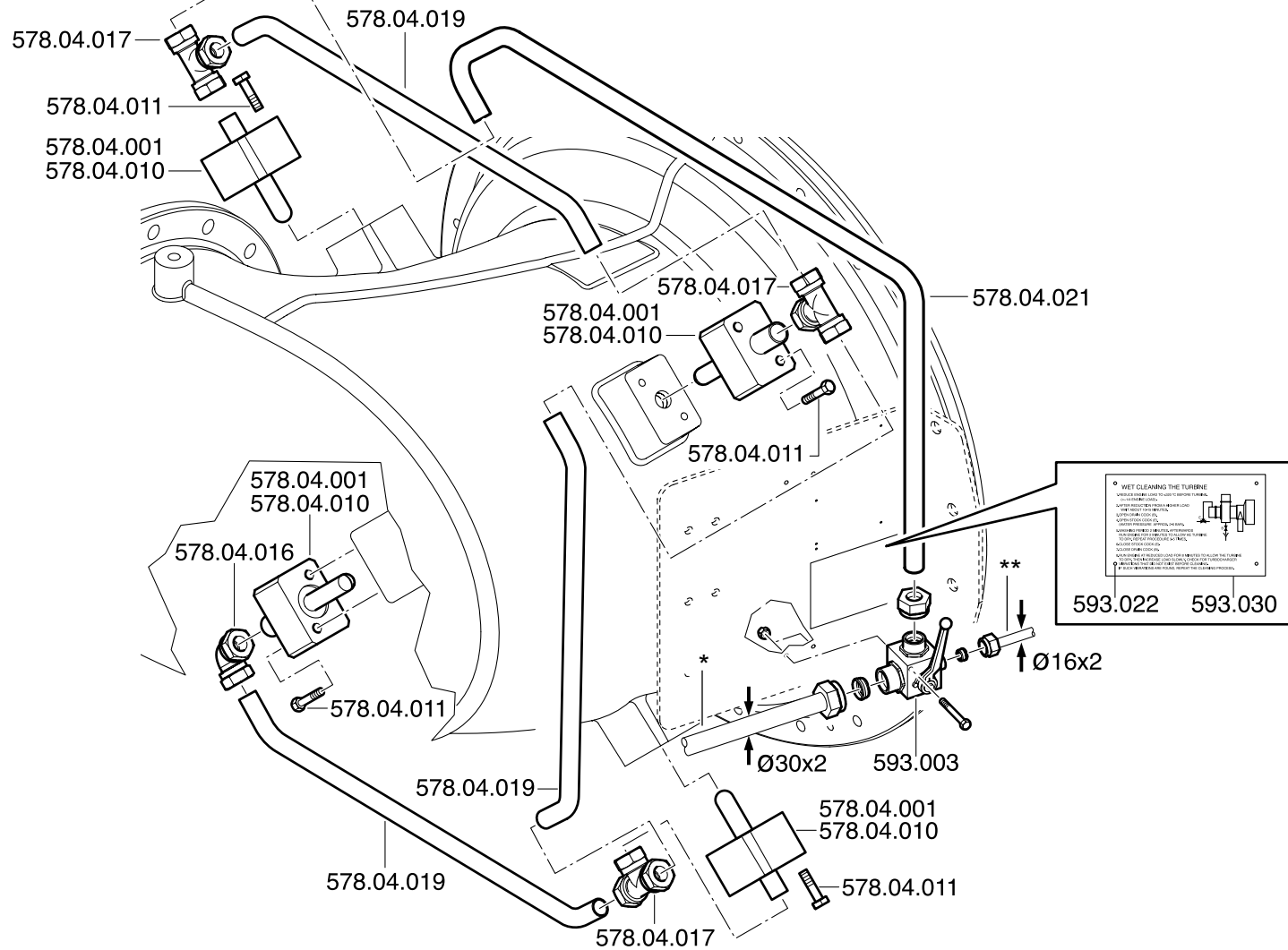
Dry Cleaning of the Turbine

578.03

Order No.	Number	Designation	Order No.	Number	Designation
578.03.000	1	Washing equipment turbine, compl.			
578.03.001	1	Container			
578.03.020	1	Screw plug			
578.03.025	1	Angular screw connection			
578.03.026	1	Pipe			
578.03.027	1	Threaded socket			
578.03.028	1	Reducing connection			
578.03.029	1	U-bolt			
578.03.030	1	Ball cock			
578.03.031	1	Ball cock			
578.03.032	3	Cap nut			
578.03.033	3	Self-tapping ring			
578.03.034	1	Threaded socket			
578.03.035	1	Plate (cleaning instructions)			
578.03.037	2	Stud			
578.03.038	2	Hexagon nut			
578.03.039	1	Flange			
578.03.041		Sealing cement, heat-resistant			



578.04.000



Wet Cleaning of the Turbine

578.04

Order No.	Number	Designation	Order No.	Number	Designation
578.04.000	1	Cleaning device for turbine, compl.			
578.04.001	4	Spray nozzle			
578.04.010	1	Sealing cement, heat-resistant			
578.04.011	8	Hexagon bolt			
578.04.016	1	Angular screw connection			
578.04.017	3	T-screw connection			
578.04.019	3	Pipe			
578.04.021	1	Pipe			
593.003	1	Ball cock			
593.022	4	Blind rivet			
593.030	1	Plate (cleaning specification)			
*	1	Water supply connection			
**	1	Sealing air			



579.03.000

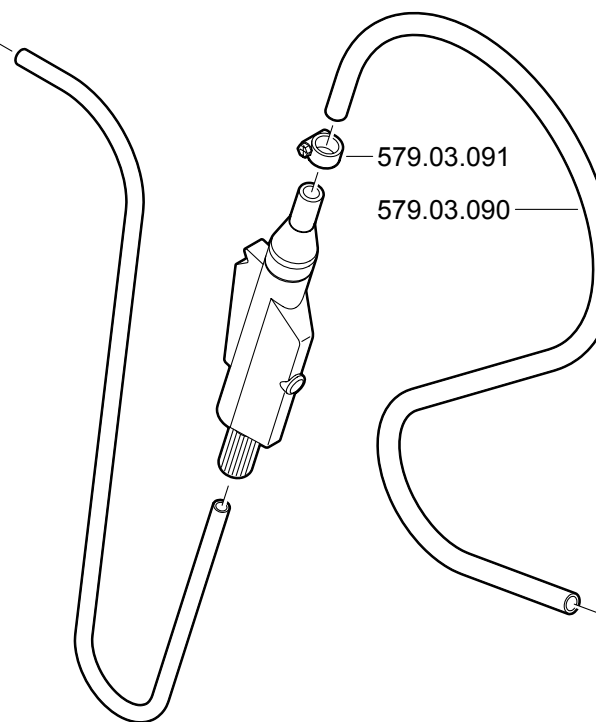
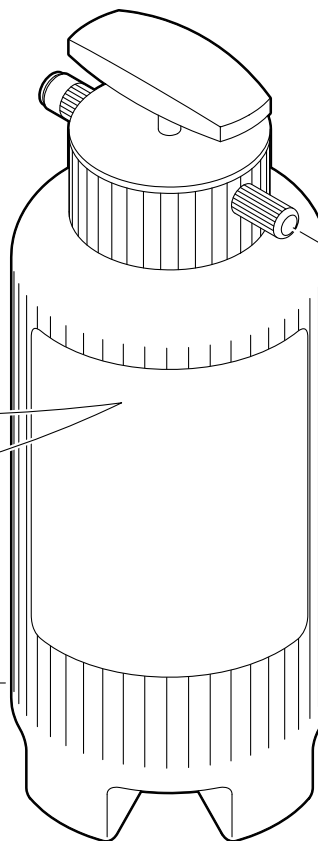
Compressor cleaning
with clean water (no sea water)
charge 2 l
operating overpressure 3 bar
Cleaning only when the engine is
hot and at full load operation
Do not stop the engine
immediately after cleaning

Admission
Press the pump slightly down and turn
left. Remove the pump. Each time before
loosening the pump, discharge the
overpressure by drawing out the safety
valve.
Fill in water up to the 2 liter marking
at the container scale.
Screw in the pump in the container,
acute the pump untill an operatingpressure
of 3 is reached in the container.
>> Red colour range at the piston
pressure gauge <<
Overpressure discharged through the
safety valve.

COMPRESSURE CLEANING
move the quick-acting clutch in the
wash-up connection for compressure.
Press the push button of the hand valve
to initiate the cleaning process. By
releasing the push button of the
spray procedure is interrupted.
Wash up time approx. 30 seconds.

579.03.179

579.03.029



579.03.091

579.03.090

579.03.091

579.03.096

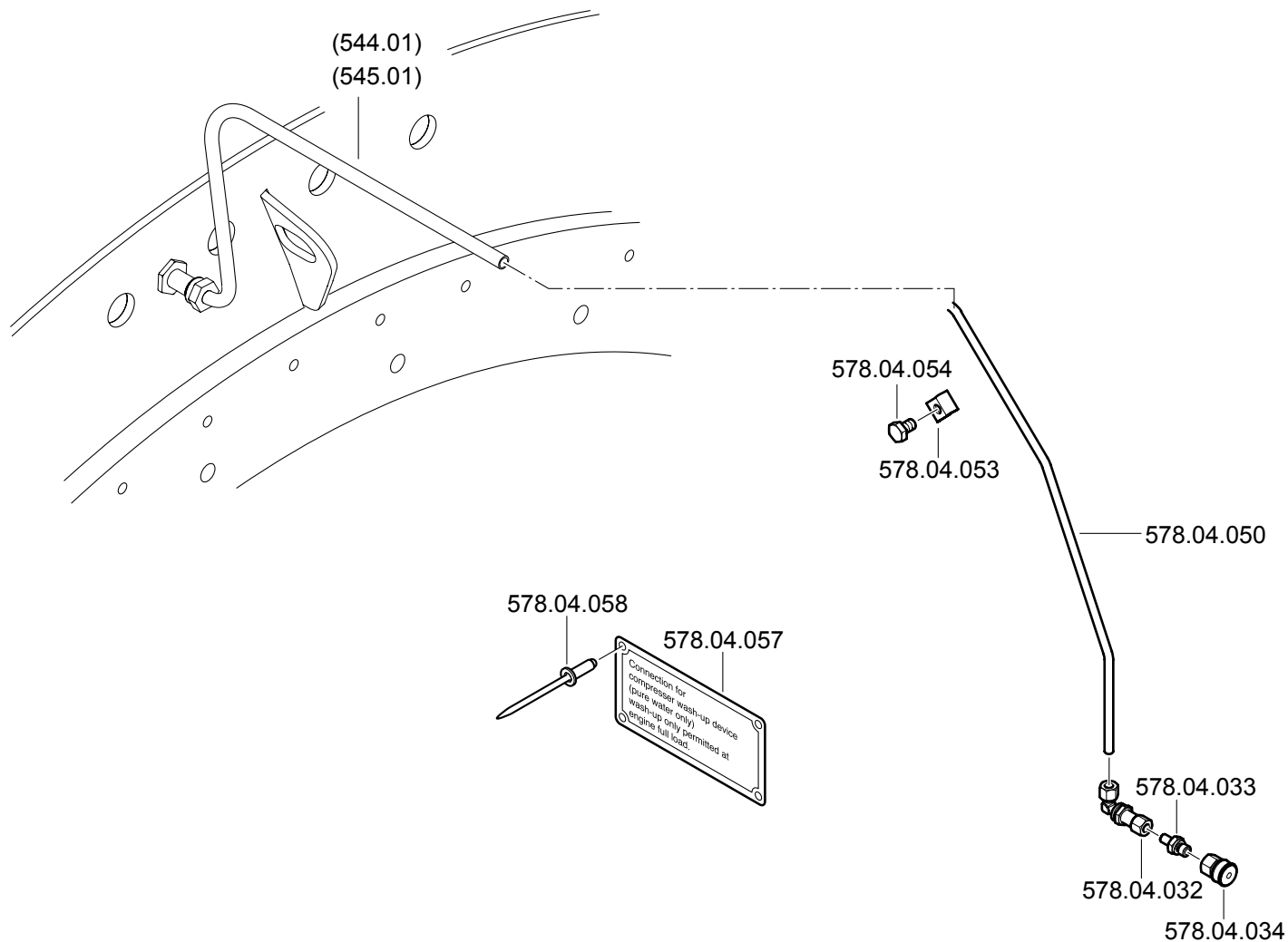


Wet Cleaning of the Compressor (pressure sprayer)**579.03**

Order-No.	Number	Designation	Order-No.	Number	Designation
579.03.000	1	Pressure sprayer, compl.			
579.03.029	1	Container			
579.03.090	1	Hose			
579.03.091	2	Hose clamp			
579.03.096	1	Socket			
579.03.179	1	Plate (cleaning instructions)			



579.04.000

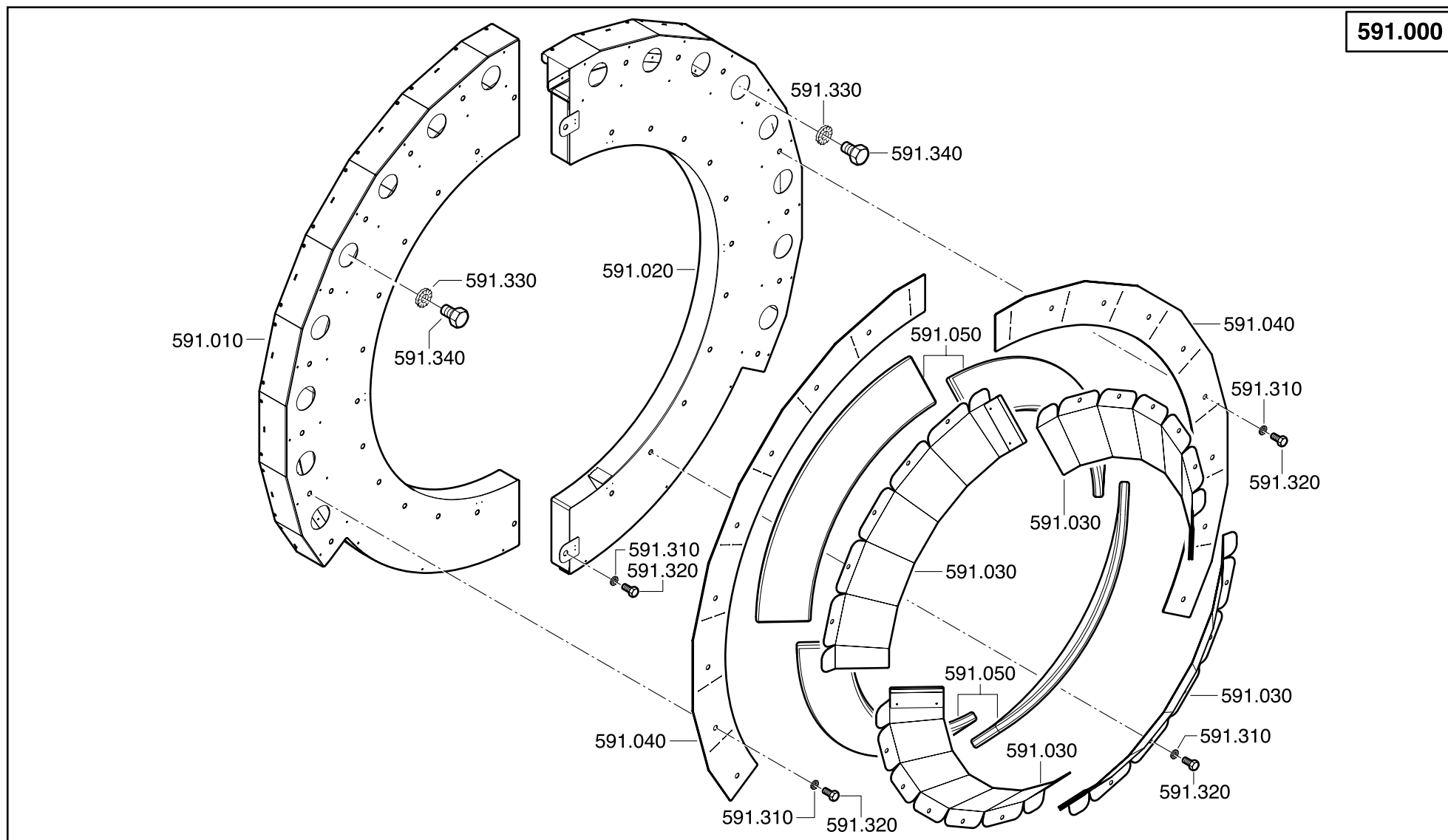


Wet Cleaning of the Compressor (pipe connection)

579.04

Order No.	Number	Designation	Order No.	Number	Designation
579.04.000	1	Pipe for compressor-washing unit, complete			
578.04.032	1	Screw connection			
578.04.033	1	Socket			
578.04.034	1	Nipple plug			
578.04.050	1	Pipe			
578.04.053	1	Pipe clip			
578.04.054	1	Hexagon bolt			
578.04.057	1	Plate (cleaning specification)			
578.04.058	4	Blind rivet			





Annular Lagging Gas-admission Casing

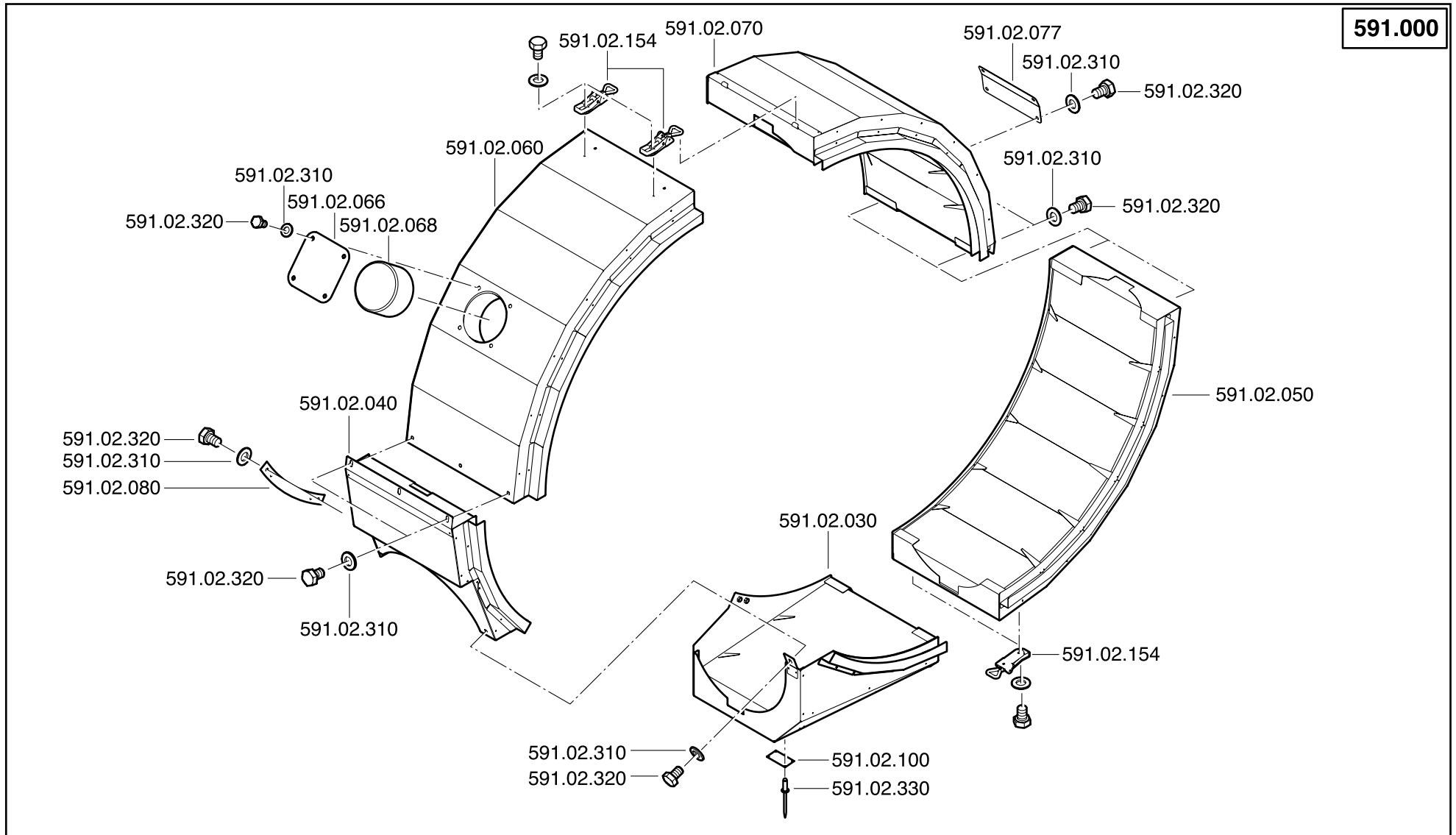
591.01

Order No.	Number	Designation	Order No.	Number	Designation
591.000	1	Ring covering, complete			
591.010	1	Ring covering half			
591.020	1	Ring covering half			
591.030	1	Jacket sheet			
591.040	2	Cover sheet			
591.050	4	Insulation			
591.310	38	Washer			
591.320	38	Hexagon bolt			
591.330	8	Lock washer pair			
591.340	8	Hexagon bolt			



Sound Insulation Compressor Casing (single outlet)

591.02



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Sound Insulation Compressor Casing (single outlet)

591.02

Order No.	Number	Designation	Order No.	Number	Designation
591.02.000	1	Sound insulation, complete			
591.02.030	1	Jacket sheet with insulation			
591.02.040	1	Jacket sheet with insulation			
591.02.050	1	Jacket sheet with insulation			
591.02.060	1	Jacket sheet with insulation			
591.02.066	1	Cover			
591.02.068	1	Insulation			
591.02.070	1	Jacket sheet with insulation			
591.02.077	1	Cover			
591.02.080	1	Strap			
591.02.100	1	Type plate			
591.02.154	4	Turnbuckle			
591.02.310	19	Washer			
591.02.320	19	Hexagon bolt			
591.02.330	4	Blind rivet			





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